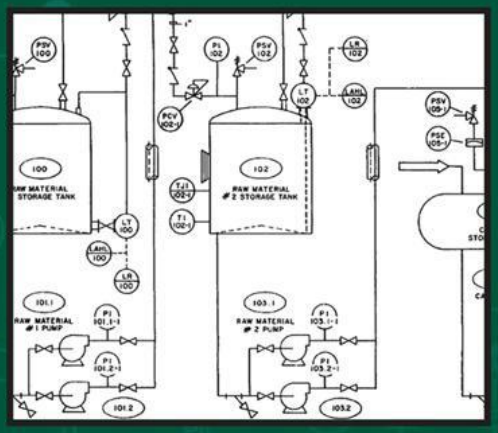
TECHNICIAN SERIES

William L. Mostia, Jr., P.E.

Troubleshooting: A Technician's Guide

2nd Edition



- Causes of failure
- Logical/analytical troubleshooting frameworks
- Safety issues
- Troubleshooting scenarios

...and more

ISA

Setting the Standard for Automation™

TROUBLESHOOTING
A TECHNICIAN'S GUIDE

2ND EDITION

William L. Mostia, Jr., P. E.

ISA TECHNICIAN SERIES



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DEDICATION

Raymond D. Molloy, Jr. (1937-1996)

The ISA Technician Series is dedicated to the memory of **Raymond** D. Molloy, Jr. Mr. Molloy was an ISA member for 34 years and held various Society offices, including Vice President of the ISA Publications Department. Mr. Molloy was a valued contributor to the ISA Publications Department for many years and led the Department in the introduction of many new ISA publications over the years.

Ray also served as President of the New Jersey Section. He was the recipient of ISA's Distinguished Society Service and Golden Achievement Award and the New Jersey Section Lifetime Achievement Award.

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1

LEARNING TO TROUBLESHOOT

Learning by doing

Apprenticeships

Mentoring

Classroom instruction

Individual study

1.1. EXPERIENCE

This chapter discusses several types of training and assistance that you can use to develop your troubleshooting skills. While some argue that troubleshooting is an art, in fact, successful troubleshooting depends more on logic and knowledge. Because of this, troubleshooting can be taught and developed. Some of the troubleshooter's skill develops naturally due to experience, but experience alone is seldom enough to produce a troubleshooter capable of tackling a wide variety of situations.

To develop a wide range of skills, a technician needs initiative, training, and assistance. To be successful in your training, you must become an active participant. You must seek out training opportunities and take responsibility for developing your skills. You cannot passively rely on your company, your supervisor, or chance to do the job for you.

Experience is the most common way technicians develop troubleshooting skills. It comes naturally with the job, and is sometimes called OJT (on-the-job training). It means getting out there and getting your hands dirty.

As a training method experience has a varied range of success. In some cases, particularly when range of experience is wide or your troubleshooting results in failure or mistakes, experience can have a lasting effect. On the other hand, if the range of experience is too narrow or if you only perform repetitive tasks, for example, experience may not teach you much. A mix of challenging and familiar tasks, though, will help you develop troubleshooting skills.

1.1.1. Information and Skills

The learning you gain from experience can be divided into two types: information and skills.

Through experience, you get information about classes of instruments and about individual instruments or systems, such as how a particular control valve works and how control valves work in general. It is particularly important to be able to generalize about classes of instruments. All control valves, for example, have components in common (such as an actuator, a stem, and a trim), which have similar functions. Knowing about these common components means that you will be familiar with the essential features of any new control valve you have to work on. If you understand the basic principles of a class of instruments, you can apply that knowledge across the board. Knowledge about specific instruments is also required because each instrument has unique features that may be pertinent to your troubleshooting task.

Skills are how you apply your knowledge to troubleshoot a particular instrument or system. Skills involve reasoning using the information available to you about the system you are troubleshooting and the techniques you have learned, such as how to calibrate or zero an instrument, how to read the power supply voltage or a particular test current, and so on.

1.1.2. Diversity and Complexity

How well experience contributes to your learning also depends on its diversity and complexity. Diversity means the range of different types of systems you have the opportunity to troubleshoot. The more different types of systems you work on, the more you gain not only a wider range of information but also a larger set of skills. Likewise, the more complex the systems that you work on, the more you can

learn. Working on complex systems requires the development of complex skill sets because complexity itself provides diversity.

1.1.3. Learning from Experience

So, how can you make the most of the experiences available to you to improve your troubleshooting skills?

•Look for opportunities to learn

•Talk to your supervisor

•Volunteer for jobs

•Volunteer to help other people

There are always opportunities for you if you want to learn. Choose work that will give you good experience. Be in charge of your training.

1.2. APPRENTICESHIPS

Apprenticeships can be of two types, formal and informal. Formal programs are done by unions or by companies. These typically involve three to five years of classroom training, hands-on experience, on-the-job training, and testing. Such training is typically very thorough, but the range may be limited because everyone gets the same training, which may not change to keep up with new instruments or may not be trained on all of the various instrument types.

Informal apprenticeships develop when an apprentice is assigned to an experienced technician for training. The success of these apprenticeships varies based on the trainer's knowledge, ability to transfer information, and willingness to do so. Apprentices who can develop good working relationships with their trainers may find this kind of instruction well worthwhile.

1.3. MENTORING

Like apprenticeships, mentoring can also be formal or informal. Many companies have formal mentoring programs in which experienced technicians serve as mentors for the less experienced. Informal mentoring happens when an experienced technician agrees to help a newer employee learn job skills. It can be in your best interest to find a mentor to help you develop your skills. Even if you cannot find a mentor, observation of how other successful troubleshooters work can be helpful. Never be afraid to learn from others.

1.4. CLASSROOM INSTRUCTION

Classroom study is the traditional way of gaining knowledge and skills. Today, a multitude of learning opportunities is available: college and community college programs, commercial courses, and courses taught by professional associations such as ISA. Company-based courses are somewhere in the middle

and tend to be more specific whereas outside courses tend to be more general. The quality and content vary, so check the course out before you sign up.

Courses with hands-on training are generally the best because most of us remember better when we do rather than when we listen or read. And classroom training alone may not be as helpful because what you are trained on may not correspond to what you work on. Always look for general principles in your training that may apply to a range of problems or instruments.

1.5. INDIVIDUAL STUDY

Finally, individual study is an important aspect of your training and your career. Programs like ISA's Certified Control Systems Technician (CCST) tests reward training at home, on the job, and in classrooms. Many of the books, videos, and computer software in ISA's publications catalog are designed for home study. Other specialized disciplines often offer home-study courses and products as well, and you can learn about them by joining other professional associations and by talking with coworkers who are members. Books and home-study courses are also available commercially. Look for ads in technical and trade magazines.

Many companies allow their technicians to attend trade shows. These can be good training opportunities because many instruments are shown in cross section, allowing you to see how the instruments are constructed. Other instruments are shown in operation and can be discussed with vendors. Reading trade magazines, most of which are free, can provide information that can help you when you are troubleshooting. Some of the free magazines are *InTech, CONTROL, Control Engineering, Personal Engineering & Instrumentation News, EC&M, Electronic Design, Sensors, AB Journal, Plant Engineering, Pipeline & Gas, Control Design, Control Solutions*, and *Hydrocarbon Processing*. Two that are available through paid subscriptions are *Measurement & Control* and *Chemical Engineering.*

1.6. LOGIC AND LOGIC DEVELOPMENT

Logic is the bedrock of troubleshooting. The use of logic permeates all aspects of troubleshooting. Yet failure to apply logic to troubleshooting represents a major shortcoming in many people's troubleshooting activities.

Where does one get proficient in the principles of logic? Unfortunately, it is not a subject that is stressed in school directly as one is expected to learn it as one goes along in learning other subjects. The closest term I have heard to address logic in school at the lower levels is development of critical thinking skills. At the college level, one can take a course in logic typically taught by the math or philosophy department but practical applications of the material as typically taught is limited. So the question remains, where does one get proficient in the principles of logic?

One approach is self-study through solving logical puzzles. There are several good books available that help the student. These are typically puzzles that involve true and false statements or reasoning about statements from which one can solve the puzzle. Some of these books are books by Raymond Smullyan — *Lady or the Tiger?* and *What is the name of this book?: The riddle of Dracula and other logical puzzles*

— and books by Norman D. Willis titled, *False Logic Puzzles*. Other puzzles that stretch your mind and require logic to solve may also serve the purpose. The idea is to get your mind working in logical patterns that you can apply to troubleshooting.

SUMMARY

The possibilities for training are virtually endless. The major training opportunities are illustrated in Figure 1-1. While some of the responsibility for the success of your training is up to your company and your supervisor, much is up to you. Take advantage of all opportunities to receive training.

QUIZ

1.The success of your training is up to

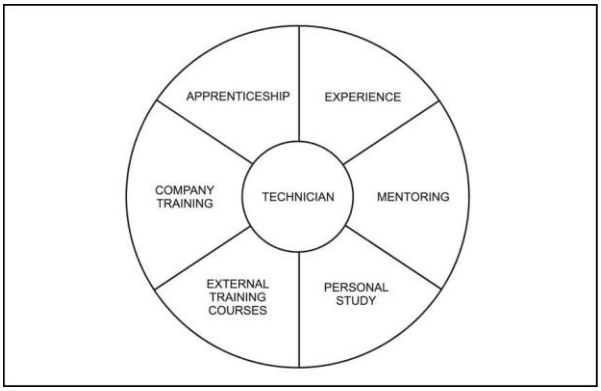
A.you

B.your company.

C.your supervisor.

D.all of the above

FIGURE 1-1 ***Training Opportunities***



2. OJT stands for

A. occupational job training.

B. on-the-job training.

C. occupational joint training.

D. none of the above

3. Mentoring is

A. guidance and assistance by a more experienced technician.

B. a form of on-the-job training.

C. classroom training by more experienced members of your group.

D. a form of correspondence training.

4. CCST stands for

A. Certified Control Service Technician.

B. Certified Contract Service Technician.

C. Certified Control System Technician.

D. none of the above

5. Experience can be divided into two areas, information learned and

A. work.

B. skills learned.

C. time on the job.

D. mistakes made.

2

THE BASICS OF FAILURES

What failure is

How hardware fails

How software fails

How environment effects failure rates

Functional failures

Systematic failures

Common cause failures

Root cause analysis

2.1 A DEFINITION OF FAILURE

Failure is the condition of not achieving a desired state or function. Everything is subject to failure—it is only a matter of when and how. Dealing with failures is a troubleshooter's business, and to troubleshoot successfully, we must first understand how failures occur.

Failures can occur due to factors such as a faulty component (hardware), an incorrect line of programming code (software), or a human error (systematic). A system can even have a functional failure when it is working properly but is asked to do something it was not designed to do or when it is exposed to a transient condition that causes a momentary failure.

Consequently, we can classify failures according to four general types:

•Hardware failures

•Software failures

•Systematic failures

•Functional failures

The troubleshooter's primary purpose in an operating plant is to find what has failed so that it can be repaired and be made available again. Keeping the process running properly is the primary concern. At its heart, this means identifying the root cause of a failure.

Failures can have internal or external causes. If the cause is internal to an instrument, that is generally the root cause; the instrument is repaired or replaced and that is the end of the problem. But the root cause may be outside the instrument itself. If a failure happens too often, the reliability of the instrument comes into question, or a common-cause failure mechanism may be involved. We will discuss these later in this chapter. If the cause is external to the instrument, or is a functional failure, a causal (cause and effect) chain may not be obvious. While we may still repair or replace the instrument, we must find the root of the problem so that we will not keep fixing the same problem. Formal root-cause analysis is discussed in section 2.8 below.

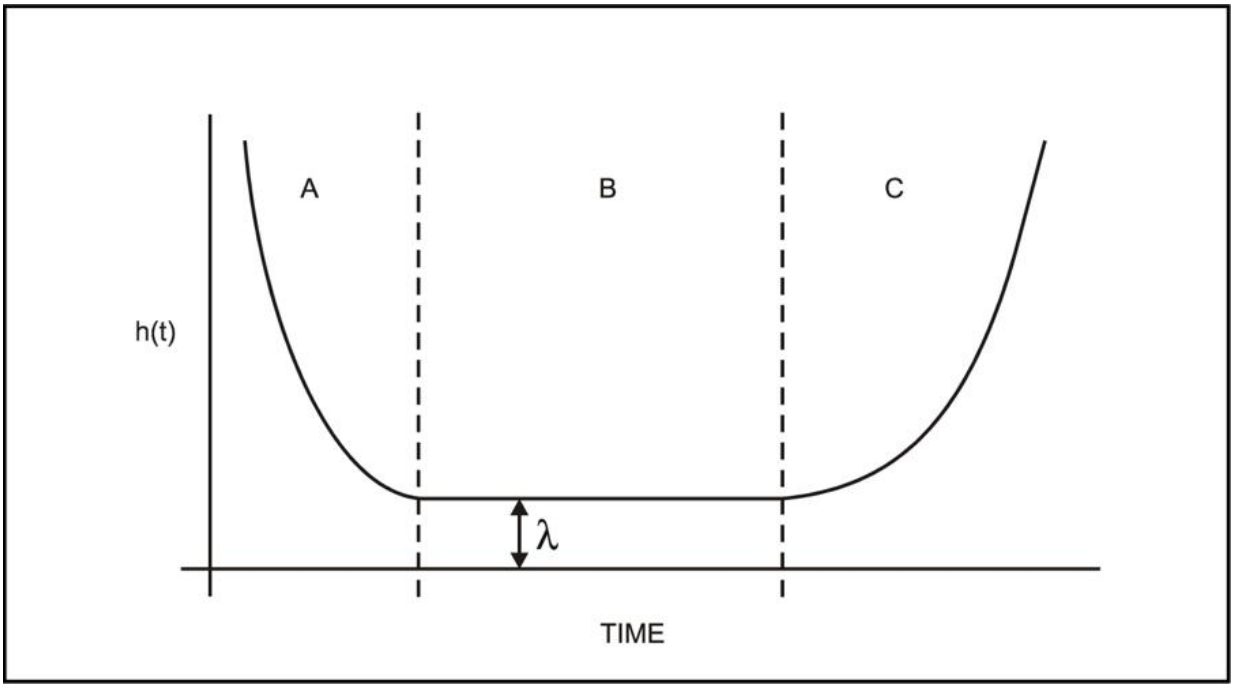
First, though, let's look at how things fail.

2.2 HOW HARDWARE FAILS

The life cycle of electronic and other types of instrumentation commonly follows the well-known bathtub reliability curve. The name comes from the curve's shape, which resembles a bathtub. The bathtub curve can be divided into three periods or phases: the infant mortality period, the useful life period, and the wear-out period. These periods are illustrated in a graph of failure or hazard rate h(t) versus time (t) in Figure 2-1. In some devices, the failure rate may be measured in units such as failures per counts, operations, miles, or rpm, rather than in time. An example of this is an electromechanical relay, for which the failure rate is stated in failures per mechanical operations and failures per electrical operation.

FIGURE 2-1

Bathtub Curve (courtesy of Control Magazine)



The infant mortality period, shown as Area A in Figure 2-1, occurs early in the instrument's life, normally within the first few weeks or months. For the user, this type of failure typically occurs during the factory acceptance test (FAT), during staging, or just after installation. Failures during this period are primarily due to manufacturing defects or mishandling before or during installation. Most manufacturing defects are caught before the instrument is shipped to you, through the manufacturer testing and burn-in procedures. Be careful of rushed or expedited shipments, though, as vendors may bypass some of their testing and burn-in procedures to satisfy your schedule. Mishandling is more difficult to control. Inspection, observation, and care before and during installation can minimize mishandling.

The second phase on the bathtub curve is the useful life period, shown as Area B in Figure 2-1. This is where the failure rate, called the random failure rate (λ), remains constant. The time length of this period is considered the useful life of the instrument. Normal failures during this period are considered to be statistically random. An instrument that fails during this period and is repaired rather than replaced effectively restores its reliability. Many times individual instruments, while repairable, are simply replaced due to expediency. So, while the instrument is non-repairable to the user, the overall system is repairable.

2.2.1 Measures of Reliability

An important concept to understand during this period is the instrument's mean-time-to-failure (MTTF), a measure of reliability of the instrument during its useful life period. The MTTF is the inverse of the failure rate during the constant-failure-rate period. The MTTF is not related to the useful life of the instrument, which is the time between the end of the infant mortality period and the beginning of the wear-out period. A device could have an MTTF of 100,000 hours but a useful life of only three years. This means that during the three years of its useful life, the device is unlikely to fail, but it may fail rather rapidly once it enters its wear-out period.

Another example illustrating the difference between MTTF and useful life is human death rates—the failure rate of a human instrument. For humans in their thirties, this rate is estimated to be 1.1 deaths per 1,000 person-years, or a MTTF of 909 years. This is much longer than our useful life, which is usually less than 100 years. In other words, in their middle years people are very reliable (subject only to the random failure rate). But past that, in their wear-out period, their reliability decreases rapidly. Another example is a computer disk drive with an MTTF of 1 million hours but a useful life of only five years. Within its useful life, the drive is very reliable, but after five years the drive will begin to wear out and its reliability will decrease rapidly. The drive with an MTTF of 1 million hours, however, would be more reliable than a drive with an MTTF of 500,000 hours with the same expected useful life.

A related measure is mean-time-to-repair (MTTR), the mean time needed to repair an instrument. MTTR has several components as shown below:

The second item, Mean time to troubleshoot the failure, is of particular interest. It is a major component of MTTR that affects the uptime or the availability of an instrument.

Mean-time-between-failures (MTBF) is a measure of the reliability of repairable equipment. It is the MTTF plus the MTTR:

MTBF = MTTF + MTTR

Many times vendors use the terms MTTF and MTBF interchangeably. If the MTTF is much larger than the MTTR, this is an acceptable approximation.

Availability is the fraction of time the instrument is available to perform its designated task. Availability is given by the equation:

$$\text{Availability} = \frac{\text{MTTF}}{\text{MTTF} + \text{MTTR}}$$

An availability of 0.99 would mean that an instrument is available 99% of the time.

To have a high mean-time-to-failure (i.e., a low failure rate) select a well-designed, sturdy instrument and apply it properly. Selecting an instrument designed and properly installed for maintainability is essential to having a low MTTR. Unfortunately, other factors such as cost, delivery, and engineering preference, can reduce availability. (That is what keeps troubleshooters in business.)

2.2.2 The Wear-out Period

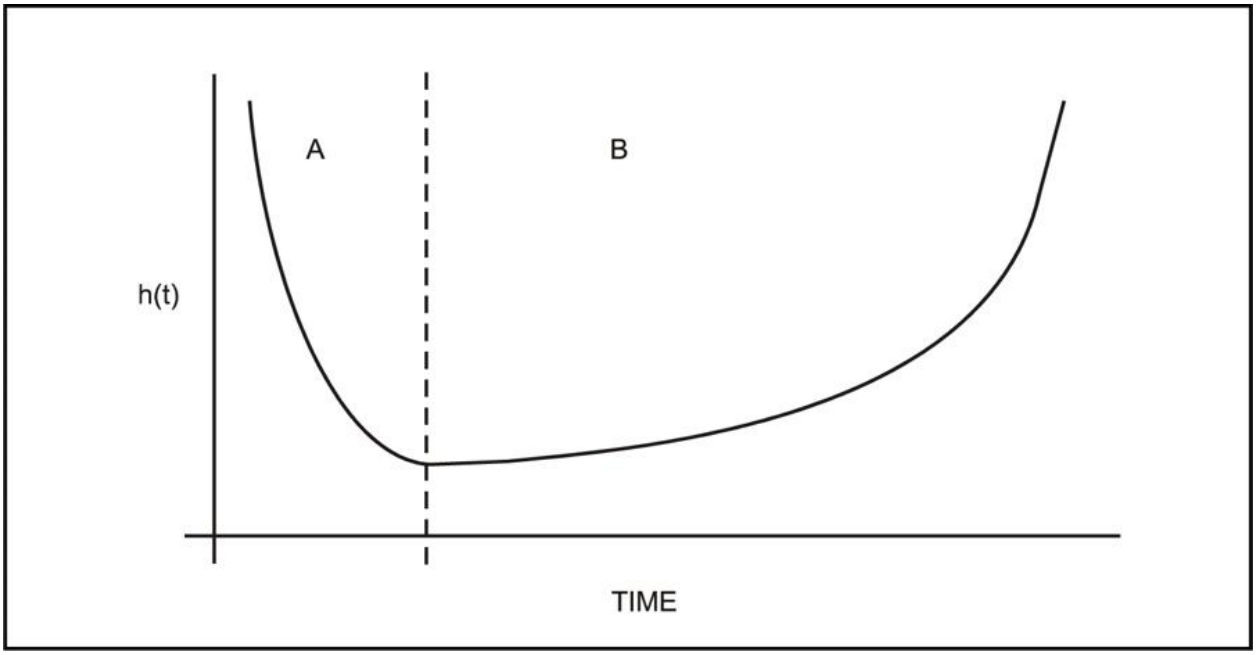
The third period on the bathtub curve is the wear-out period shown as Area C in Figure 2-1. This is where the instrument is on its last legs; it is wearing out. Detecting the beginning of this period is a key to knowing when to replace rather than repair an instrument, before it becomes a maintenance hog. Because the instrument as a whole is wearing out during this phase, it makes more sense to replace it than to repair individual components.

Mechanical equipment with rotating or moving parts begins wearing out immediately after it is installed. Such equipment typically has only the infant-mortality phase (A) and the wear-out phase (B), though the

wear-out phase for mechanical equipment should have a shallower slope than for the electronic instrument's wear-out phase. The failure curve for mechanical equipment is shown in Figure 2-2.

FIGURE 2-2

Mechanical failure curve (courtesy of Control Magazine)



Catastrophic failures (such as an instrument being run into by a forklift truck, or struck by lightning) are not considered in the bathtub curve, nor are failures due to human error or abuse. While these types of failures cannot always be prevented, they can be minimized.

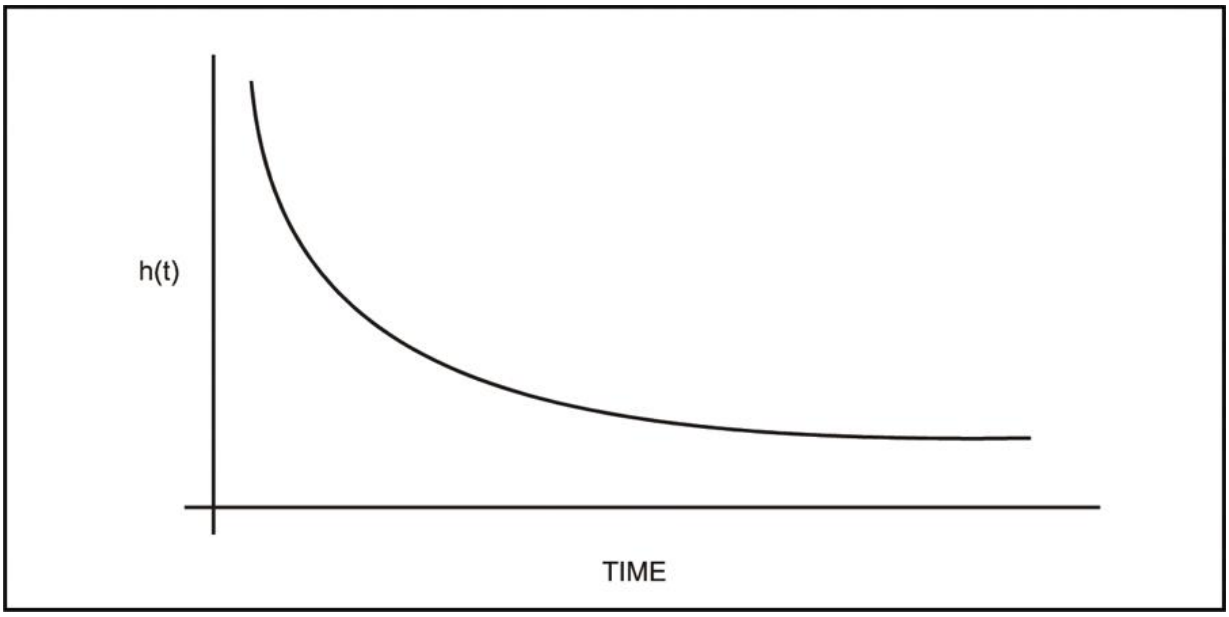
2.3 HOW SOFTWARE FAILS

To reduce failures, software should be written to meet specifications correctly and completely and then thoroughly tested. Software failures in an industrial setting are not considered random. They occur due to errors during the design and coding of the software. They can also be introduced during changes of procedures and equipment.

Generally these failures do not manifest themselves immediately because the manufacturer tests system software, and most errors are discovered during this testing. Once in use, however, users put stress on the software, and additional errors may be found. Software designed and generated by users follows the same general failure path. Typically, then, the failure rate of software over time decreases—the more it is used, the more likely it becomes that errors will be found and fixed. A graph of the typical software failure rate versus time is shown in Figure 2-3.

FIGURE 2-3

Software Failure Curve (courtesy of Control Magazine)



Failures in manufacturers' software are not always corrected in a timely manner, which worsens the failure curve. Some manufacturers wait until their next software revision to correct errors, do not tell users about errors until asked, or do not admit to the error at all. Some errors become new features of the software. A feature is something that has utility and in this case, was not considered in the original design but was coded in by accident. In some cases, the software error is corrected, but new errors are introduced during the fix. New errors can also be introduced when enhancements are made to the software. This means that trusted software might become unreliable after revision. Always keep backup copies of software in case the previous version needs to be restored.

2.4 ENVIRONMENTAL EFFECTS ON FAILURE RATES

If an instrument fails while operating in its designed operating range, the failure rate should follow the bathtub curve. The key here is in its designed operating range—a condition that is more rare than you would like. Failure rates are affected by stresses due to misapplication or abuse of the instrument that were not anticipated in its design. The most common stresses are ambient temperature, ambient and process corrosion, exceeding process conditions, and abuse.

All instruments have strengths and weaknesses, and operation inevitably applies stresses to them. If an instrument is overspecified, so that it is much stronger than the application it is used for, reliability improves and the failure rate decreases. If the stresses applied to an instrument exceed its strengths or find a weakness, it may malfunction or fail. If stresses exceed an instrument's designed operating conditions, the instrument's failure rate increases and the failure curves discussed above will shift or be distorted. The causes of these failures are not intrinsic to the instrument itself. Replacing the instrument will not solve the problem, only postpone it until the next failure due to excessive stress.

2.4.1 Temperature

A common stress is ambient temperature. For electronic instruments and electrical equipment, a rule of thumb is that for every 10°C the temperature rises over the normal operating temperature for the

equipment, the failure rate doubles. This is based on Arrhenius's Equation, which is used to model electronic components. One version of this equation is:

$\lambda = e(E/kT)$

where

For more information on temperature effects on failures, consult the military handbook on reliability, MIL-HDBK-217.

2.4.2 Corrosion

Another environmental effect is corrosion. It can take the form of ambient corrosion, which is caused by improper selection of the instrument or the enclosure to protect the instrument, or exposure of surfaces to corrosive elements due to abuse, improper closure, or damage. Or it can involve process corrosion, which occurs when the wrong materials are selected for the wetted parts of the instrument (those exposed to the process). These may include both exposed metal parts and the instrument's sealing parts (such as gaskets, O-rings, and seals). Changes in operating conditions or process materials can also cause process corrosion.

2.4.3 Humidity

Ambient humidity or moisture can also be detrimental to instruments. Condensation can lead to corrosion, in some cases producing electrical short circuits. Field instruments used in areas where the ambient temperature changes from day to night are subject to breathing (air moving in and out of an instrument), which can cause condensation inside them. This often occurs in high-humidity areas, and can be combated with instrument air and nitrogen environmental purges.

2.4.4 Exceeding Instrument Limits

Exceeding instrument limits means exceeding the process temperature, pressure, or another physical property for which an instrument was designed, and it can damage or weaken instruments. Many things can cause instrument limits to be exceeded: selecting the wrong instrument; transient process conditions not considered during instrument selection; or changing process conditions due to process design changes, clearing of bottlenecks, and increased rates.

2.5 FUNCTIONAL FAILURES

Failure is the condition of not achieving a desired state or function. Failure can also be defined as the inability to perform a desired function. This definition says nothing about what caused that inability. What if there is nothing wrong with the instrument? What if it was just asked to do something it was not capable of doing? This type of failure is called a functional failure.

Many times functional failures occur in the field, but when the suspect instrument is taken to the shop, it checks out. Examples are instruments calibrated to the wrong range and instruments that are too small or too big (a control valve, for example). Often, functional failures can also be caused by associated equipment. For example, a transmitter's failure to respond might be caused by plugged lines that feed it. Nothing is wrong with the transmitter; it simply is not getting the process pressure. Another example might be a low supply voltage.

In one plant a reactor blew its relief valve to the flare before a transmitter-based detection system opened the reactor dump valves. The transmitter was removed and found to be fully functional. Further troubleshooting found that the transmitter's dedicated power supply output was only 40V instead of 70V (a 10-50 mA system), and the transmitter using this voltage could only go up to 36 mA, short of the 40 mA required to trip the dump valves. It was a classic functional failure of the transmitter to read the correct pressure even though it was fully functional.

2.6 SYSTEMATIC FAILURES

Systematic failures are due to human error and are not random. They are errors due to design mistakes, errors of omission or commission, misapplication, improper operation, or abuse. These are not just engineering errors—they can occur throughout the instrument's life cycle. Some examples of human errors are specifying the wrong materials for a process transmitter, operating a piece of process equipment above its design temperature and the specified temperature of its associated instruments, and leaving the screws loose on a NEMA 4 (weatherproof) enclosure door, exposing the inside to ambient conditions.

One example of systematic failure occurred in the northern part of the United States, where a contractor building a plant was careful to specify the upper temperatures on all the instruments. But, because the contractor forgot to consider the lower temperature limit (an error of omission), the first winter caused numerous instrument failures.

These types of failures can be hard to spot because the root cause is not the instrument itself. Physical examination of the instrument, reviewing the documentation, determining the ambient and process conditions, and looking at the instrument nameplate information can provide clues. But the cause of a systematic failure is not always obvious.

2.7 COMMON-CAUSE FAILURES

Sometimes more than one failure results from a single cause. Such common-cause failures can occur in a redundant system, where a single component failure causes the redundant system to fail. Common-cause failures can also come from a single cause, such as corrosion, that causes multiple instruments to fail. In a single system they are typically easy to spot, but common-cause failures of multiple instruments can be trickier. Record keeping and good observation can be invaluable in such cases.

Typical common-cause failure sources are shared components, power quality, grounding, ambient temperature, ambient corrosion, ambient humidity, and manufacturer defects (where all the instruments have the same bad component, for example). In redundant systems, common-cause failures can be due to failure of common switching elements, common power supplies, or failure of redundant channels due to a common cause. Human error is the root of many common-cause failures.

One example of a component common-cause failure occurred in a tried and true pneumatic instrument that had a spinning rotor, where a purchasing agent of the manufacturer (seeking to save money) substituted a component material without checking with engineering. The spinning rotor in this

instrument began to disintegrate shortly after installation. This caused numerous failures of the instrument, much to the manufacturer's embarrassment.

2.8 ROOT-CAUSE ANALYSIS

This brings us back to the question of the root causes of failure. Again, internal failure of an instrument usually reveals itself quickly. But when dealing with external causes of failure, more investigation may be needed.

External failure may be transient or continuous. If transient, finding the cause may be very difficult if not impossible without additional failures, as well as additional monitoring and diagnostics. If the cause is continuous and if it causes immediate failure, we should be able to find it through troubleshooting. Failure of a continuous but deteriorating nature often requires more information (and probably more failures) before the root cause can be determined.

To meet such demands, the technique of root-cause analysis (RCA) was developed. Root-cause analysis is a logical, structured process used to find the cause of a problem. RCA is usually a team effort, sometimes by a multidisciplinary team. RCA generally starts by finding the immediate cause and then making it an effect, then listing all the possible causes of this effect and analyzing them to find the second-level cause. Once that cause is determined, the process is repeated again and again until the root cause is found. RCA is like a backward tree, where we climb down the limbs to find the root cause.

Another metaphor is the causal chain, where each link depends on the previous one. The causal chain may be several links long and may be conditional (X and Y must be true to make Z true). There is no easy formula for learning to perform RCA—it requires practice and experience.

Though there is no substitute for practice, several commercial systems can help facilitate root-cause analysis. Four such systems available in the late 1990s included Kepner-Tregoe (KT); REASON® from Decision Systems, Inc.; Apollo from Apollo Associated Services; and TapRooT® from Systems Improvements, Inc.

SUMMARY

Everything fails eventually, and finding the cause of failure is a big part of troubleshooting. Understanding failure mechanisms is important when the cause of the failure is not readily apparent. Failures can take different forms, including hardware and software failures. A failure can be functional, due to misapplication or abuse. Systematic failures result from human error. Failures from a single cause can affect multiple instruments or channels and lead to longer and more complex cause-and-effect chains.

QUIZ

1.Failures that occur early in an electronic instrument's life are

A. infant mortality failures

B. wear-out failure.

C. common-cause failure.

D. systematic failure.

2.Software failures are

A. systematic failures.

B. not random.

C. decrease over time.

D. all of the above

3.Mean-time-between-failures (MTBF) is

A. the same as mean-time-to-failure.

B. a measure of reliability of a repairable instrument.

C. how long an instrument will last.

D. none of the above

4.Systematic failures are

A. the same as common-cause failures.

B. failures in the useful life of an instrument.

C. due to human error.

D. the same as functional failures.

5.Common-cause failures are due to

A. human errors.

B. failure of a shared or common element in a redundant system.

C. multiple failures in a system due to a common cause.

D. both B and C.

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3

FAILURE STATES

Overt and covert failures

Failure direction

Directed failure states

What instrument failures indicate

3.1 OVERT AND COVERT FAILURES

In the previous chapter we talked about failures in general. In this chapter we will discuss several ways of classifying failures: overt and covert, unpredictable and directed, and several types of directed failures, in which the instrument itself detects the failure and directs it toward a particular end state.

Failures can be overt, which means they are self-revealing: they announce themselves as a failure to perform a function that is monitored by another device or by plant personnel. An example of this might be a level-control valve installed on the inlet of a tank that is designed to shut when it fails. If the level decreases, an operator or low-level alarm detects the failure. Many instruments have directed failure modes that make failures more obvious, such as fail-closed or fail-open. In continuous control systems such as basic process control systems (BPCS), many failures are self-revealing because they are continuously monitored by operators or alarm systems.

In demand systems, such as safety systems, failures are not always so obvious. These systems only operate when requested or demanded. In these systems, and occasionally in continuously operated systems, failures can lie in wait and fail at what seem the most inopportune times. These are called hidden, covert, or latent failures. Such failures often appear after troubleshooting another failure, after a demand is placed on the system, or during routine testing. Testing is the most common way that latent failures are found and defeated.

Latent failures can be confusing when they are combined with another failure: A failure that has nothing to do with the problem you are troubleshooting can lead you down the wrong path. It may also seem that two failures have occurred simultaneously and must somehow be related, even though they are not.

3.2 DIRECTED FAILURES

Directed failures are designed to fail in a certain way when motive power is lost or a diagnostic detects a failure. The most common directed failures are designed to occur upon loss of instrument air or electrical power. Some input devices also have a directed failure mode. The most common are up-scale or down-scale burnout on thermocouples. Although equipment may have these directed modes, life is not that simple—the same equipment can also have unpredictable failure modes.

3.2.1 Failure Direction

Four basic failure directions are fail-safe, fail-dangerous, fail-known, and fail-unknown (fail-I don't care).

Upon failure, a fail-safe instrument forces the system to a safe state. This is most commonly associated with control valves and wiring but can apply to other design situations. One example of this is a fail-close valve, where the safe state for the process is for fluid flowing through the valve to be stopped. Since some instruments can be powered by both electric current and instrument air, there can be two failure directions, depending upon which power source fails. Another example is circuits wired so that they trip when they lose power, commonly called de-energized-to-trip or fail-safe wiring. Upon loss of power, these circuits drive the process to the safe (tripped) or no-voltage state. This type of fail-safe wiring protects against damage from loss of power by driving the loop or system to a safe state. Fail-safe failures are generally self-revealing.

A fail-dangerous instrument fails in a manner that moves toward a dangerous state. In a continuous system this generally happens immediately; in a demand system this might be a latent failure that, when subjected to the demand, makes the system fail to function and a dangerous situation occur. A fail-dangerous latent example might be a plugged measurement connection on a high-level alarm. An overt example might be a control valve that when failed-open allows a reactor to run away.

The fail-known state is used when safety is not involved but a known failure state has been designed into the instrument or system. Generally the state is chosen so that it will be easily noticeable.

The fail-unknown state occurs when the failure in any direction does not cause a dangerous situation. This failure direction applies generally to loss of motive power.

3.3 DIRECTED FAILURE STATES

Many times instrument systems are designed to fail in a certain (directed) manner when particular conditions occur. The following are some of the directed failure states commonly specified or designed into instrument systems.

•*Fail-close (FC):* Seen most commonly on control valves, fail-close means that the valve closes upon loss of motive force (air, electricity, hydraulic) or signal.

•*Air fail-close (AFC):* Seen most commonly on control valves, it means that the valve closes upon loss of air. See Figure 3-1 for an example of an air fail-close valve.

•*Fail-open (FO):* Seen most commonly on control valves, it means that the valve opens upon loss of motive force (air, electricity, hydraulic) or signal.

•*Air fail-open (AFO):* Seen most commonly on control valves, it means that the valve opens upon loss of air. See Figure 3-1 for an example of an air fail-open valve.

•*Fail-last state (FL):* Seen in motorized and double-acting valves; it means that the instrument fails in its last state upon loss of motive force or signal.

•*Fail-last good state (value):* Seen on inputs to computers or PLCs (Programmable Logic Controllers), the last state is maintained when diagnostics detect an input failure. The same may apply to maintaining an output upon a detected failure.

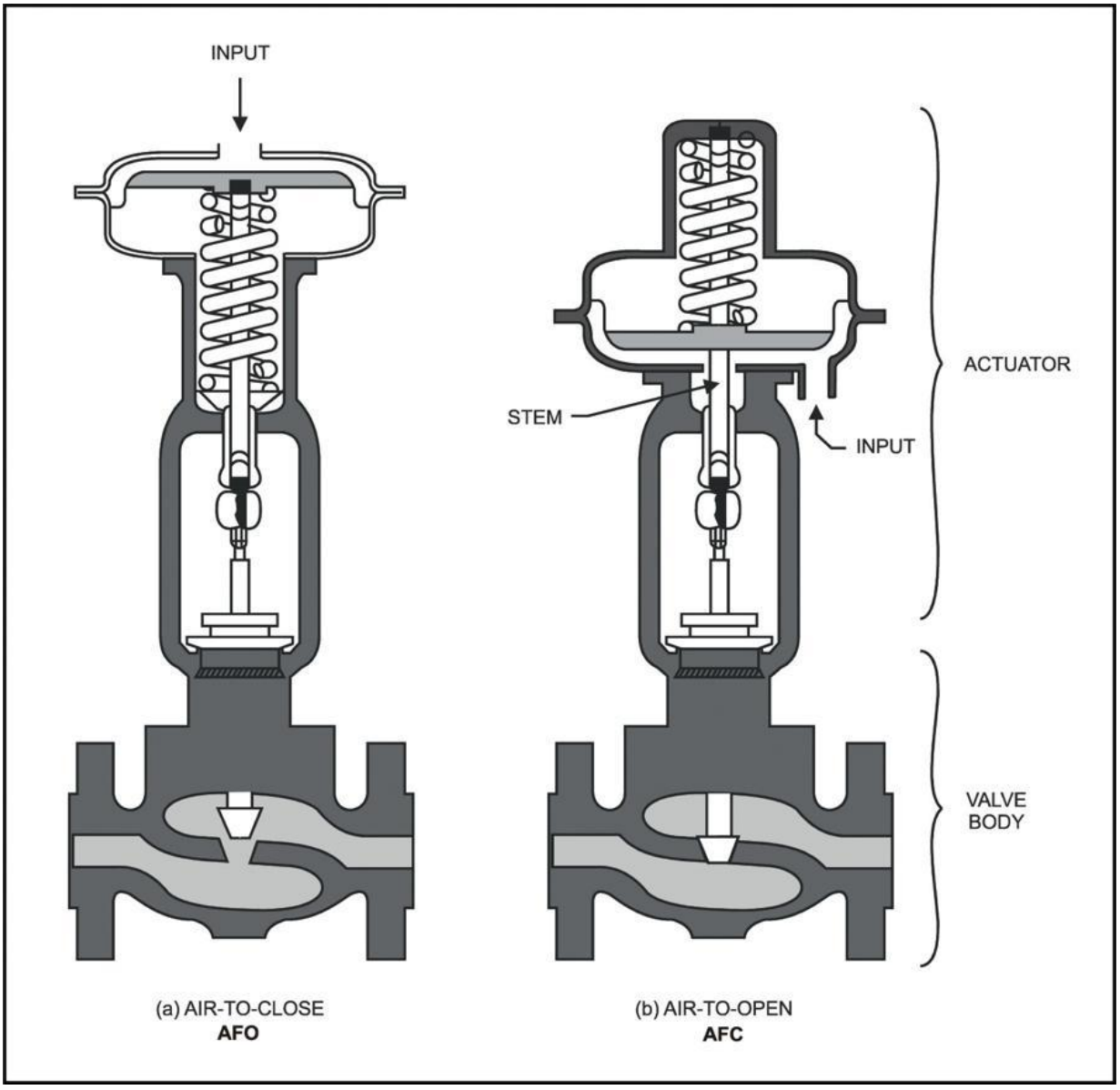
•*Fail-safe state (value):* Seen on inputs to computers or PLCs, the instrument goes to a predetermined safe state when diagnostics detect an input failure. The same may apply to maintaining an output upon a detected failure.

•*Up- or down-scale burnout:* Used with thermocouple or RTD inputs, this means that when an open thermocouple or RTD is detected, the instrument fails in a predetermined way—either up- or down-scale.

•*De-energized state (DE):* This describes the state into which wiring or an energized component will force the system when power fails. Also, it is typically shown on solenoids with arrows to indicate the state they assume upon loss of power.

•*Fail-unknown (I don't care):* No predetermined directed failure state exists.

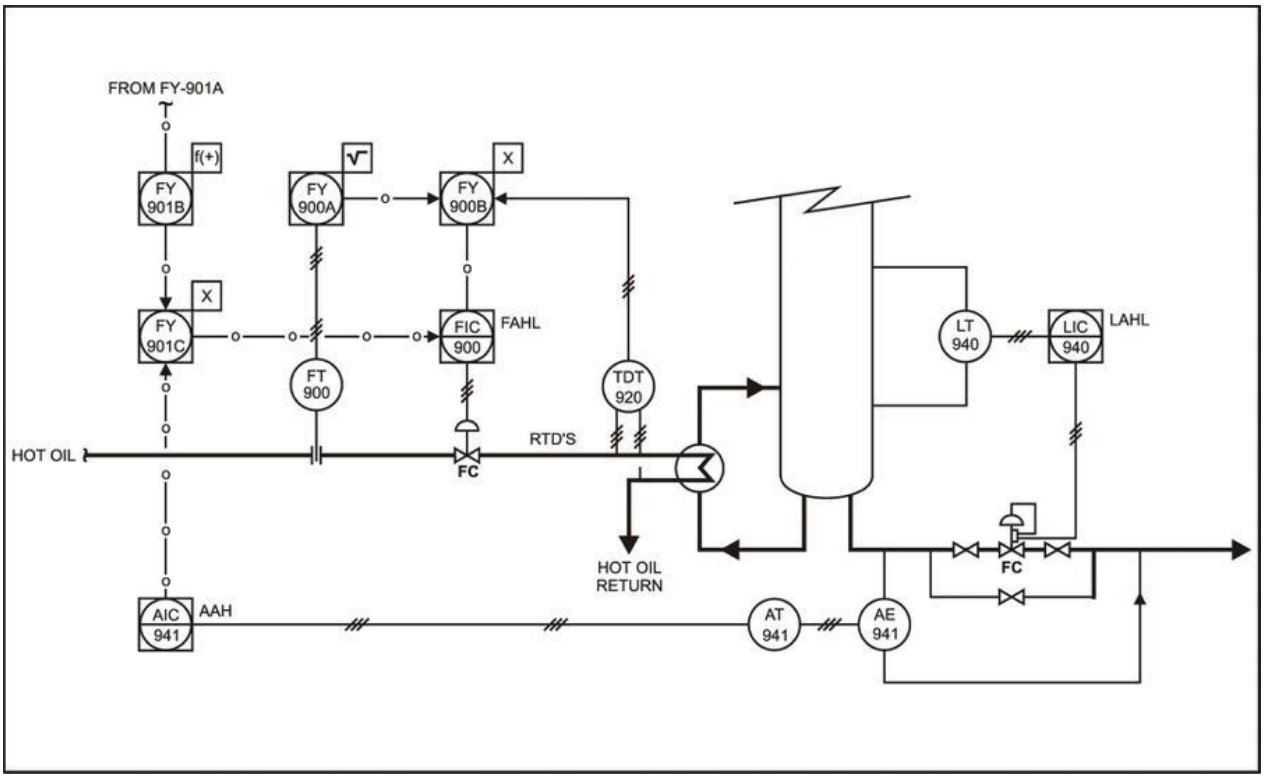
FIGURE 3-1 ***Air Fail Positions on Globe Valve***



3.4 WHAT FAILURE STATES INDICATE

When we encounter a directed failure, we may not initially be able to tell why the failure occurred. For example, the fact that a valve has failed closed does not imply that it is strictly a valve failure. If the valve is a fail-close valve, the valve may have lost its motive power or its signal may have gone to zero. Information about final control elements and failure modes should appear on the instrument's loop drawing and on the piping and instrument diagram (P&ID), and must be taken into account when troubleshooting. Input failure modes should be indicated on loop drawings. An example of a directed failure state indicated on a P&ID is shown in Figure 3-2.

FIGURE 3-2 ***Piping and Instrument Diagram***



RELEVANT STANDARD

•ISA-5.1-1984 - R1992 — Instrumentation Symbols and Identification.

SUMMARY

Instrument failures can be classified in a number of different ways. Instruments can fail safely, fail dangerously, in a known state, or in an I don't care state. The failure can be self-revealing or overt, or it can be latent or covert.

The failed state in which you find an instrument is not always the actual failure. It may be in that state because it was directed to that state, which may be due to another failure, unrelated to the instrument that has stopped operating. Always review the applicable loop drawings to see if there are any directed failure states before beginning to troubleshoot the problem.

QUIZ

1.Fail-safe is when the instrument fails

A.in a manner that brings the process to a safe state.

B.up-scale.

C.in the last state.

D.in the last safe state.

2.For instruments, AFC means

A.automatic frequency control.

B.air fail–close.

C.always fail closed.

D.both B and C

3.Instrument failure modes should be shown on

A.wiring diagrams.

B.P&IDs.

C.loop drawings.

D.both B and C

4.Up-scale burnout is typically associated with

A.fire detection instruments.

B.thermocouples and RTDs.

C.control valves.

D.none of the above

5.Latent failures are the same as

A.fail-safe failures.

B.self-revealing failures.

C.overt failures.

D.covert failures.

REFERENCES 1.Goble, W. L. *Evaluating Control Systems Reliability*. Research Triangle Park, NC: ISA, 1992.

4

LOGICAL/ANALYTICAL TROUBLESHOOTING FRAMEWORKS

Logical/analytical troubleshooting frameworks

Specific troubleshooting frameworks

How a specific troubleshooting framework works

General or generic logical/analytical frameworks

How a general or generic troubleshooting framework works

Vendor assistance advantages and pitfalls

Why troubleshooting fails

4.1 LOGICAL/ANALYTICAL TROUBLESHOOTING FRAMEWORK

A framework underlies a structure. Logical frameworks provide the basis for structured methods to troubleshoot problems. But following a step-by-step method without first thinking through the problem

is often ineffective. We need to couple logical procedures with analytical thinking. To analyze information and determine how to proceed, we combine logical deduction and induction with knowledge of the system and then sort through the information we have gathered regarding the problem.

Often a logical/analytical framework does not produce the solution to a troubleshooting problem in just one pass. We usually have to return to a previous step and go forward again. We may have to do this several times.

Even after we have gathered a large amount of information, this iterative process can tell us that we need more. Sometimes a single measurement can send us back up the framework to a previous step. We can thus systematically eliminate possible solutions to our problem until we find the true solution. For example, we might think that a blown fuse is causing a problem, but when we replace the fuse it blows again. This means that we will have return to a previous step in the troubleshooting process and investigate further.

Logical/analytical frameworks can be divided into two types:

•Specific frameworks

•General or generic frameworks

4.2 SPECIFIC TROUBLESHOOTING FRAMEWORKS

Specific troubleshooting frameworks have been developed to apply to a particular instrument, class of instruments, system, or problem domain. For example, frameworks might be developed for a particular brand of analyzer, for all types of transmitters, for pressure control systems, or for grounding problems. When these match up with your system, you have a distinct starting point for troubleshooting. Otherwise, the starting point will generally be determined by the problem description and information-gathering process.

Such frameworks typically come in several formats:

•Tables

•Flowcharts or trees

•Procedures

For example, Figure 4-1 shows a table for troubleshooting a magnetic flow meter. You could also have a table to troubleshoot a problem domain of pneumatic transmitters in general, as shown in Figure 4-2. Figure 4-3 illustrates a problem domain troubleshooting flowchart or tree.

FIGURE 4-1 ***Magnetic Meter Troubleshooting Table***

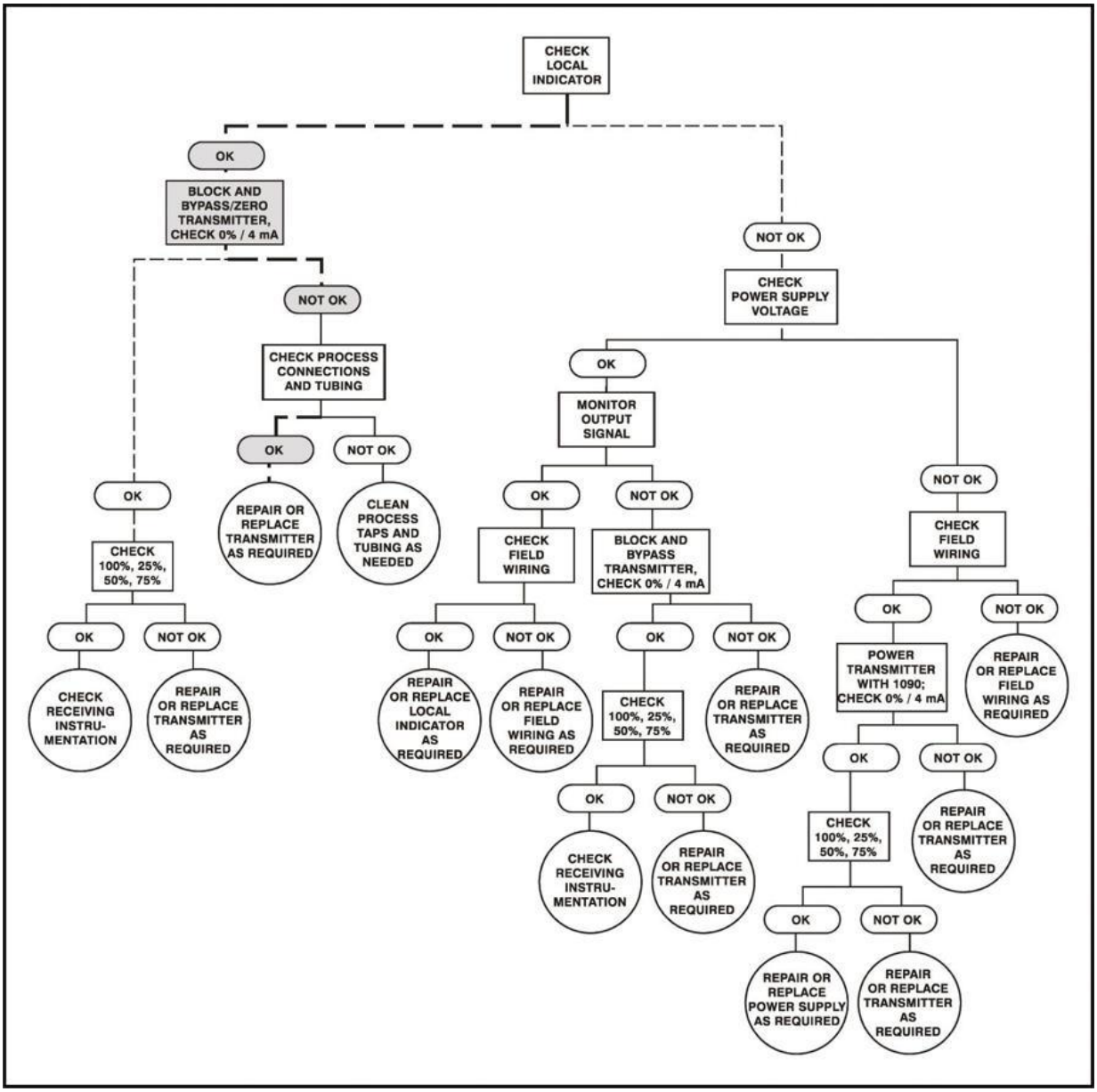
SYMPTOM	POTENTIAL CAUSE	CORRECTIVE ACTION
Coil drive open circuit displayed.	Faulty terminal connection.	Isolate the break (faulty connection). Perform: Test B—flowtube coil.
Indicated flow equals half of expected flow.	One signal is being drawn to ground, or is open.	Perform: Test D—electrode shield resistance. Consult your vendor's service center for further instructions.
Indicated flow is erratic.	A less than full flowtube or a non-homogeneous process fluid.	You may need special transmitter features to process the signal correctly.
	Improper grounding.	Make sure the electrode and coil drive shields connect to both the flowtube and the transmitter. Perform: Test D—electrode shield resistance. Perform: Test E—positive-to-negative electrode.
	An inherently noisy process fluid.	Contact your vendor for information regarding the high-signal magnetic flowmeter system.
Reverse flow detected.	Inverted connections at one of the four terminal sites.	Reconnect terminal sets correctly.
	Flow direction is opposite of flowtube arrow.	Reverse the wiring at flowtube terminals 18 and 19; there is no need to invert flowtube.
No flow indicated.	The valves, positioners, or actuators of the physical piping are not properly set.	Perform: Test A—electrode shield voltage. Perform: Test D—electrode shield resistance. Perform: Test E—positive-to-negative electrode
Insufficient process fluid conductivity.	Process is a hydrocarbon.	Perform: Test E—positive-to-negative electrode.

FIGURE 4-2 ***Typical Pneumatic Transmitter Troubleshooting Table***

SYMPTOM	PROBABLE CAUSE
No output	Bent flapper. No air supply; plugged restrictor (very common). Corroded control relay or components. Dirty control relay seats. Flapper is away from the nozzle due to freezing, improper adjustment, bent "C" flexure, or transmitter has been dropped. Leak in the feedback bellows. Leak in the nozzle circuit. Leak in the sensor pressure circuit. Disconnected or broken links in a motion balance transmitter.
Partial output	Plugged low-pressure leg on a dP cell. Worn control relay parts. Partially plugged supply screen or filter. Burr on the flapper assembly. Hole in the flapper assembly. Damaged feedback bellows. Worn capsule diaphragms. Warped or distorted "C" flexure or "A" flexure on a dP cell. Wrong range-sensing unit. Pin hole leaks in the control relay diaphragm.
Full output	Plugged nozzle. Ballooned capsule diaphragm. Loose nozzle lock nut. Blocked control relay vent. Sensing capsule impacted with process solids. Flapper assembly distorted or bent.
Zero shift diaphragms	Dirty flapper assembly set point capsule problems: coating, fatigue, warped. Temperature changes: either ambient or process temperatures. Process static pressure changes. Worn zero or span adjustments. Flapper is "dimpled" on the surface. Pin hole leak in the flapper. Flashing and/or condensate on either leg of a dP cell installation.
Output oscillates	Liquid in the feedback bellows (water or oil, etc.). "C" flexure lock nut loose. Close-coupled pneumatic system. Loss of capsule fill fluid. Hole in the feedback bellows. Loose bleed/vent valves. Flashing due to pressure variations.

FIGURE 4-3

Flowchart or Tree Troubleshooting Framework



Company-developed troubleshooting procedural frameworks typically appear in formal maintenance procedures. They are text-oriented but may also contain table, flowchart, or tree formats. Figure 4-4 shows an example of a procedural framework.

FIGURE 4-4

Procedural Troubleshooting Framework

PRESSURE TRANSMITTER TROUBLESHOOTING PROCEDURE

PURPOSE: This procedure is design to troubleshoot process pressure transmitters from the process connection to connection to a controller or DCS system.

TRANSMITTER IS NOT RESPONSIVE BUT NOT ZERO OR 100%

1. Verify problem by looking at historical (trend) records.
2. Verify field indicator (if available) or field signal matches control room reading.
3. If so, check to see that the:
   a. Process taps are not blocked off and are clean.
   b. Transmitter functions properly
4. If not, check to see that the:
   a. Transmitter functions properly
   b. Signal to controller is correct
   c. Controller functions properly

TRANSMITTER IS AT HIGH(> = 100%) OR LOW LIMIT(< = 0%)

1. Verify problem by looking at historical (trend) records.
2. Verify field indicator (if available) or field signal matches control room reading.
3. If field indicator agrees and reading is below zero, check to see if there is loop power
   a. If only one transmitter is effected and power is not present:
      1. Check loop fuse. If blown, replace; if blows again check loop components and wiring for electrical fault
      2. If not blown; check loop wiring and fusing back to power supply
      3. Check power supply
   b. If multiple loops are effected, check power supply fusing and power supply.
4. If field indicator agrees and signal is present, check that the transmitter is functioning properly. If signal is zero, check to see if transmitter is blocked and bled.
5. If field indicator does not agree, check controller input is functioning properly

TRANSMITTER READS INCORRECTLY

1. Determine if the reading is slightly or grossly incorrect.
2. If slightly incorrect,
   a. Check calibration.
   b. Check the transmitter is functioning properly
   c. Check transmitter and controller parameters
3. If grossly incorrect, check to see if:
   a. Process connections are clean.
   b. Transmitter is functioning
   c. Controller input is functioning.

TRANSMITTER SIGNAL OSCILLATING

1. Verify problem by looking at historical (trend) records.
2. Verify field indicator (if available) or field signal matches control room reading.
3. If so, check and see if the process is oscillating; if so, a process problem.
4. If not, check transmitter for proper functioning.
5. Check transmitter damping
6. Check input card.
7. Check DCS or controller filtering/damping.

TRANSMITTER IS SLUGGISH

1. Check that the process taps are clean.
2. Check transmitter damping.
3. Check DCS or controller filtering/damping.

TRANSMITTER HAS INTERMITTENT PROBLEMS

This is a difficult problem and no series of questions will lead to a solution for all problems of this type. The following questions can provide clues that can lead to possible solutions.

1. Check and see if there is any indication of process problems via trend recordings at the same time.
2. Check and see if any other loops are effected.
3. Check and see if there is any time dependence.
4. Check and see if there is any operational dependence.
5. Check and see how signal wires are routed to see if it can be EMI.
6. Check grounding.
7. Check to see if there is any corrosion in the loop equipment.
8. Is the weather involved?
9. Are certain operators or engineers on duty when it happens?
10. Did any power problems occur?

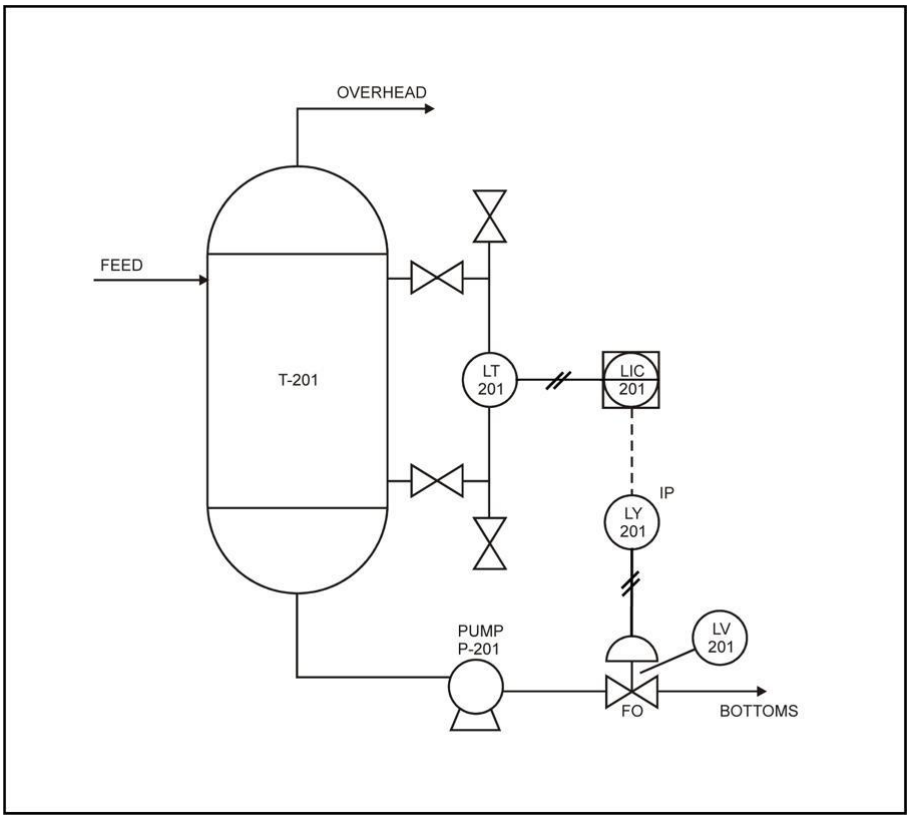
Modern equipment may offer software-based frameworks that implement the table, flowchart, or procedural frameworks, or that have a conversational framework, where the software asks the troubleshooter questions and leads the way to a solution. Such frameworks may be supplied by the instrument system vendor or may be developed in-house. Some third-party software can be customized to your plant's systems or may come with a configuration for a particular vendor's equipment. A few systems based on artificial intelligence, known as expert systems, are also on the market.

4.3 HOW A SPECIFIC TROUBLESHOOTING FRAMEWORK WORKS

Look at the troubleshooting framework shown in Figure 4-3 (steps highlighted) and the loop in the P&ID shown in Figure 4-5. Here the operator reports that the level in Tank 201 is too high, and automatic control loop LIC201 is not responding to the problem. The level on the DCS (Distributed Control System) is 60% (which is also the set point), and has not changed in quite a while. The operator reports that he has placed the loop in manual and the situation is under control.

FIGURE 4-5

Tank 201 P&ID



Here, based on the operator's description, since the DCS appears to be getting a signal and it is constant, the control system is not making any changes. Field examination of the local indicator would appear to be the first step, with further testing of the problem as necessary, starting in the field. Using the tree framework in Figure 4-3, take the following steps:

•Step 1: Check the local indicator. The local (field) indicator shows the same level as the DCS: OK.

•Step 2: Block and zero the transmitter, and check for 0%/4mA. The transmitter does not zero, but a check of the process connections indicates that they are clear: not OK.

•Step 3: The level transmitter appears to be the problem and is replaced. Problem solved.

In the above example, examination of the faulty transmitter indicates that the problem is with its electronics. A check of the maintenance records indicates that the last time the transmitter was worked on was three years earlier,

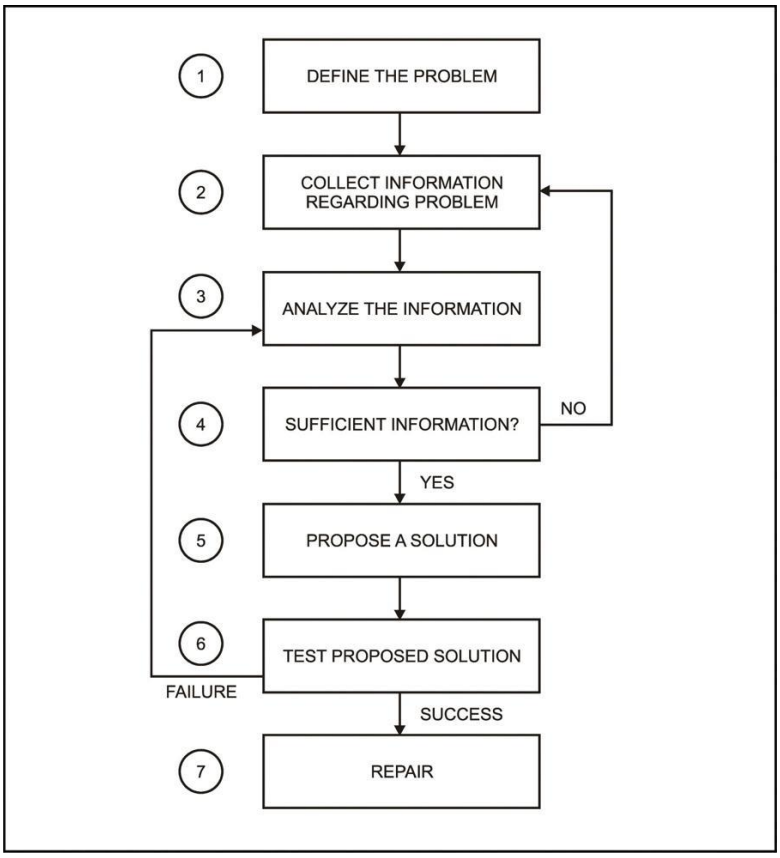
so the troubleshooter decides that the problem is level transmitter and that no more general solution is required. The final steps here would be to notify the operator, make sure field tagging and instrument configuration are correct, update maintenance documentation, and clear permits.

4.4 GENERIC LOGICAL/ANALYTICAL FRAMEWORKS

Since we do not always have a specific structured framework available, we need a more general or generic framework that will apply to a broad class of problems. Figure 4-6 depicts this as a flowchart, and this section suggests a series of steps to follow when using it.

FIGURE 4-6

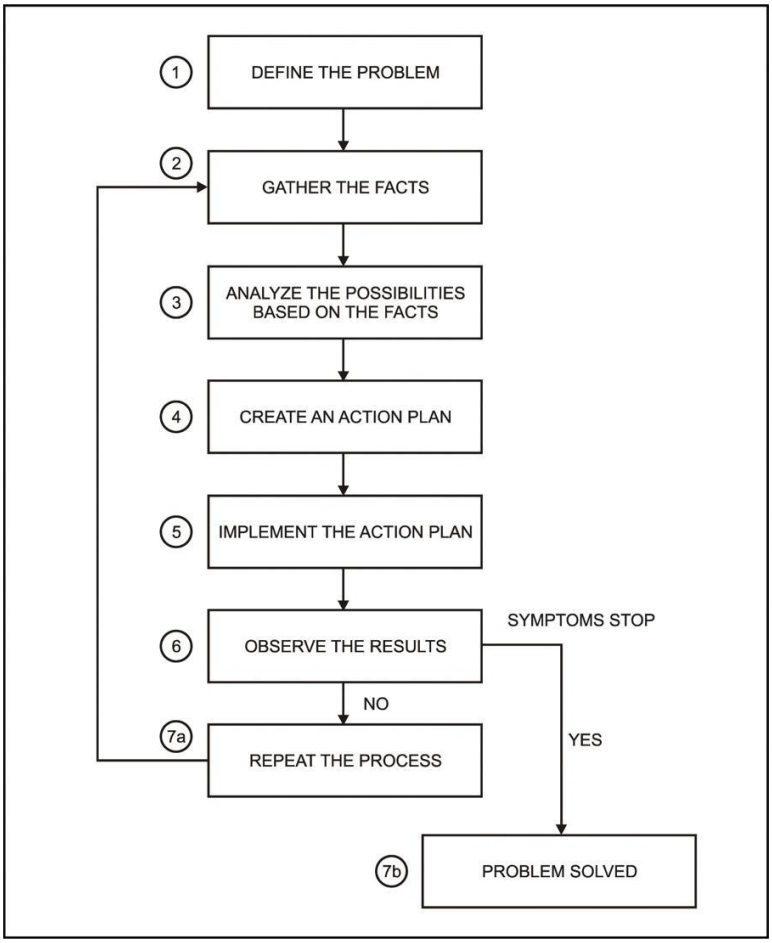
General Troubleshooting Flowchart #1



While the framework shown in Figure 4-6 is logical and efficient, it is not the only one available. Figure 4-7 shows an example of another general framework. In such frameworks, the logical path is essentially the same, but the semantics are different, or they are customized to a particular general problem domain. In some cases, a limited scope-specific framework (one that only applies to part of the system you are troubleshooting) may be used in conjunction with the more general framework.

FIGURE 4-7

General Troubleshooting Flowchart #2



The frameworks shown in Figures 4-6 and 4-7, while efficient, do leave out some important safety-related tasks associated with or related to the troubleshooting process. These associated safety tasks are discussed in Chapter 6, but a few important points should be made here:

•Always communicate with the operator in charge before you start.

•Get the proper permits.

•Get the appropriate permit to work if the work is in a hazardous area.

•Always communicate your actions with the operator in charge as you work.

•Always make sure what you are doing is safe for you, your fellow workers, and your plant.

4.5 A SEVEN-STEP PROCEDURE

The following seven-step procedure is a generic, structured approach to troubleshooting.

4.5.1 STEP 1: Define the Problem

When people report problems to you, the facts may be incomplete, unclear, or buried in too much information. Typically, someone reports the problem as he or she sees it, and may add the impressions of other operators and people on the same shift. Some problems occur on the night shift, graveyard shift, or on the weekend, and you will only have what was written down in the shift log or what was passed on from shift to shift.

4.5.1.1 SIMPLE PROBLEMS

For simple problems, identifying the problem could involve merely discussing it with an operator. For example, the operator might tell you that the temperature transmitter on tower 301 has gone to zero and stayed there, flow loop F101 is oscillating, or alarm LSH 501 will not clear.

4.5.1.2 TRANSIENT AND COMPLEX PROBLEMS

For transient conditions and complex problems, reports of symptoms may range from clear to vague. They may not be concise or even correct. You may be told too much or too little, and what you are told can be inconsistent or biased.

4.5.1.3 COMMUNICATION

When defining the problem, listen carefully and allow the person reporting the problem to you to provide a complete report of the problem as they see it. The report generally will consist of symptoms and specific observations, based on preliminary attempts to troubleshoot the problem or a working knowledge of the system. After listening carefully, ask clear and concise questions. Avoid high-level technical terms or technobabble. A troubleshooter should speak the language of the person reporting the problem. This means understanding the process, the plant physical layout, instrument locations and process functions as they are known in the plant and the dialect of abbreviations, slang, and technical words commonly used in the plant.

4.5.1.4 BIAS

The person reporting a problem may introduce biases into the description. When there is no clear cause for a problem, these biases can become apparent. For example, one common bias is that it is always the instrument's fault, when it may in fact be a process or operational related problem; in this case, instruments just report the information as they see it. Another bias may be toward a particular instrument or system, perhaps one that has caused problems in the past or that someone else has criticized.

You may get a subjective assessment because the person delivering the report believes, for whatever reason, that some particular piece of equipment has caused the problem; in fact it may be something else. Operators may believe trusted instruments to the exclusion of other information. Here you must learn to separate the signal from the noise and be sure that your own biases do not add further interference.

4.5.1.5 DEGREE OF GENERALITY

When you define the problem, consider the degree of generality with which it is reported. For example, an operator may say that a control valve has failed shut, thus giving you a definite hardware target to investigate. But if the operator says that the level in tank 201 isn't working right, which is more vague, it could mean a number of problems—even something to do with the process itself. Transient problems only occur once in a while, such as a control workstation lockup that only occurs on third shift. The problem may also be invisible, such as one reported by diagnostic software. For example, a diagnostic program on a communication link might only report that the error rate on a particular link is higher than normal.

Part of a good troubleshooter's skill is sorting through the reports to get to a good definition of the actual problem. The problem description provides the starting point for collecting data. If you do not know where to start, you can find yourself collecting more data—or the wrong data—or lost in the data you have collected. It is like navigating: if you don't know where you are starting from, you are probably lost.

4.5.2 STEP 2: Collect Information Regarding the Problem

Once a problem has been defined, you should then collect additional information. This step may overlap with Step 1, and for simple problems these two steps may even be the same. For complex or sophisticated problems, though, collecting information is a more distinct stage.

Develop a strategy or plan of action for collecting information. This plan should include determining where in the system you will begin to collect information. For instance, if the person reporting the information has narrowed down the problem to a component or subsystem, start gathering information there. If not, then the information gathering must start at a higher or more general level and work down from there. Information gathering typically moves from general to specific. If you get on the wrong path, however, you will have to move back to general and then back down to specific. In other words, you are continually working to narrow down the problem domain (scope).

4.5.2.1 SYMPTOMS

The information you gather typically consists of symptoms (what is wrong with the system) as well as what is working properly. Primary symptoms are directly related to the cause of the problem at hand. Secondary symptoms are downstream effects, that is, not directly resulting from what is causing the problem.

4.5.2.2 CHARACTERISTICS AND PARAMETERS

We also need to know what is right with the system—what is working, the timing of the breakdown, and what may have changed in the system since it last worked. Finally, you should gather information about system configuration, such as the actual components, the design, and system parameters. Specific information is typically available from the instrument loop drawings. Other drawings, such as instrument wiring schematics, motor schematics, electrical wiring diagrams, and so on, may also provide information.

4.5.2.3 INTERVIEWS AND DATA

Typically, a large part of your information gathering will be in the form of interviews with the person who reported the problem and with any other people who have relevant information. Then you review the instrument's or system's performance from the control system's faceplates, trend recorders, summaries, operating logs, alarm logs, paper recorders or charts, and look at any system self-diagnostics.

4.5.2.4 INSPECTION

Next, you may inspect an instrument that is suspected of being faulty or other local instruments (such as pressure gauges, temperature gauges, sight glasses, and local indicators) to see if there are any indications that might shed light on the matter.

4.5.2.5 DOCUMENTATION AND HISTORY

After this, you will typically turn to loop drawings, P&IDs, electrical drawings, manuals, maintenance records, and system built-in documentation, reviewing it for additional information. The old saying, When in doubt, read the manual, is always worth remembering.

Historical records can also provide useful information. If your facility has a manual or computerized maintenance management system, it may contain information regarding the failed system or ones like it. Some facilities keep loop files on each instrument loop that can contain useful troubleshooting information. And do not forget the people who work there. Check with others who have worked on the instrument or system; your fellow technicians may keep personal records that may be of use. Ask if anyone has seen the problem before. Learn to share information.

4.5.2.6 BEYOND THE OBVIOUS

If there are no obvious answers, testing may be in order. Here, communication with operators becomes very important. For control loops, testing usually involves putting the control loop in manual or bypassing the final control element; working on a control loop while it is in automatic can be disastrous. If the loop is a shutdown or safety loop, put the loop in bypass and follow your plant's safety system bypass administrative controls. Plan

your testing to ensure it is done safely and gets you the information you need with a minimum of intrusion. The first preference should be nonintrusive testing, where you do not have to open up (disconnect) any wires, remove components, or manipulate any process connections.

When you test by manipulating the system, plan to test or manipulate only one variable at a time. If you alter more than one, you might solve the problem but be unable to identify what fixed the problem. Always make the minimum manipulation necessary to obtain the desired information. This minimizes the potential upset to the process.

4.5.3 STEP 3: Analyze the Information

Once you have collected information, you must start analyzing it to see if you have enough to propose a solution. Begin by organizing what you have collected, then applying external knowledge and logical principles to sort through it. By external knowledge we mean knowledge beyond the information gathered in Step 2. This can mean such things as the basic principles, other basic science and

engineering principles, system knowledge, vendor information, personal experience, and principles of logic.

Information can be organized in different ways, depending on the path you take in the troubleshooting. You could organize the information by location, by parts of the system being investigated, by identifying causes and effects, by weeding out extraneous information, by identifying information that is time- or event-dependent, or by identifying what is fact and what is opinion. For example, if the problem occurs only on the night shift, the operational information gathered might be organized into the plant activities that only occur on night shift and those that do not occur on night shift.

You then can analyze the problem by reviewing what you already know and the new information you have gathered, connecting causes and effects, exploring causal chains, applying if/then and if/then not logic, applying the process of elimination, and applying other relevant analytical methods.

Knowledge and understanding of a system does not happen by accident. It comes from prior work, reading, word of mouth, and training. Here are some suggestions for how to become knowledgeable about a system:

•First, be observant when working on an instrument or system—it can help you in the future. Try to identify key knowledge about how a system works or is put together.

•Sometimes team meetings or bull sessions examine difficult problems, maybe even the very problem you are troubleshooting. Though information coming out of a bull session should be taken with a grain of salt, listening carefully can help.

•Never turn down training. It is a good way to learn about an instrument or system, what can go wrong, and how to fix it. Always do the classroom training exercises, particularly the hands-on ones.

•Understand cause and effect within the system. Learn the basic principles by which the system you are troubleshooting operates. Knowing things about a system is not much use if you do not understand how the system works. If you cannot match information you have collected with the operation of the system you are troubleshooting, it will not make sense. Make sure you have this basic knowledge before you jump in, or you may find yourself drowning.

Understanding the system allows you to distinguish between primary and secondary information or symptoms. For example, a compressor might shut down due to high discharge pressure, which results in several other alarms also coming on. The primary symptom is the high pressure, and the secondary symptoms are the subsequent alarms. Another example might be a measurement loop for which the signal goes to zero. Further investigation finds a fuse blown, but the blown fuse may be a secondary symptom to an overload, a transient, or a fault. Further review is necessary to determine what the primary cause is and whether it needs to be protected against.

4.5.3.1 CASE-BASED REASONING

Probably the first analysis technique that you will use is past experience you have had with the same problem. If you have seen this situation or case before, then you know a possible solution. Note that we say a possible solution: similar symptoms sometimes have different causes and hence different solutions. Keep personal notes or records to help you remember previous cases.

4.5.3.2 SIMILAR TO ANALYSIS

Compare the system you are working on to similar systems you have worked on in the past. For example, a pressure transmitter, a differential pressure transmitter, and a differential pressure level transmitter are similar instruments. All transmitters that have process taps (connections) have similar problems. All motors have similarities. Different PLC brands often have considerable similarities. RS-485 communication links are similar even on

very different source and destination instruments. Similar instruments and systems operate on the same basic principles and have potentially similar problems and solutions.

4.5.3.3 WHAT, WHERE, WHEN ANALYSIS

This type of analysis resembles the Twenty Questions game. You ask questions about what the information you gathered may tell you. These are questions such as:

•What is working? (For example, does the system have power? A signal at the DCS? At the field indicator?)

•What is not working?

•What is a cause of an effect (symptom) and what is not?

•Where does the problem occur?

•Where does it not occur?

•When did the problem occur?

•When did it not occur?

•What has changed?

•What has not changed?

Given a problem with a loop with no signal, you may look at the blown fuse indicator and see no indication, then say the cause of the problem is not a blown fuse. Conversely, you may look at the indicator and see it lighted, then say the problem may be due to a blown fuse. Or the blown fuse may be a secondary symptom; a short to ground, for example, could be the primary cause.

A big question can be whether you are dealing with a new system, with an existing system, or with an existing system that has been modified. The answer influences what kind of information you collect or where you look. For example, if you are troubleshooting a new system, you could suspect any of the wiring or instruments. For an existing system, where the wiring worked before the problem and nothing in the wiring has been changed, improper wiring can generally be ruled out. When troubleshooting a modification that does not work, you would normally suspect the modified part of the system, though of course other parts of the system could have been damaged in the process of modification.

What, where, when analysis is essentially a process of elimination and is not always linear. Take care to not get off the path to the solution of your problem.

4.5.3.4 PATTERNS

Symptoms can sometimes be complex and can be distributed over time. Looking for patterns either in symptom actions or lack of action or in time of occurrence can sometimes help in the analysis of symptoms. Symptom patterns can be repetitious (i.e., occurs only on night shift), connected to normal events (i.e., every time a large motor is started or when a piece of equipment is operated), connected to a specific event (i.e., every time the pressure is greater than 100 psig), connected to specific operator action, or connected operational events.

4.5.3.5 BASIC PRINCIPLES

For example, electrical current can only flow certain ways, Ohm's and Kirchhoff's Laws always work, and mass and energy always balance. Some examples – the sum of the voltages around a loop always equals zero, and the level in a tank is based on the following equation:

Rate In – Rate Out = Rate of Accumulation

You can apply basic principles like these to analyze data to determine problem areas or when determining what is a process problem and what is an instrument problem.

4.5.3.6 THE MANUAL

When in doubt, read the manual. The information you have gathered can be used in conjunction with the instrument's or system's manual. It may have information on circuits, system analysis, or troubleshooting that can lead to a solution. It may also provide voltage, current, or indicator readings, test points, and analytical procedures. Often manuals use troubleshooting tables or charts to assist you.

4.5.4 STEP 4: Determine Sufficiency of Information

When you are gathering information, how do you know that you have enough? Can you determine a cause and propose a solution to solve the problem? This is a decision point for moving on to the step of proposing a solution.

At this step you can evaluate the information as a whole or piecemeal, by adding a single piece of information and then reviewing the whole again. If this proves insufficient, add another piece of information, and so on. Keep doing this until you have sufficient information.

4.5.4.1 THE DIRECT PROCESS

Sometimes Steps 2, 3, and 4 may lead a troubleshooter directly to a solution. The direct process is generally based on three methods: experience, historical documentation, and use of the manual. But be careful to not jump the gun and move on to testing a solution before you have firmed up all your facts. There may be costs associated with testing the solution that can be prevented with a little forethought.

First, experience is the quickest means of troubleshooting problems. If you have seen a particular problem before, you already know the solution. The more experience you have, the better your chances that this will occur. Not all experience is equal, however. Ten years of mediocre experience may not equal one year of excellent experience. Good experience will have you working on such things as unfamiliar systems, complex systems, difficult systems, poorly documented systems, sophisticated systems, and other challenging work. Avoid repetitive or unchallenging work.

Second, your facility may have documented a prior experience with the problem at hand, and the solution. Historical documentation may be in a plant maintenance management system or kept manually in a loop file or equipment log. The troubleshooter may also keep personal records on troublesome instruments or systems.

And third, read the manual. You would be surprised how many times people do not read it and consequently do not know the simplest things about their systems. Common problems and their solutions are usually described in the manual or given in tables or flowcharts. Test points and

procedures are commonly provided.

4.5.4.2 THE ITERATIVE PROCESS

More commonly, Steps 2, 3, and 4 work in concert, in a repetitive or iterative fashion, to help you find a solution. Here you collect information, analyze it, and decide if you have enough to propose a solution. If not, then you go back to Step 2 to collect more information, and so on, until you have a proposed solution. This iterative process may work when you have lots of information, or even just a single piece of information such as a test voltage or current. It is illustrated in Figures 4-6 and 4-7 by the paths that return to previous steps.

Now you will need a logical process to make this iterative procedure successful. Several approaches, such the linear approach and the divide and conquer method, may apply.

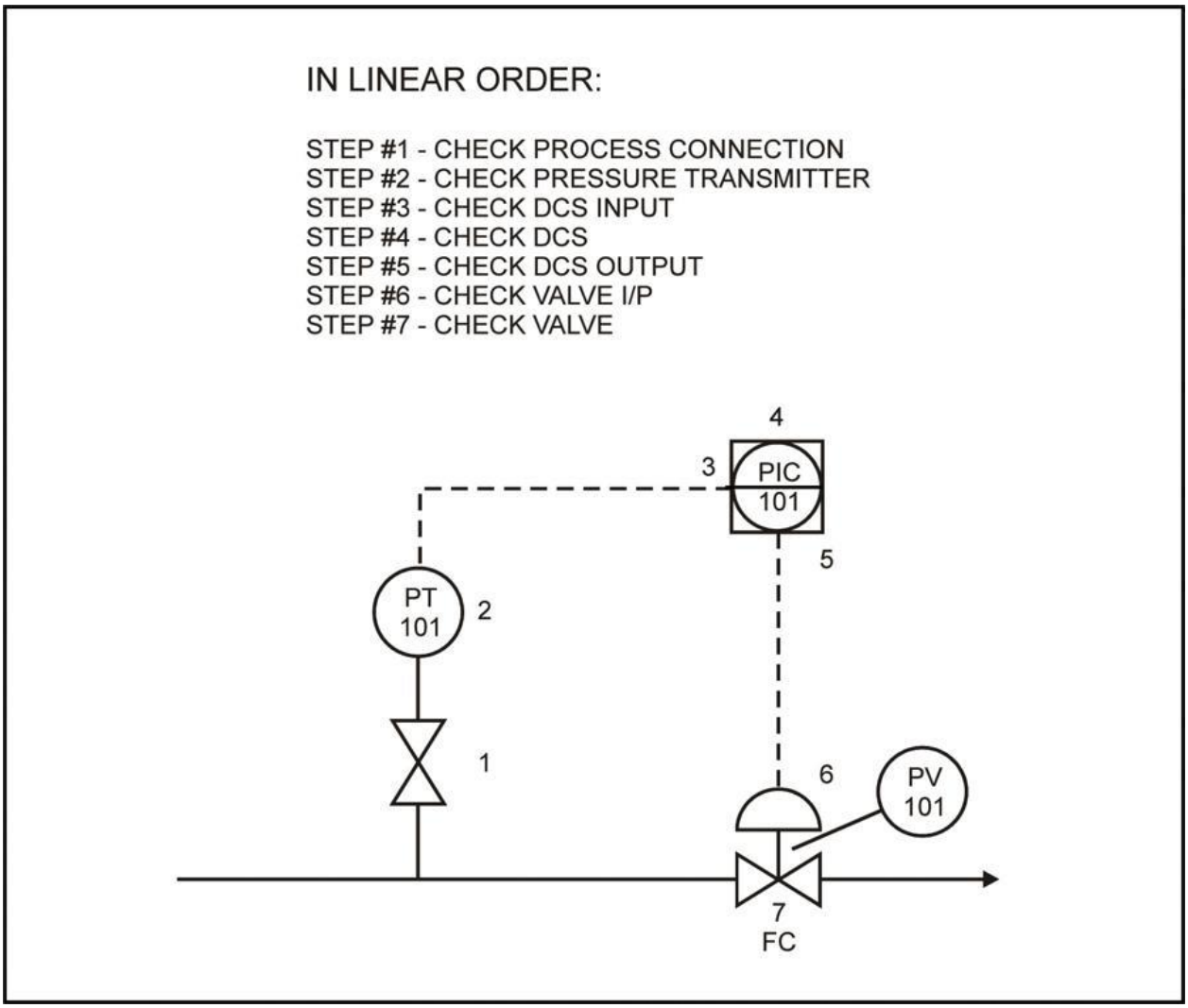
First, try the linear or walk-through approach. This is a step-by-step process (illustrated in Figure 4-8) that you follow through a system. The first step is to decide on an entry point. If the entry point tests correctly, then you test the next point downstream in a linear signal path. If this test point is all right, then you choose the next point downstream of the previous test point, and so on.

Conversely, if the entry point is found to be bad, choose the next entry point upstream and begin the process again. As you move upstream, each step narrows down the possibilities. Branches must be tested at the first likely point downstream of the branch.

For example, suppose a pressure signal is reading incorrectly in the DCS. The DCS signal is not zero, so there appears to be a signal from the field (and consequently the problem is probably not loss of power), but the signal is dead—it does not move when compared to the signal prior to the problem. There is no field indicator, so you choose an entry point at the process pressure tap where the pressure sensor connects to the process. You find this tap to be clean. Next, you check the pressure transmitter, which also checks out correctly, as does the power. The next thing you check is the DCS input card, which tests bad.

You can also use the linear approach when a likely entry point is not known by choosing a point of entry (generally either one end of a loop or circuit or the middle) and working from there.

FIGURE 4-8 ***Linear Troubleshooting Approach***

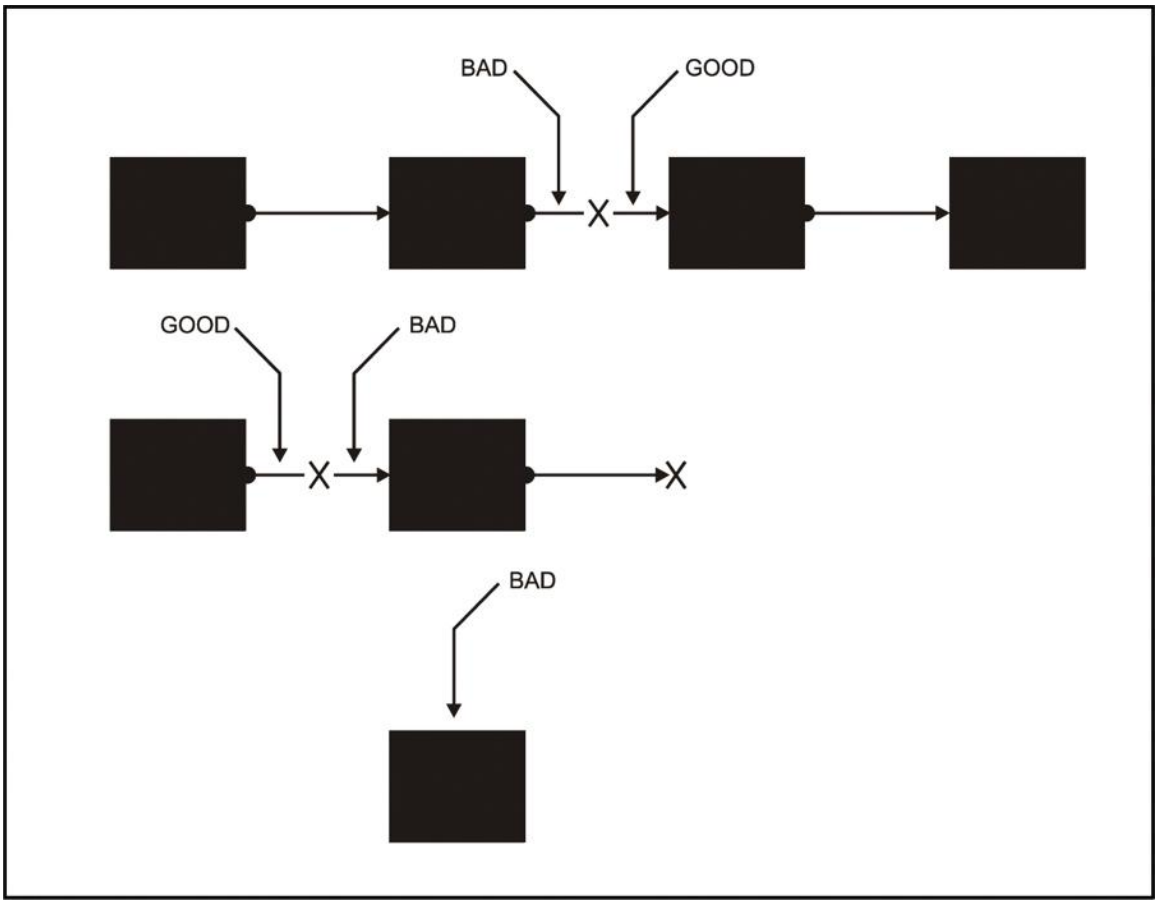


Second, the divide and conquer approach (illustrated in Figure 4-9) is a common technique that the electronics industry uses to test systems. You choose a likely point, or the midpoint of the system, and test it. If it tests bad, then the upstream section of the system contains the faulty part. The upstream section is then divided in two parts and the system is tested at the dividing point. If the test is good, the downstream section contains the bad part and is then divided in two, and so on until the cause of the problem is found. If on the other hand, the test is bad, then the upstream section contains the bad part. It is divided into parts and tested again, and so on until the bad part is found.

RELEVANT STANDARD

•ISA-5.1-1984 - R1992 - Instrumentation Symbols and Identification.

FIGURE 4-9 ***Divide and Conquer Troubleshooting Technique (courtesy of Control Magazine)***



4.5.5 STEP 5: Propose a Solution

When you believe that you have determined the cause of the problem, propose a solution. In fact, you may propose several solutions based on your analysis. Usually the proposed solution will be to remove and replace (or repair) a bad part. In some cases, however, your proposal may not offer complete certainty of solving the problem and will have to be tested, or another, more certain solution tried instead.

If you have several possible solutions, propose them in the order of their probability of success. If this is roughly equal, or other operational limitations come into play, use other criteria. You might propose solutions in the order of the easiest to the most difficult. Sometimes in a plant that cannot shut down operations, the right approach is proposing what you can do without a shutdown. In other cases, there may be cost penalties (in such things as labor, consumable parts, and lost production) associated with trying various solutions; you may propose to try the least costly option. Most of the time, however, you will try to find a compromise between the above criteria.

Do not try several solutions at once. This is called the shotgun approach and will confuse the issue. Management will sometimes push for a shotgun approach due to time or operational constraints, but you should resist it; with a little analytical work, you may be able to solve the problem and meet management constraints at a lower cost. With the shotgun approach, you may find that you do not

know what fixed the problem, and it will be more costly both immediately and in the long term. If you do not know what fixed the problem, you may be doomed to repeat it.

Do not rush to a compromise solution proposed by a committee, either. Consider the well-known Trip to Abilene story, illustrating the group think approach that is the opposite of synergy. In the story, some people are considering going to Abilene, though none of them really wants to go. They end up in Abilene, though, because everyone in the group figures that everyone else wants to go to Abilene. This sometimes occurs in troubleshooting where a committee gets together to assist the troubleshooter and the committee gets sidetracked by a trip to Abilene. This is sometimes caused by having high-level people or strong-willed people involved.

4.5.6 STEP 6: Test the Proposed Solution

Once a solution or combination of solutions have been proposed, they must be tested to see if your analysis of the problem is correct.

4.5.6.1SPECIFIC VERSUS GENERAL SOLUTIONS

Be careful of specific solutions to more general problems. At this step you must determine if the solution needed is more general than the specific one proposed. In most cases, a specific solution will be repairing or replacing the bad instrument. For example, you replace a bad transmitter, which solves the transmitter problem.

But what if replacing the transmitter only results in the new transmitter going bad? Suppose a transmitter with very long signal lines sustains damage from lightning transients. The specific solution would be replacing the transmitter; the general solution might be to install transient protection on the transmitter as well. Or suppose you have a systematic failure where the transmitter was specified with the wrong diaphragm material, which has corroded out; it will only corrode out again after a period of time. Replacing the failed transmitter with the correctly specified transmitter is the more general solution.

4.5.6.2 THE ITERATIVE PROCESS

If the proposed and tested solution is not the correct one, then return to Step 3, Analyze the Information. Where might you have gone astray? If you find your mistake, then move on to propose another solution. If not, move back to Step 2, Collect Information. It is time to gather more information that will lead you to the real solution.

4.5.7 STEP 7: The Repair

In the repair step, implement the solution you have proposed. In some cases, testing a solution results in the repair, as when replacing a transmitter both tests the solution and repairs the problem. Even in this case, there will generally be some additional work to be done, such as tagging, updating the database, and updating maintenance records, in order to complete the repair.

If the repair includes replacement, take care on modern microprocessor devices that you transfer the complete configuration of the failed instrument to the replacement. Otherwise, you may solve the immediate problem but introduce a new problem with a faulty configuration. If the repair involves a programming change, take care that you do not introduce a bug. All programming changes must be fully

tested. Communication with operators and termination of permits will also be necessary to complete the work.

Document the repair so that future troubleshooting is made easier. This is particularly true if the problem is unusual or unique. In repairing an instrument or system, take care not to damage it or leave it in a manner that might cause problems later. For example, be sure to tighten all terminal screws and installation bolts, replace and secure all covers, and check for damaged insulation, screws, or bolts.

4.6 AN EXAMPLE OF HOW TO USE THE SEVEN-STEP PROCEDURE

The following is an example of the use of the seven-step troubleshooting procedure outlined in section 4.5.

4.6.1 STEP 1: Define the Problem

The reporting party (an operator) states that the level in tank 201 (illustrated in Figure 4-5) is too low and the automatic control loop LIC201 is not responding to the problem. He has the situation under manual control at this point. This is a fairly good problem definition. It identifies the process vessel and the control loop involved, what the control loop is not doing correctly, and how the operator is compensating for the problem.

4.6.2 STEP 2: Collect Information Regarding the Problem

Next, you query the operator:

Q: How did you know you had a problem?

A: I had a low-level alarm and the level sight glass on the tank confirmed the low level.

Q: How are you determining the level?

A: Joe is watching the sight glass on the tank and keeping me informed on the radio.

You also collect other data. For example, you determine that the trend is flat with no noise.

4.6.3 STEP 3: Analyze the Information

It appears that since the level transmitter did not sense the level decrease but has a signal with no noise, the problem is with the field transmitter loop.

4.6.4 STEP 4: Determine Sufficiency of Information

Upon analysis, the likely candidates are the process taps or the transmitter. You need to do some testing in the field to gather more information. You go back to Step 2, get appropriate permits, and tell the operator that you will be doing field testing.

In the field, testing the process taps indicates the taps are clean. The loop's local indicator indicates the same as the DCS. You then repeat Step 3 and analyze the new information: Since there are two indications of a signal (power and a signal) and the process tap is clean, it appears that the transmitter is the problem. You then repeat Step 4 to determine whether there is now sufficient information. You decide that there is.

4.6.5 STEP 5: Propose a Solution

Your proposed solution is to replace the transmitter. One is in stock.

4.6.6 STEP 6: Test the Proposed Solution

The transmitter is calibrated, configured, and installed. Indicator and DCS now agree with the sight glass. Consideration is given to the possibility that a more general solution is required. A check of the maintenance records indicates that the last time there was a problem with this transmitter was four years ago, and there were no indications of transient or abuse problems. So a specific solution replacing the transmitter appears to be appropriate.

4.6.7 STEP 7: Repair

The level transmitter is already installed. Field tagging is verified to be in place on the new transmitter. (If this was a modern microprocessor-based device, the full configuration of the old transmitter would be transferred to the new transmitter.) Maintenance records are updated. The operator is notified of the repair completion and permits are cleared.

4.7 VENDOR ASSISTANCE ADVANTAGES AND PITFALLS

Sometimes it is necessary to involve vendors in troubleshooting, either directly or by phone. Manufacturer's service personnel can be very helpful, but some are quick to blame other parts of the system when they cannot find anything wrong—in some cases before they have even checked out their own system. Help desks can also give you good information, but be wary of statements like We've never seen anything like this or You're the first person ever to report this to us.

All information has a subjective aspect to it. When trying to identify a problem, you have to strip away subjective elements and get to the meat of the situation. Do not let vendors off the hook when trying to solve a problem just because they say it is not their equipment. Ask questions and make them justify their position.

On the other hand, you will sometimes get vendor service representatives who are really on the ball. Learn from them. Ask them questions, even if your questions do not have a direct relationship to the immediate problem (wait until after they have finished troubleshooting your problem). Get them to explain their equipment. Many times they are more than happy to do that for someone who is interested. Sometimes they will even provide information or documentation not in their manuals.

4.8 WHY TROUBLESHOOTING FAILS

The most common reasons troubleshooting fails are

•Lack of knowledge

•Failure to apply an organized approach to data gathering

•Choosing the wrong entry point

•Dimensional thinking

4.8.1 Lack of Knowledge

Lack of knowledge can become apparent at all stages of the troubleshooting process. The skill of recognizing the need for additional knowledge is a key to being successful at troubleshooting. There are basically three areas of knowledge that may be needed:

•Information about the function of the instrumentation system

•Information on how it accomplishes its function

•Information about this particular system

To be successful in troubleshooting a system, you must first understand what the system does and how it does it. Understanding what the system measures, controls, or accomplishes can separate process, operational, and instrumentation functions and give you a clearer picture of what each part does. In some cases it is not necessary to understand how the particular system works, but to understand how that *class* of systems works. By reading the manual and understanding the basic principles, the details of a particular system can be filled in. For example, understanding how one brand of control valve works can allow you to work successfully on another brand.

Information about the particular system includes how the system is used, how it is hooked up, and how it is powered. This information usually comes from drawings, plant documents, and vendor documents. The primary drawing is generally the loop drawing, as illustrated in Figure 4-10. This drawing shows the wiring and provides references to other drawings and documents for the instruments. In more complicated systems, additional wiring diagrams and system drawings may be used.

4.8.2 Failure to Gather Data Properly

When a troubleshooter is not organized, gaps in the information will make it impossible to get a clear picture of the problem. Organization can help assure that you gather enough information to lead to the correct path. Otherwise, the troubleshooting path you take may be the wrong one. Even if you have all the information organized, but fail to analyze it properly, the same thing happens.

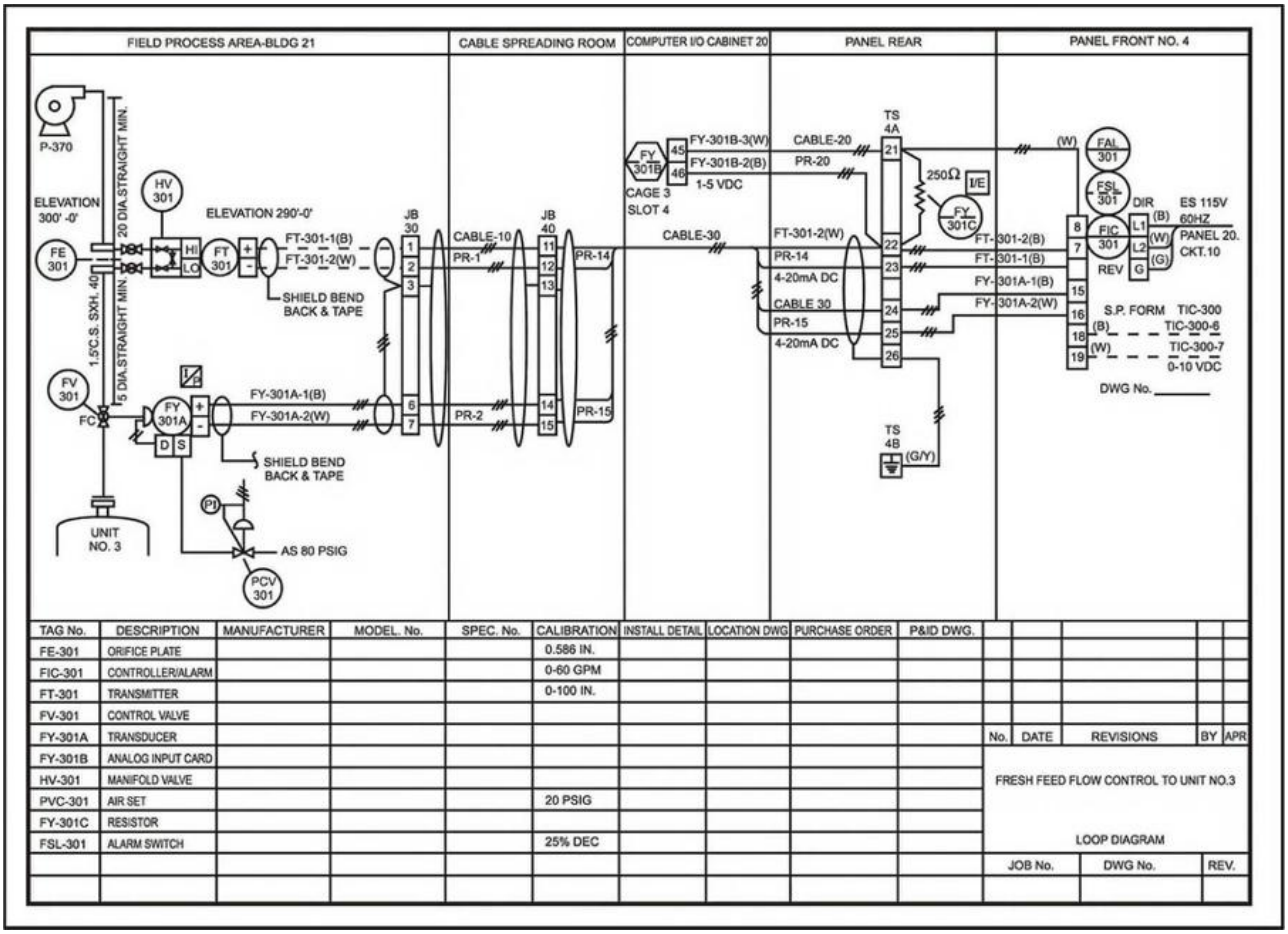
Look for information that relates to the problem you are troubleshooting. Discard information that clouds the issue. Look not only for facts, but also for causal relationships and patterns.

4.8.3 Failure to Look in the Right Places

Instrument systems can be simple, complex, or somewhere in between. Complexity is determined not only by the instrument's system construction, but how it communicates, both internally and externally. How we think about instrument systems is influenced by their complexity.

FIGURE 4-10

Loop Drawing Sample



When troubleshooting, you must be able to look at both the big picture and the little picture. Concentrate on a particular device or part of a system as the culprit (the little picture) and you may fail to realize that something else is affecting that device, making it appear to be the culprit (the big picture). This commonly happens when the process causes the problem or there is coupling between systems.

A complex system may overwhelm you with the amount of data available. You must typically divide the system up logically into smaller systems and deal with each in turn. How we think about a system is also influenced by how much we can determine about it using our senses, and what we must measure. Consider the simple pneumatic control system shown in Figure 4-11. Much of this loop can be comprehended using our senses—we can see links, levers, and flapper nozzles move, feel and hear air pressure, and so forth. We can also move many of these elements ourselves and see the effects. Most of the components are easily examined by the naked eye when taken apart.

If we consider the loop as an electronic loop, as in Figure 4-12, we have many of the same components, but the signal components are wires. Now we cannot see or feel much and must use measuring devices to understand what is going on. In a microprocessor-based digital system (Figure 4-13), we still have the wires, but we can see even less of what is going on, even with test equipment. With digital communications, we cannot measure the signal but must use a protocol analyzer to see what is going on. In digital systems, information is transferred via continuous analog values (voltage and current) to transient digital signals (zeros and ones) at high speed. Note that the drawings are very similar, but the actual internal operation is quite different.

FIGURE 4-11

Pneumatic Control Loop

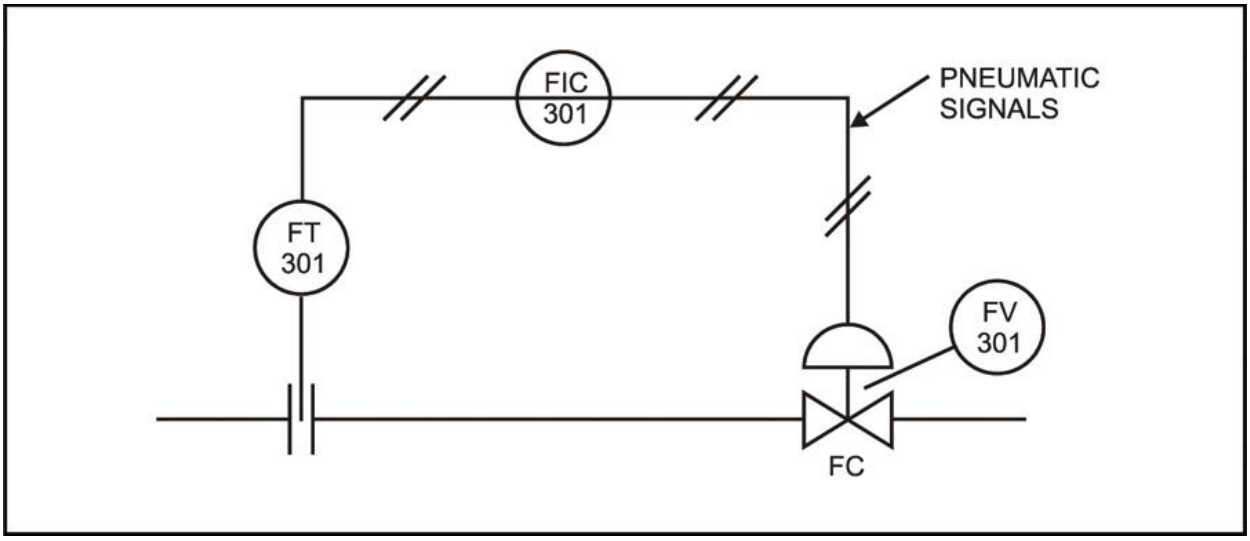


FIGURE 4-12

Electronic Analog Control Loop

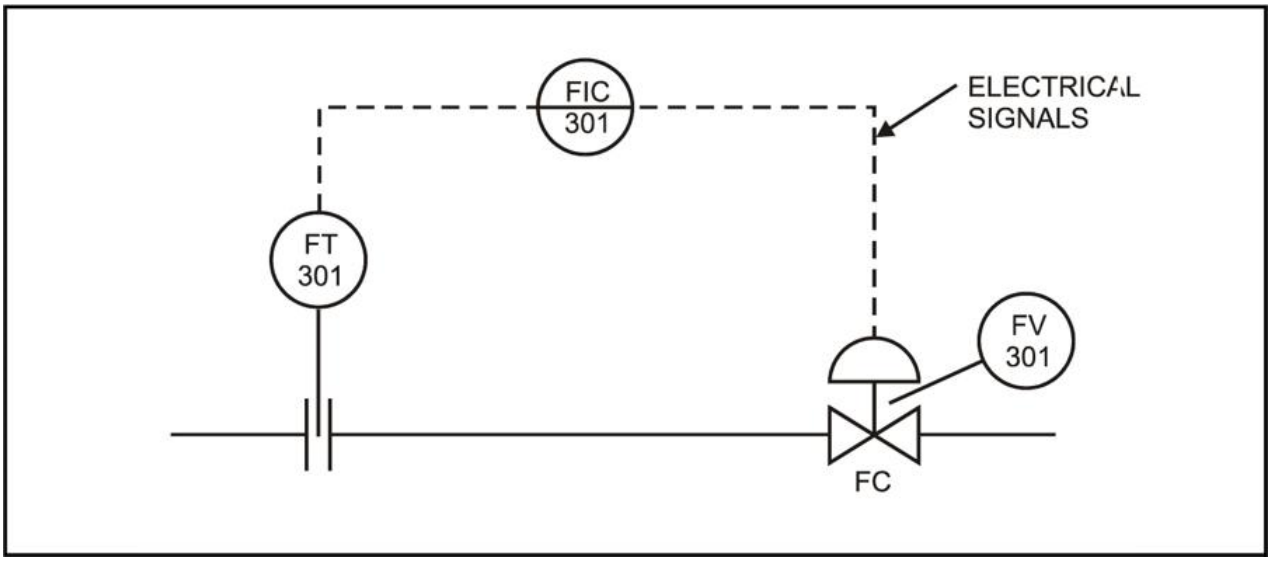
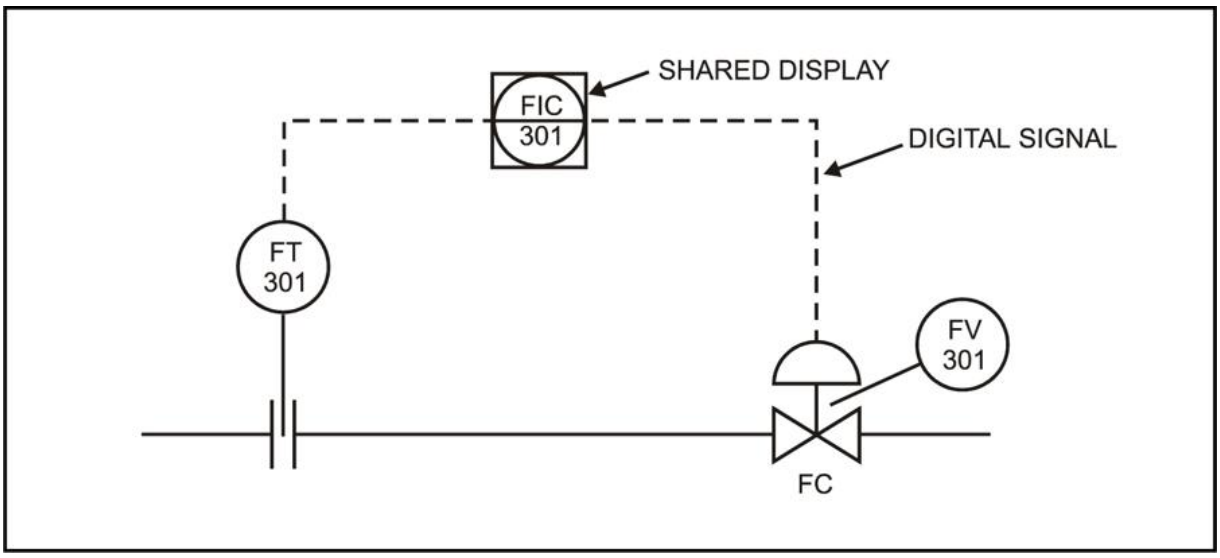


FIGURE 4-13

Digital Control Loop



As you can see, as instruments have developed, they have moved through different levels of abstraction. The first instruments were mechanical: we could see and touch much of what they were doing. Then with pneumatic instruments we could see and touch some of what they were doing, but had to get some information using a pressure gauge. With electrical instruments, we could see less of what they were doing, and certainly not touch them, but we could read what they were doing with a volt-ohm meter (VOM); they still were made of individual components. With electronic instruments, we could see little of what they were doing inside, and individual components gave way to integrated and digital circuits and measurements; individual signals became a lot harder to see.

Digital instruments use transient 1 and 0 signals to represent information; data is transferred at megahertz rates. In computer programming, the level of abstraction now defines the level of language used: at the lowest level are the 1s and 0s; then comes the assembly language level; then high-level programming languages (such as FORTRAN, Visual Basic, and C). Many times we deal with configurable devices and cannot even see or get to any of the programming that is going on.

Consider a simple system of a 4-20 mA level transmitter that sends a signal to a DCS. We can easily measure the 4-20 mA signal. But if the transmitter is using fieldbus or a proprietary digital signal, reading the transmitter's signal is not as easy, and conceptualizing what is going on is more difficult. On an even more abstract level, once it is inside the DCS the signal disappears in a maze of digital signals. The signal may interact with other signals, other information, and software constructs; it may eventually come out on a CRT screen, or influence the position of a control valve through a controller.

Each of these levels in the development of instruments represents levels of abstraction, and each changes the way we conceptualize the instruments at that level. We must also know what abstraction level to think about when we begin troubleshooting. The wrong level will lead us astray. Troubleshooting requires the ability to think at the different levels of abstraction in which instrumentation operates. We can no longer rely on analog measurements or digital snapshots alone, but must be able to think about how data flows through systems. To do this we must have a method or framework that allows us to approach and analyze problems logically.

4.8.4 Dimensional Thinking

People sometimes think one-dimensionally, along a line or in one direction. Sometimes the line is straight, or logical, and the next step flows from the previous. At other times, illogical steps produce a line that is not so straight. Many times people troubleshoot a system unsuccessfully; then another person asks a couple of questions about the problem and the solution jumps right out. This happens when we have all the information necessary to solve the problem but are not looking in the right place. This can be the result of one-dimensional thinking.

We can think in three dimensions. By analogy, consider driving a car: when you just concentrate on driving down a road, you are thinking in one dimension; when you navigate using a map, you are thinking in two dimensions. If in addition to your navigating, you are worried about whether your old car will keep running as you ascend the high mountain pass that the road crosses, you are thinking in three dimensions. A common symptom of dimensional thinking is looking for something and later finding it right in front of you. A common saying that applies is that you can't see the forest for the trees. The opposite also sometimes applies when you are overwhelmed with too much data, or you are looking at the system at too high of a level. Data analysis is often one dimensional—we follow a path through the data, ignoring other possibly useful data. Stepping back and taking a break can help get you out of the one-dimensional trap.

If you are thinking in two dimensions, you can still miss things that are not on the same level of abstraction. This is like a military patrol looking for enemies that looks forward, sweeps right and left, and looks for tracks and disturbed bushes to no avail; but they fail to look up in the trees where snipers lurk. They've forgotten the third dimension.

Four-dimensional thinking involves time. Events in a system occur at various times. You may be at the right place in three dimensions but at the wrong time. When system failures occur can be important when analyzing troubleshooting data.

Five-dimensional thinking is not a linear process—it is at the level of intuition, subconscious understanding, or gut feeling. This can lead you astray at times, or it can be the source of good ideas. This is part of the art of troubleshooting. Some intuition comes with experience and training, and some is a talent or a habit of mind, but it should not be ignored. It can save you when all other methods fail. We will discuss this type of thinking in the next chapter.

SUMMARY

Troubleshooting begins by following a logical, analytical framework—the guide or map to troubleshooting. It is much like going on a trip to a place you have never visited. If you do not have a good map, you may never get to where you want to go.

QUIZ

1.Troubleshooting can be defined as the means used to determine why something is not working or not performing its designated task or function.

A.TRUE

B.FALSE

2.A framework is

A.the basis by which to build a structure.

B.used to frame a picture.

C.a method of troubleshooting.

D.none of the above

3.Iterative processing is

A.repeating the same step in a process.

B.thinking about something in two different ways.

C.returning to a previous step in a step-by-step process and repeating the subsequent steps.

D.all of the above

4.A specific troubleshooting framework is

A.a troubleshooting framework that can be used for a specific vendor's equipment.

B.a troubleshooting framework for a specific class of instrumentation.

C.a troubleshooting framework for a specific type of instrumentation.

D.all of the above

5.Which of the following are types of troubleshooting frameworks?

A.a troubleshooting table

B.a troubleshooting book

C.both A and D

D.a troubleshooting tree

6.Plant dialect is

A.operator slang.

B.plant abbreviations.

C.both A and B

D.none of the above

7.The reasoning method you use when you have seen the exact problem before is

A.similar to analysis.

B.case-based reasoning.

C.linear analysis.

D.the iterative process.

8.Dividing a system in successive halves until you find the problem is called

A.divide and conquer.

B.the linear approach.

C.the halving system.

D.remove and conquer.

9.Once you think you know what instrument is causing the problem, you should

A.replace the instrument.

B.test to see if the instrument is in fact causing the problem.

C.repair the instrument.

D.put a new instrument in.

10.Troubleshooting can fail for which of the following reasons?

A.lack of knowledge

B.failure to look in the right place in the system

C.thinking in two dimensions for a three-dimensional problem

D.all of the above

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5

OTHER TROUBLESHOOTING METHODS

Substitution and fault insertion

Remove and conquer

Circle the wagons

Trapping

Consultation

Intuition and out-of-the-box thinking

5.1 WHY USE OTHER TROUBLESHOOTING METHODS?

The previous chapter discussed logical/analytical frameworks for troubleshooting. While these frameworks work most of the time, some problems require less systematic techniques to complement the logical frameworks. Normally, you will begin to use these other techniques only after the logical

analysis has failed to suggest a viable solution. Defining the problem, gathering information, and performing analysis still take place when you use these methods.

You may need to approach troubleshooting from a different point of view because a system may be too complex or sophisticated to troubleshoot with the knowledge available to you. This can occur with microprocessor-based equipment consisting of multiple components (such as a PLC or a DCS). Sometimes manufacturers provide only limited information about what goes on inside the equipment. Maybe the problem is transient in nature, or is in a complex system with communication links between components and multiple power systems and grounds, as in a multiple variable-speed drive system.

5.2 SUBSTITUTION METHOD

The substitution method is troubleshooting by substituting a known good component for a suspected bad component. For modularized systems or those with easily replaceable components, substitution may reveal the component that is the cause of the problem. First, define the problem and gather and analyze as much information as you can. Note that these steps are no different than the initial steps in the structured framework methodology. Then select a likely replacement candidate and substitute a known good component for it. If the problem goes away, you have at least found a partial solution. Then evaluate to see if a more general solution is needed. For example, if the component can be repaired on-site, either troubleshoot it further to find the lower-level cause of the problem or return it to the manufacturer for analysis.

By substituting components until the problem is found, the substitution method may find problems where there is no likely candidate, a group of candidates, or even a vague area of suspicion. One potential problem with modular substitution, though, is that a higherlevel cause can damage the replacement component as soon as you install it. This may confuse the issue if the failure is immediate as you will generally have the same symptoms after the replacement. If the failure is not immediate, this will give you a clue that the real cause of the problem is external to the module. The use of this method can raise the overall maintenance cost due to extra module cost and the cost of inventory of replacement modules. Even more problematic are cases in which the higher-level cause does not damage the replacement right away.

Another form of this method is to substitute or insert a known good signal or value into a system to see where a problem comes from. If you insert a known good signal and the downstream part works properly, then the problem is upstream. The converse is also true.

5.3 FAULT INSERTION METHOD

Sometimes you can insert a fault instead of a known good signal or value and see how the system responds. For example, when a software interface keeps locking up, you may suspect that the interface is not responding to an I/O timeout properly. You can test this by inserting a fault—an I/O timeout. Another example of fault insertion would be inserting a bad value into a point in a computer program to see how the program responds. A third example might be inserting a transient into a system, such as a simulation of a voltage sag.

5.4 REMOVE AND CONQUER METHOD

For loosely coupled systems that have multiple independent devices, removing devices one at a time may help to find certain problems. For example, if a communication link with ten independent devices talking to a computer is not communicating properly, you might remove the boxes one at a time until the offending box is found. Once the problem device has been detected and repaired, the removed devices should be reinstalled one at a time to see if any other problems occur.

The remove and conquer technique is particularly useful when a communication system has been put together incorrectly or exceeds system design specifications. For example, there might be too many boxes on a communication link, cables that are too long, cable mismatches, wrong cables, impedance mismatches, or too many repeaters. In these situations, sections of the communication system can be disconnected to see what happens.

Remove and conquer can also work for grounding problems. To detect whether a system is grounded in two places, try lifting a ground to see if things get better. (If you are lifting a safety ground, take care that you are protected while doing this.) One common problem for which this method is useful is when a shield is grounded in two places, causing a ground loop; if the regular shield ground is disconnected and the system improves, then another ground connection on the same shield may be the problem.

A similar technique, add back and conquer, means removing all the boxes and adding them back one by one until you find the cause of the problem. For example, on a new communication system like the one mentioned above, the boxes were removed one at a time and replaced, but no offending box was found. But when all the boxes were removed and added back one at a time, the troubleshooter found that there were too many devices for the computer's port to support. This could also have been detected if the devices were removed one at a time and not replaced and a point was found where the system worked.

5.5 "CIRCLE THE WAGONS" METHOD

When you believe that the cause of a problem is external to the device or system, try the "circle the wagons" technique. Draw an imaginary circle or boundary around the device or system; then see what interfaces (such as signals, power, grounding, environmental, and EMI) cross the circle. Then isolate and test each boundary crossing. Obviously, if you do not identify all the boundary crossings, you may miss the one causing the problem. Often this is just a mental exercise that helps you think about external influences, which then leads to a solution. Figures 5-1 and 5-2 illustrate this concept.

FIGURE 5-1 ***Circle the Wagons—Single Box Example***

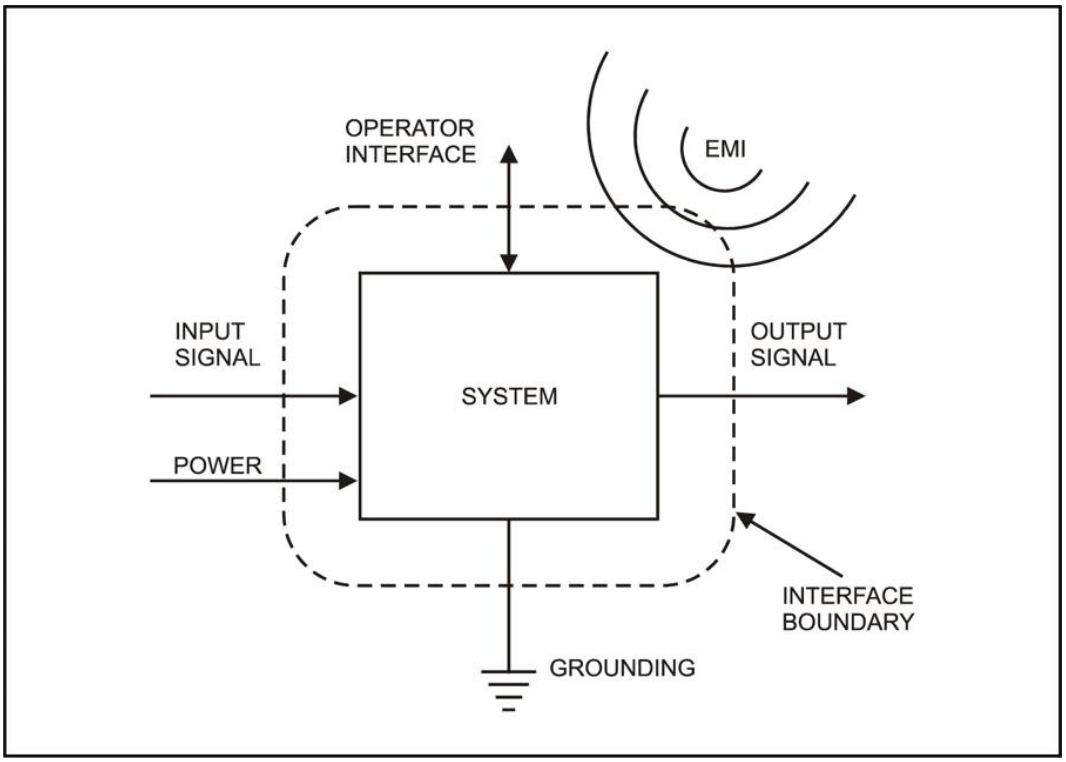
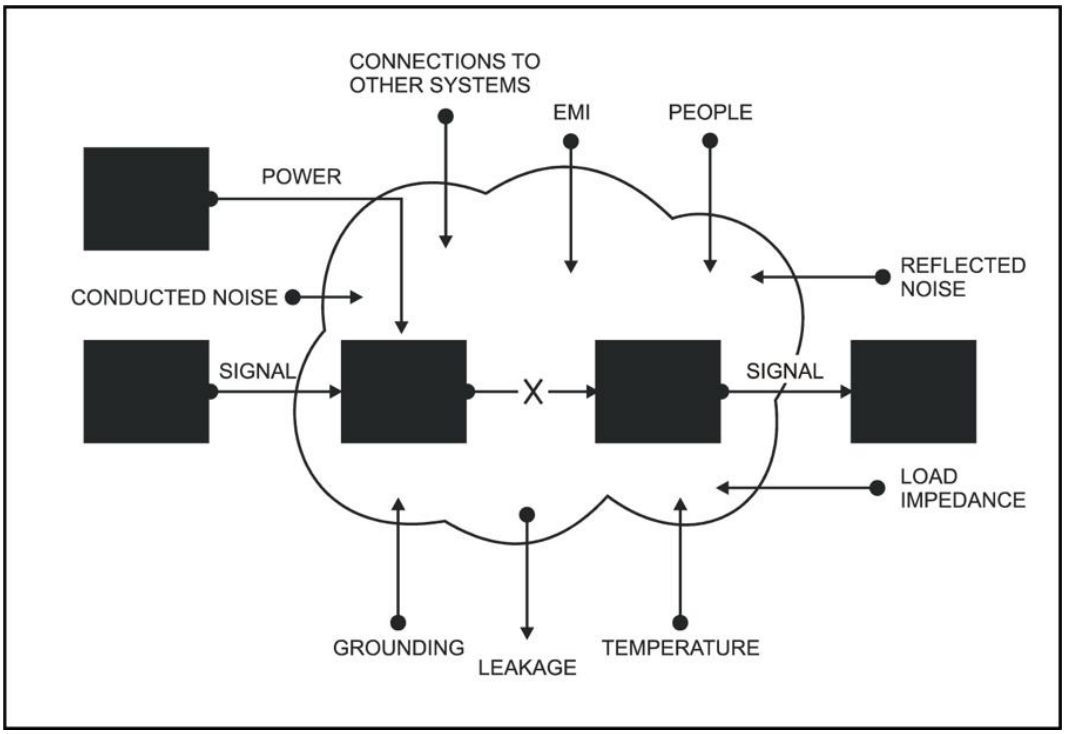


FIGURE 5-2

Circle the Wagons—Multiple Box Example (courtesy of Control Magazine)



5.6 TRAPPING

Sometimes the event that causes the problem is not alarmed, or is a transient, or happens so fast the system cannot catch it. This is somewhat like having a mouse in your house. You generally cannot see it, but you can see what it has done. How do you catch the mouse? You set a trap.

In sophisticated systems, you may have the ability to set additional alarms or identify trends to help track down the cause of the problem. For less sophisticated systems, you may have to use external test equipment or build a trap (see Figure 5-3). Power monitors such as the Dranetz 658 (see Figure 5-4) are often used for power problems. A storage scope may also be used to trap transients. Portable data loggers can be connected to monitor variables over time and dump the results into a computer, where the information can be graphed or evaluated. Examples of these include the AEMC Simple Logger, which can monitor voltage, current, or temperature, or the HOBO series from Onset Computer Corporation, which can log several different variables.

FIGURE 5-3

Example of a Signal Trap (courtesy of Dranetz BMI)

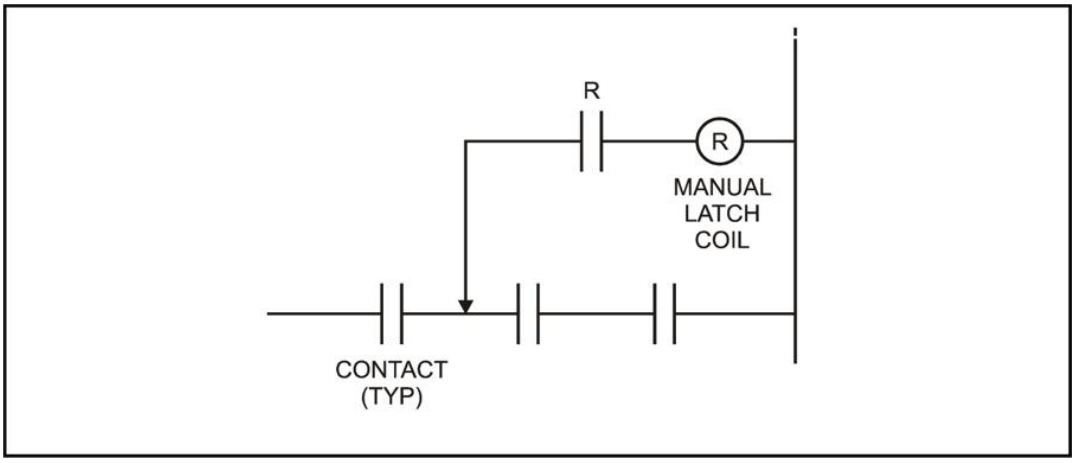
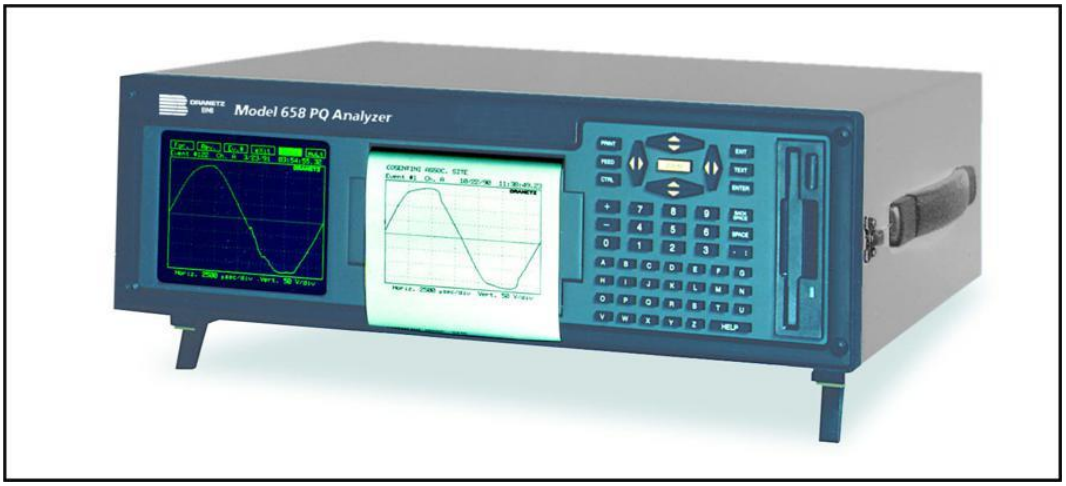


FIGURE 5-4

Dranetz Model 658 (courtesy of Dranetz BMI)



If software is involved, you may have to build software traps that involve additional logic or code to detect the transient or bug. In programs that use languages like FORTRAN or BASIC, a debugger may be available, or you can add print statements to print out intermediate values to detect the problem. If you are the programmer, consider putting in diagnostic print statements and having a software switch or switches turn them on and then print the results to a log file. You can use multiple levels of diagnostics with different switches to trap in different places.

5.7 COMPLEX TO SIMPLE METHOD

Many control loops and systems may have different levels of operation or complexity with varying levels of sophistication. One troubleshooting method is to break systems down from complex to simple. This involves finding the simple parts that function to make the whole. Once you find the simplest non-functioning part, you can evaluate the non-functioning part or, if necessary, you can start at a simple known good part and rebuild the system until you can find the problem.

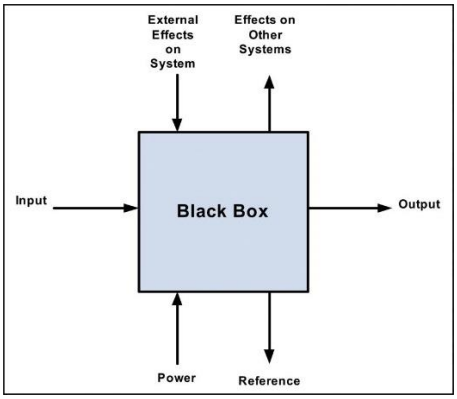
A common example of this is a varying or oscillating control loop. By placing the control loop in manual (moving from automatic control [complex] to manual control [simple]), one can determine if the automatic part of the system (commonly the tuning) is causing the problem or if the problem is being caused by the process or other external inputs. Another example is a cascade loop where you have a master loop and a slave loop. Cascade loops are commonly used to isolate variations in the slave loop measured variable from causing variations in the master loop measured variable (the desired control variable). Breaking down this kind of loop involves breaking the cascade by placing the master loop in manual or breaking the loop at the slave controller to see if the problem goes away, which can tell you which loop or if the process is causing the problem.

Computer control, either cascade or direct digital control, can be troubleshot sometimes by breaking the computer link, though this may be done for you as it is typically the first thing an operator does when he has a control problem in this type of system.

Hierarchical systems are another type of system that can be sometimes troubleshot using this method by isolating the different hierarchical levels from each other and reconnecting to find the problem.

Another example of this method is breaking down a complex system into sub-units that have defined inputs and outputs and verifying each sub-unit's functionality. A sub-unit can typically be broken down into a black box representation as shown in Figure 5-5.

FIGURE 5-5 ***Black Box View of Troubleshooting***



5.8 CONSULTATION

Consultation, also known as the third head technique, means that you and the equipment operator use a third person, perhaps someone from engineering or an outside consultant, with advanced knowledge about the system or the principles for troubleshooting the problem. This person may not solve the problem but may ask questions that make the cause apparent or that spark fresh ideas for you. This process allows you to stand back during the discussions, which sometimes can help you distinguish trees from forest.

5.9 INTUITION

Intuition can be a powerful tool. What many people would call troubleshooting intuition certainly improves with experience. During troubleshooting or problem solving, threads of thought may develop, one of which may lead you to the solution. The more experience you have, the more threads develop during the troubleshooting process.

Can you cultivate intuition? Experience suggests that you can, but success varies from person to person and from technique to technique. During troubleshooting you normally tend to narrow your focus; if you widen your focus instead and don't dismiss ideas too soon, you may spark good ideas. If a problem lasts more than one day, try to program your mind by giving it a specific task during relaxation or at bedtime. For example, tell your mind to think about whether grounding is the problem, and see what ground-related ideas develop. Try some kind of relaxation technique; once you are relaxed, let ideas flow until one comes along that can help. Experiment with different techniques and see what works for.

5.10 OUT-OF-THE-BOX THINKING

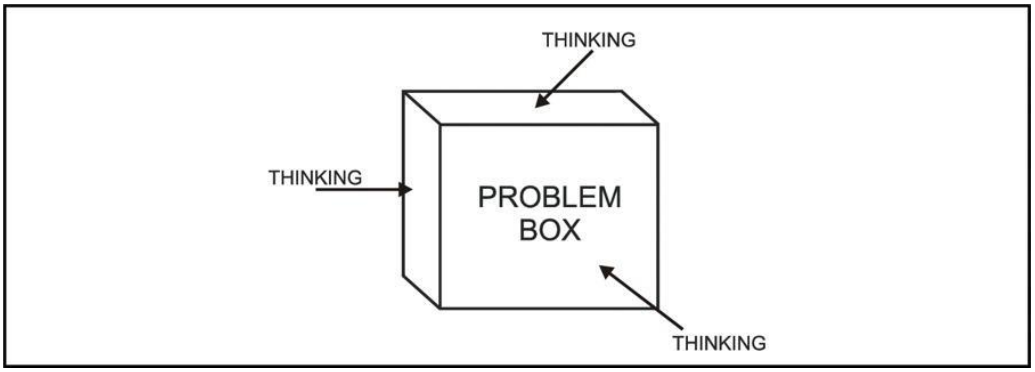
Difficult problems may require different approaches beyond normal troubleshooting methods. The term out-of-the-box thinking was a buzzword for organizational consultants during the 1990s. Out-of-the-box thinking means approaching a problem from a new perspective, not being limited to the usual ways of thinking about it.

The problem in using this approach is that our troubleshooting perspective is generally developed along pragmatic lines (i.e., what has worked before) and changing can sometimes be difficult. This can, however, be a developed skill. There are books available to help develop the skill, and you should keep an eye out for different troubleshooting perspectives as your experience develops and adapt them to your troubleshooting style. You will find people who have this skill; learn from them. Never be too proud to learn a skill from someone else!

An example might be an oscillating control valve. The signal is steady and there is no impressed noise, so you assume the valve positioner is causing the problem and replace it. To your surprise, the oscillation continues. What else could cause the oscillation? You have concentrated on a hardware solution. But what about the process? Could it cause the problem? For example, could an upstream pump be cycling? If you also consider the process side, you may reach the solution sooner.

FIGURE 5-6

Out-of-the-Box Thinking



How can you practice out-of-the-box thinking? How can you shift your perspective to find another way to solve the problem? Here are some questions that may help:

•Is there some other way to look at the problem?

•Can the problem be divided up in a different way?

•Can different principles be used to analyze the problem?

•Can analyzing what works rather than what does not work help to solve the problem?

•Can a different starting point be used to analyze the problem?

•Are you looking at too small a piece of the puzzle? Too big?

•Could any of the information on which the analysis is based be in error, misinterpreted, or looked at in a different way?

•Can the problem be conceptualized differently?

•Is there another box that has similarities that might provide a different perspective?

SUMMARY

Sometimes a logical/analytical framework will not help to find the cause of a problem and another method must be used to supplement the framework. But even using techniques like thinking out of the box, we can still use problem definition, fact-gathering, and analysis to define and narrow the problem.

QUIZ

1.The substitution method

A.inserts a bad signal for a good signal.

B.divides the system in half and tests one half.

C.substitutes a known good component into the system.

D.none of the above

2.The remove and conquer method

A.removes a component and puts another one in.

B.removes modules until the problem goes away.

C.removes all the modules and adds them back in one at a time.

D.all of the above

3.The fault-insertion method

A.inserts a fault to see if a problem can be duplicated.

B.inserts a good signal to see if the system works.

C.inserts a bad component.

D.both A and C

4.Which method checks all the external interfaces to a system?

A.trapping

B.out-of-the-box thinking

C.circle the wagons

D.substitution

5.Which method uses outside people?

A.substitution

B.out-of-the-box thinking

C.circle the wagons

D.a third head

6

SAFETY

Troubleshooting safety practices

Human error in industrial settings

Plant hazards faced during troubleshooting

Troubleshooting in electrically hazardous areas

Protection, procedures, and permit systems

6.1 GENERAL TROUBLESHOOTING SAFETY PRACTICES

This chapter covers the general troubleshooting safety practices used in industrial and petrochemical facilities. Part of the discussion will be background information to help you understand what is required for these areas. We will also look at work-tagging, permitting, and work procedures. Though this may be a review of procedures and practices similar to those of your workplace, the intent of this chapter is to emphasize safe troubleshooting, and it never hurts to review the basics. If anything here contradicts standards or practices at your workplace, follow your company's rules. You may, however, wish to discuss any differences with your supervisors.

First, let us consider two basic questions: What does safety have to do with troubleshooting? And why is troubleshooting different from normal maintenance?

Troubleshooting exposes you to hazards that other maintenance activities do not. Troubleshooting often requires interaction with or work around energized circuits and running machinery. Troubleshooting may require interaction with active control loops and safety systems. It also has many of the same dangers that any activity in an industrial facility has—working in hazardous areas, around hazardous materials and hot pipes, on dangerous equipment, at heights, in constricted areas, and with equipment that operates continuously. Paying proper attention to safety when troubleshooting is essential to your own safety, your fellow workers' safety, and the safety of your facility.

There are many different types of hazards in an industrial facility. Some are apparent or identified, but some are invisible. Failure to perform any job safely is a no-win situation for both the technician and the company. While many safety practices depend on the hazards involved, certain general safety practices apply to all work.

The following safety practices apply to most work situations:

•Study the job carefully to determine all the hazards present and see that all safeguards and safety equipment are provided to protect you, other persons in the job area, and the equipment during the performance of the job.

•Before beginning work, acquaint yourself with the applicable safety regulations, standard operating procedures (SOPs), standard maintenance procedures (SMPs), and facility practices.

•Plan safety into each job. Orderliness, organization, and good housekeeping are essential to your safety and to the safety of others.

•Examine all the tools and protective, safety, test, and personal protective equipment (PPE) before using them; make sure that they are in good shape and that they are adequate for the intended work.

•Provide the proper safeguards, such as danger signs, roped-off space, and firewatches, needed to protect persons close to, but not engaged in, the work being done.

•Consider the result of each act. There is no reason for you to perform an unsafe act to accomplish your work or to take chances that will endanger yourself or others.

•Ensure that there is adequate lighting, access, egress, and work space for the intended work.

•Satisfy yourself that you are working under safe conditions. The care exercised by others cannot always be relied on.

•Always double-check that the equipment you are going to work on is the equipment you are supposed to work on.

•Always know where safety showers, eye baths, fire extinguishers, respirators, fire blankets, self-contained breathing apparatus (SCBA), and other safety equipment are located.

•Never pass up training in safety.

•When in doubt, ask. Never take safety for granted.

•If you are involved in the design or review function, remember that safety must be designed into an installation from the very beginning.

•Take your time to do the job safely; shortcuts are an accident waiting to happen.

•Remember that safety is a state of mind, coupled with conscious action and maintained by continuous vigilance, and you are the person responsible for your own safety.

6.2 HUMAN ERROR IN INDUSTRIAL SETTINGS

All humans make errors. Many errors have nothing to do with safety, but those that do can lead to an accident. These are the errors we must try to prevent. Remember, under small errors lurk a large error.

Human errors that affect industrial settings can be classified into six general categories: slips or aberrations, lack of knowledge, over- or undermotivation, impossible tasks, mindset, and errors by others.

6.2.1 Slips or Aberrations

Slips or aberrations occur despite your best intentions to do the task safely. This can come from inattention or distraction. It often occurs on routine tasks in which you may have put yourself on autopilot. An outside-the-plant example of autopilot might be when you get in your car and it seems to drive itself home from work without much attention from you. Have you ever planned to go shopping after work but missed your turn because you were driving on autopilot? As a troubleshooter, you should beware of being on autopilot on routine troubleshooting or simple tasks. Inattention can also come from thinking about one thing while doing another thing. Humans have a very difficult time doing two tasks at once. Always concentrate on the task at hand.

Distraction can come from other activities occurring nearby. Again, always concentrate on the task at hand. That can sometimes be hard to do when there is a guy nearby breaking up concrete with a jackhammer. If you cannot concentrate on the job at hand, try to remove the distraction or do the task at a time when the distraction is not there.

6.2.2 Lack of Knowledge

Lack of knowledge is generally due to lack of training or experience. It may also come from failure to ask questions or from making assumptions. There is a saying, Assumptions are the mother of all screwups. Lack of knowledge can also come from faulty learning, such as learning incorrectly from on-the-job training. If you do not feel you have enough information, experience, or training to do a task safely, do not do it. Talk to your supervisor and get some help. Be wary of generalizing too much from your training: if you do not fully understand the system you are working on, you may reach incorrect and dangerous conclusions about what needs to be done. Seek out good teachers, and never turn down training or retraining.

Lack of knowledge can also come from incomplete or faulty job instructions or from faulty information about the troubleshooting problem. One has to consider all the information given in a job and ensure that all is considered carefully in regards to safety.

6.2.3 Overmotivation and Undermotivation

Overmotivation can come from being overzealous in doing tasks assigned to you. Do not do unsafe things just to complete the job quickly. It may give you some spare time, but habitually doing repairs in an unsafe manner will eventually cause an accident. The same goes for getting praise from your boss for being fast; it just is not worth it if you have to do something dangerous to get it.

Undermotivation can come from problems at home or at work. When you have to do a task, leave these problems at the door of the shop. Do not let them make you take a safety risk.

6.2.4 Impossible Tasks

Sometimes a task may be not humanly possible or may require you to do something unsafe. Examples of this are valves that are located where they cannot be operated safely, access at unsafe heights or in constricted areas, or jobs located too close to operational hazards. For these kinds of tasks, step back and figure out how the task can be done safely. If you cannot figure it out, get help. Do not take unnecessary risks.

6.2.5 Mindset

Sometimes humans believe something to be true, even if there is evidence to the contrary. For example, one technician stuck his hand into an electronic device and received a shock. He stepped back and said, There can't be voltage there, stuck his hand back in, and received another shock. This was an actual true-life example witnessed by the author. Another case of dangerous mindset is the statement We've been doing this for twenty years and nothing has ever gone wrong. Reportedly, this was once said at a plant the day before it blew up. Always look at your troubleshooting tasks as if you have never done them before, and then ensure that you are doing it safely.

6.2.6 Errors by Others

Many times you will work with a partner or around other people. Their errors affect your safety as well as their own. Always be aware of other peoples' actions. This is like defensive driving but in troubleshooting we might call it "defensive working"—not a bad idea for any kind of work.

6.3 PLANT HAZARDS FACED DURING TROUBLESHOOTING

An industrial facility poses many hazards, some obvious and others hidden. Many times you can get away with taking a risk, a shortcut, or a chance, but sooner or later such actions will catch up with you. You can be safe in your troubleshooting activities only if you act to ensure your safety.

Hazards come in all shapes and forms. The following is a general though by no means a complete discussion of some of the hazards you may face while troubleshooting.

6.3.1 Personnel Hazards (Electrical)

Troubleshooting often requires working on or around energized circuits. This exposes the troubleshooter to additional hazards over normal de-energized maintenance. Troubleshooters must know which circuits are energized and what voltages are present in the area that they will be troubleshooting. The assumption of a single power source for an instrument or electrical device must be taken with a grain of salt, as it is not uncommon to have more than one power source for an instrument. For example, a speed or level switch may have a 120 VAC contact closure output powered from one source, but also a 120 VAC power supply from a different source.

If you have to work in close proximity to hazardous high voltage-level energized circuits during troubleshooting, consider the use of personal protective equipment such as gloves or a flash suit, guards, and insulating mats or covers. Always use the proper tools and never stick your hand where it might encounter an unknown or unseen hazard. If you don't know what is there, don't stick any part of your body there!

Electrical personnel hazards come in two general forms: electrical shock and physical damage resulting from the release of electrical energy.

6.3.1.1 ELECTRICAL SHOCK

An electrical shock can be defined as a flow of electrical current through a living organism. The results of an electrical shock can range from mild annoyance to burns, organ damage, and death. When a person touches an electrically energized surface and also provides a return path for the electricity to flow through the body, an electrical shock occurs. The physical effect of the electrical shock depends on the kind of electricity (AC or DC), the physical flow of the electricity (where it flows), the magnitude (how much electricity flows), and the duration of exposure (how long it flows). Here are some rules to keep in mind about these effects:

•AC electricity is, in general, more damaging for the same level of direct current flow.

•If the physical flow of electricity is from the right index finger to the right elbow, the effect for the same magnitude of electrical current will be

less than if the flow is from right index finger to the left index finger (in other words, across the heart). At power line frequencies (50–60Hz), the body becomes a volume conductor, and any electrical current flowing in the trunk of the body will spread out so that some of it will flow through the heart.

•The effects of a small amount of current, such as 1mA (a barely perceptible shock), are different from those of 80mA (in the range that can cause ventricular fibrillation).

•Duration of a current of 20mA for 1 second may be safe, whereas 20mA for 10 seconds may be hazardous.

Another known effect is called the no-let-go effect, where muscle contractions due to an electrical shock cause the person to grab the live wire or surface and not be able to let go. Care must be exercised in dealing with this situation, as rescuers may be shocked if they touch a person being shocked.

One of the rules of working on energized circuits is called the one-hand rule, which means using only one hand and keeping the other hand in your back pocket, for example. This rule reduces the risk of a significant across-the-heart shock.

The physical path of electricity is determined by the point of contact with the energized circuit and the point of contact with the electricity's return path, typically a contact with ground or grounded metal or conductive parts or surfaces. The magnitude of the electrical shock current is determined by Ohm's law:

$$\text{Shock Current} = \frac{\text{Contact Voltage}}{\text{Body Resistance}}$$

At power line frequencies, the body's resistance is primarily a function of surface contact resistance, that is, skin resistance. The surface contact resistance is affected by skin moisture, the type of skin contact (such as direct skin contact or penetration), and the area of contact. At power line frequencies, the body is essentially resistive, while at frequencies of 300 Hz and above, the skin effect comes into play; the body's impedance increases and less current will flow. In most industrial countries, the safe voltage is generally considered to be around 30V RMS. The National Electrical Code (NEC) 110-27a requires that guarding be used at 50V and above. Women tend to have lower reaction values than men. The effect of electrical current on the human body is given Table 6-1. Since the values for currents in Table 6-1 are a function of many things, such as body weight, body resistance, current path, and duration, the values given are on the conservative side of reported values.

TABLE 6-1

Effects of Electrical Current on Human Body

CURRENT	EFFECT
0-1 mA	Imperceptible

1-3 mA	Perceptible
3-5 mA	Annoyance
6-9 mA	"No-Let-Go"
20 mA	Asphyxiation
75 mA	Ventricular fibrillation
4-10A	Cardiac arrest, burns

Note in Table 6-1 how currents of a few milliamps can cause serious problems, if not death. And remember that current level alone is not the only determining factor; duration of exposure also determines the effects of the electrical shock. The longer the exposure, the greater the risk of damage.

When you receive an electrical shock, in addition to internal damage you can suffer thermal burns or physical damage at the entry and exit points. Injury can also occur from muscle contractions or response to the shock such as jerking back and hitting your arm on something. Even more severe injury can result from being flung bodily from the shock point.

6.3.1.2 DAMAGE FROM THE RELEASE OF ELECTRICAL ENERGY

Touching an energized surface with tools or other conductive materials can cause a fault (provide a path to ground); faults during the failure of electrical equipment can also cause serious equipment damage and bodily injury. This damage can be caused by several forces, including electrical arcs, high temperatures, explosive forces, and shrapnel. Electrical equipment can fail during the touching of the energized surface, and during normal or abnormal equipment operation.

Remember that electricity can release substantial energy should you touch it or be in the presence of a fault or malfunction. Arcs can have temperatures up to 35,000°F (19,450°C); they can cause fatalities at distances greater than ten feet (3 meters), and first-degree burns at distances up to forty feet (12 meters). The explosive force of vaporizing copper is equivalent to that of dynamite. While troubleshooting instruments normally involve low voltage levels, sometimes troubleshooters work in or around much higher voltages. Always respect electricity or you may not live to regret it.

When you are operating electrical equipment, as when you open or close a circuit breaker or switch, you should not take for granted that it will operate correctly, or that it will not fail during operation. For example, it is a bad practice to stand in front of a circuit breaker or switch while you operate it. Should it explode during operation, you would take the brunt of the explosive force and shrapnel. For example, one foreman stood in front of a faulty 480VAC disconnect while he switched it. Fortunately, he survived to wish he had not.

6.3.2 General Practices When Working With or Near Energized Circuits

In dealing with energized electrical equipment, the following general practices are time-tested and prudent:

•First and foremost, know what voltage and current levels are available and their location. Know what you are dealing with.

•All work should be done according to the proper standards. (See the list of relevant standards below.) Your site's work practices should reflect the requirements in these documents.

•Always double-check to ensure that you have identified the correct electrical equipment or circuit. A number of years ago, a fatality occurred when an experienced electrician, who was in a hurry, locked out one motor and then worked on a different motor. Unfortunately, he did not check to see if voltage was present before beginning work. Carelessness, as in this case, was fatal.

•Know the personal protection equipment (PPE) for the job that you are going to perform, and use it. Use other protective equipment as appropriate; if you think you need it, it is appropriate. Never leave guards off for the sake of convenience.

•Know and follow your facility's lock-out/tag-out (LOTO) procedure carefully. Your facility must have a formal lock-out/tag-out procedure per OSHA requirements. This should not be limited to just electrical sources of energy. This type of procedure is detailed in NFPA 70E.

•Your facility should have a procedure for working on or around 150V or greater to ground. Again, know this procedure and use it.

•If it is necessary to interrupt your troubleshooting work—even to go to lunch—when you return be sure to verify that all circuits are de-energized before beginning work again. Know the procedures for troubleshooting that lasts more than one shift.

•Remember that de-energized equipment can still bite. Grounding straps or discharge switches may be required to discharge all the stored electricity in some equipment.

•After your work is finished, make a thorough check of the circuits and equipment to verify that the work has been completed, that the connections are tight, all guarding devices are in place, and that the wiring is properly installed. Do this before re-energizing the equipment or circuits. Remove all bypasses or forces put in place for the purpose of troubleshooting. After the power has been applied, verify proper operation of the equipment.

6.3.3 Static Electricity Hazards

Static electricity is not normally an electrical shock hazard, even though a static electric shock may hurt or be uncomfortable. It can, however, be a source of ignition. Where the static electricity is generated by the flow of such things as nonconducting flammables, dust, powders, and pellets, there may be the possibility of a fire or explosion. Grounding is commonly used to minimize this, though it is not the cure for all types of static electricity problems; for example, it will not typically fully discharge static electricity

accumulated on insulating materials. Grounding can, in addition, serve as a discharge path, and the resulting spark can be a source of ignition.

6.3.4 Mechanical Hazards

Mechanical hazards come in all shapes and forms, such as sharp edges, head knockers, and trip hazards. Head knockers and trip hazards are commonly caught and removed or identified during safety audits. Sharp or pointed edges or corners are less commonly caught. Many times during troubleshooting you will be working on things that are not normally visible, and consequently it is less likely that these hazards will have been caught and corrected. If you find them, have them corrected so that the next technician (who might also be you) will not encounter the hazard. Always evaluate where you walk or climb, and where you stick your hands or other parts of your body. Some potential mechanical hazards are temporary, such as a dropped bolt or tool, items that have been laid down, or moving equipment such as cherry pickers or cranes. Housekeeping around your work area is important to maintain safety. Always be aware of what is going on around you. Other people working nearby may create safety hazards. Always make people nearby aware that you are working in the same area.

6.3.4.1 ACCESSIBILITY AND HEIGHTS

Many times in industrial facilities, the designs of installations do not take maintainability into consideration, and the instruments that you must troubleshoot may not be located where they are readily accessible. This may require you to use a ladder or to climb out on a pipeway. Falling is a major cause of accidents. When working on inaccessible instruments, always follow your facility's rules for such procedures as tying-off ladders and wearing harnesses. If necessary for safe work, have a scaffold built or use a bucket truck or manlift. Your facility should have scaffold inspection procedures to insure that the scaffolding is adequate. Always make sure any scaffolding you are using appears to be safe before you use it, and that it has a record of current inspection. Wherever work is required at a height greater than a certain point, normally about eight feet (2.44 meters), you may need to follow a hoisting or working-at-heights procedure and/or get the associated permit. Also remember that if you are working at height, people below you who may be at risk from your dropped tools or other items. In these cases, it may be appropriate to rope or tape off the area.

6.3.4.2 MOVING EQUIPMENT

Instrumentation on rotating and moving equipment has become more prevalent and sophisticated. Many times maintenance personnel are required to troubleshoot or perform other maintenance activities on equipment that can move, is moving, or is near moving equipment. Great care must be taken to insure that this is done safely. There should be standard maintenance instructions (SMIs) or procedures (SMPs) to cover working on moving equipment.

If the troubleshooting is to be done while the equipment is not moving but still energized, care must be taken that it does not inadvertently begin moving. Generally, the technician should not be near the moving portions of the equipment, or in contact with them, during these activities. If it is necessary to be near the moving parts, make sure guards or barriers are in place to prevent contact with them.

If the equipment is moving during troubleshooting, contact or near contact with moving or rotating parts should be handled very carefully. Barriers should be used if necessary to prevent this contact and

allow safe troubleshooting of the equipment. Great care should be taken if personnel safety interlocks or guards are to be bypassed to allow troubleshooting near the moving parts. In most cases this should not be allowed, but if it is allowed, the safety of the troubleshooter must be assured. In addition, administrative controls (such as permits, SMPs, SMIs, and SOPs) should not be bypassed for the sake of expediency.

If the equipment you are working on is not energized or running, you should still take care that it is not turned on inadvertently. Follow lockout/tag-out procedures (LOTO). Do not count on someone else's locks. Also, since the moving equipment can, in many cases, be remote from the part of the system you are troubleshooting, care must be taken not to injure someone who is around the remote equipment.

6.3.4.3 EYE HAZARDS

Eye hazards are also potential dangers while troubleshooting. As a minimum precaution, safety glasses (preferably with side shields) should be worn. Use monogoggles and face shields as appropriate for situations requiring more eye protection.

Always know where the nearest eye bath is located. It is not a bad idea to test the eye bath, though some facilities have alarms on the eye bath operation and the board operator should be notified in these cases prior to test. Do not take your glasses off for the sake of convenience. Take absolutely no chances with anything that gets in your eyes—use the eye bath, report it immediately, and get your eyes checked.

6.3.5 Stored Energy Hazards

Many safety accidents are caused by the release of stored energy. We have already discussed electrical energy, but there are other types of stored energy, such as pneumatic pressure, process gas pressure, liquid pressure, hydraulic pressure, and spring energy. All of these should be accounted for under a lock-out/tag-out procedure, but some may not be. Always be sure of where energy can be stored in the system you are troubleshooting, and take care that it is not released in an unsafe manner. In general, gas pressure is more dangerous than liquid pressure. Do not take anything apart without knowing where energy might be stored in the system.

6.3.6 Thermal Hazards

Thermal hazards generally come from hot materials, heat tracing, or hot pipes. Vaporizing regulators in analyzer sample systems can be another source. Temperatures over about 115°F can cause damage. Very cold surfaces or materials can be just as dangerous as hot ones. Always know the properties of the materials you are dealing with and working around. Even though the thermal hazards in your facility are probably already identified, you should still be careful where you stick your hands, or what you allow your body to touch.

Hot material can be a potential hazard when you are blowing down an instrument, such as a pressure transmitter or D/P cell. Under normal operation, the instrument impulse lines can be cool even though the process is hot, until the impulse lines are blown down. Then the impulse lines will

become hot, and hot liquid or gas may be sprayed about. This sort of hazard frequently involves steam and hot water lines. Do not vent through the transmitter because it may not be suitable for the process

fluid temperature. Blowing down high-pressure liquids that will be vaporized upon release can cause extremely cold surfaces, which can be dangerous.

There may also be some instances during troubleshooting where you touch hot electrical or electronic components such as transformers, resistors, or heaters. In some cases, mechanical components that are hot due to friction can be a hazard. Remember, always be careful where you place your hands and what you allow your body to touch.

Be careful when dismantling an instrument under pressure. This is a common issue when dealing with valves where you are working on the actuator. A number of fires and explosions have occurred in the past when technicians dismantled a valve top work and released large quantities of hazardous gases due to the design of the valve actuator, mounting, and body arrangement.

6.3.7 Chemical Hazards

Chemical hazards should normally be contained within the piping and vessels of your facility, but as we all know, things leak and a mistake during operations or maintenance can release potentially hazardous materials. Be aware of the chemical hazards in your facility. Remember that some chemicals become hazardous at an exposure level of a few parts per million or even billion. Chemical hazards can be breathed, absorbed through the skin, or cause surface irritations on the skin or eyes. Some may take a while to make themselves apparent.

All facilities must have Material Safety Data Sheets (MSDS) on-site that describe the hazards and other safety information for the chemicals in the facility. Since you will be dealing with instruments that contain potentially hazardous chemicals, you must take care that you deal with the hazard properly. Always wear the proper personal protective equipment. If, during troubleshooting, you must blow down an instrument, or do some other service that may expose you to the process chemicals, take care to wear proper protective gear, and follow your established maintenance practices to minimize exposure.

One of the most common hazards is an instrument taken out of service for troubleshooting in the shop. Once in the shop, a different technician may troubleshoot the problem before fixing it. If that technician is unaware of the service it performs, exposure to a chemical hazard may occur. Control valves are a common type of equipment through which this can happen. All instruments removed from service should be identified as to the service they perform and whether they have been cleaned or not. Each instrument should be cleaned before being put back in service.

Venting through the vent ports on transmitters should be done with care, as you may damage the transmitter or expose yourself to a hazard. Also, you may not realize where the vent is pointed. A young instrument man was exposed to hazardous chemicals while troubleshooting a transmitter with a more experienced technician, who used the transmitter's side vents to vent 10% caustic—directly onto the young man's chest. Fortunately, there was a safety shower nearby and no harm was done.

Always be aware of special hazards, such as hydrogen sulfide (H2S), that may be in the area where you are troubleshooting. Many times there will be special alarms or lights indicating the hazard is present. Always be aware of your surroundings and its hazards.

RELEVANT STANDARDS

•NFPA-70E, Standard for Electrical Safety Requirements for Employee Workplaces.

- NFPA-70, The National Electrical Code (NEC).
- NFPA-101, Life Safety Code, Electrical Safety Requirements for Employee Work.
- OSHA Code of Federal Regulations: Title 29, Chapter XVII, Part 1910, Subpart S, Electrical.

6.4 TROUBLESHOOTING IN ELECTRICALLY HAZARDOUS (CLASSIFIED) AREAS

Because troubleshooting is often done in the field, and many times in electrically hazardous (classified) areas, we must be able to troubleshoot instrument systems safely in such areas. This requires a basic understanding of the methods used to put instrumentation in hazardous areas. Not only must we be able to open and interact with instrument circuits, but we must also return the equipment to its original safe state. (Obviously, if the instrumentation is mechanical or pneumatic, this is not an issue.)

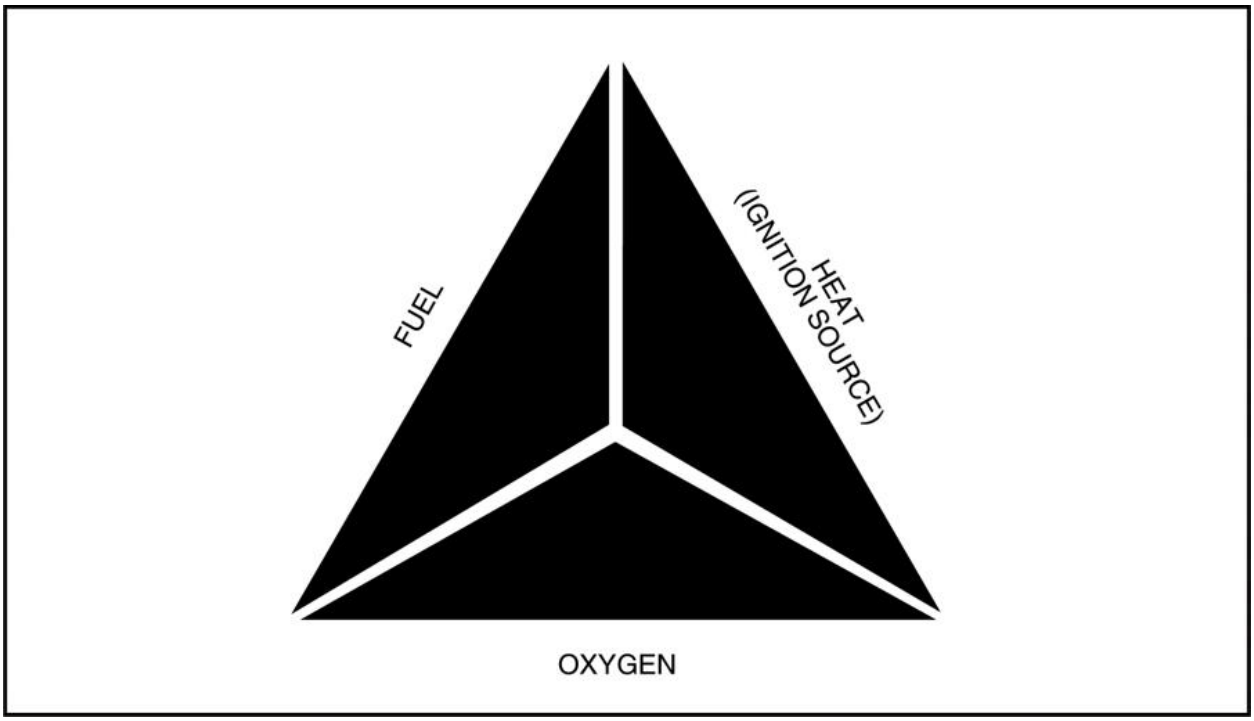
Three components must be present to have a fire or explosion: (1) an oxidizer such as atmospheric oxygen, (2) a flammable material in an ignitable mixture, and (3) a source of ignition (see the fire triangle, Figure 6-1). Since oxygen in the air makes one side of the triangle readily available, and

facilities that handle flammable or combustible materials can provide fuel, another side, all that is required to cause an explosion or fire, is the third side—an ignition source. Electrical equipment and maintenance activities can provide the required source of ignition to complete the fire triangle.

What follows is a brief review of the principles of area classification and the methods or means of placing electrical equipment in these areas.

FIGURE 6-1

Fire Triangle



6.4.1 Classification Systems

This section provides an overview of NEC Article 500 division-based and Article 505 zone-based classification systems.

6.4.1.1 NEC ARTICLE 500 TRADITIONAL DIVISION CLASSIFICATIONS

Since electrical equipment can serve as a source of ignition, methods that prevent this from happening must be used to allow the safe installation and operation of the electrical equipment in areas where flammables could be present. The first requirement is to determine the nature and extent of the flammable material hazard. The National Electrical Code (NEC) is the method used by the National Fire Protection Association to classify area hazards in regard to placing electrical equipment in the classified area. (See the list below for relevant standards.) NEC Article 500, Hazardous (Classified) Locations, defines area classification. Articles 501-555 further explain the requirements for the use of electrical equipment in hazardous (classified) areas.

The traditional (division style) NEC Article 500 area scheme classifies each hazardous area according to its class, group, and division. Areas that are not classified are considered unclassified or nonhazardous.

The class designator identifies the physical nature of the hazard. The class designators are:

•Class I—Where flammable gases or vapors, flammable or combustible liquids are processed, handled, or stored.

•Class II—Where combustible dusts are processed, handled, or stored.

•Class III—Where easily ignitable fibers or flyings are processed, handled, or stored.

The group designator identifies the physical properties of the hazard. Some of the physical properties of interest are maximum explosion pressure, pressure piling effects, and the maximum experimental safe gap (MESG)—the maximum experimentally safe gap through which explosive gases can be vented from an enclosure and not cause a fire outside the enclosure. The group designators with representative chemicals are:

A: Acetylene

B: Hydrogen, ethylene oxide, propylene oxide

C: Ethylene, acetaldehyde, carbon monoxide, methyl ether

D: Gasoline, methane, ethane, propane, propylene

E: Combustible metal dusts

F: Combustible carbonaceous dusts that have more than 8% total entrapped volatile or have been sensitized by other materials so that they present an explosion hazard: carbon black, charcoal, coal, coke dust

G: Combustible dusts not included in groups E or F: corn, wheat, polypropylene, polyethylene

Note that Groups A-D are Class I chemicals, while E-G are Class II chemicals or materials. Class III does not have any groups.

The division designator identifies the probability and the extent that the flammable or combustible mixture will exist in the area at any given time.

The division designators are:

•Division 1—The flammable or combustible mixture can exist under normal conditions. Normal conditions can include those occurring during regular maintenance activities and during regular chemical releases that occur during normal operations.

•Division 2—The flammable or combustible mixture can exist under abnormal conditions. Abnormal conditions can include conditions such as malfunctions, pipe rupture, and equipment leaks that do not normally occur.

Once the probability part of the division designator has been determined, the physical extent of the classified area must then be determined. The extent is the physical boundaries, both horizontal and vertical, where the flammable mixture can exist. In general, there must also be transitional areas. For example, a Division 1 area cannot generally transition into non-hazardous without there being an intervening Division 2 area. The area classification for your facility should be on an area classification drawing or some other document indicating the area classifications and their extent.

The traditional hazardous area classification hierarchy is illustrated in Figure 6-2.

6.4.1.2 ZONE-BASED AREA CLASSIFICATION

Another area classification method introduced in the 1996 NEC, Article 505, is a classification scheme similar to that used in Europe. The groups are different from the traditional groups (A-D). These groups are Group IIA (equivalent to Group D), Group IIB (equivalent to Group C), and Group IIC (equivalent to Groups A and B). This scheme uses zones instead of divisions so that, for example, they are designated Zone 0, 1, and 2 instead of Division 1 and 2. A primary difference is that Division 1 is now split into two zones, 0 and 1.

•In Zone 0, the flammable mixture is considered to be present all the time, or for long periods of time.

•In Zone 1, the flammable mixture is considered likely to exist under normal operating conditions. It may exist frequently due to maintenance activities, where failure of equipment would cause the simultaneous release of a flammable mixture and create a source of ignition. It may also exist adjacent to a Zone 0 area where a flammable mixture would be communicated, if not prevented, by a positive barrier.

•Zone 2 is equivalent to Division 2.

Class II and III areas are not currently covered at this time by this method. Class II zone areas will be covered in the near future and will be designated by placing a 2 in front of the normal Zone 0, 1, 2, i.e., Zone 21 for a dust zone 1 area. **The primary reasoning behind this new method is harmonization with international standards and reduced cost of equipment due to Zone 1 equipment means being cheaper than Division 1 equipment.**

As with the division designator, the physical extent of the classified zone area must also be determined. The zone classification hierarchy is illustrated in Figure 6-3.

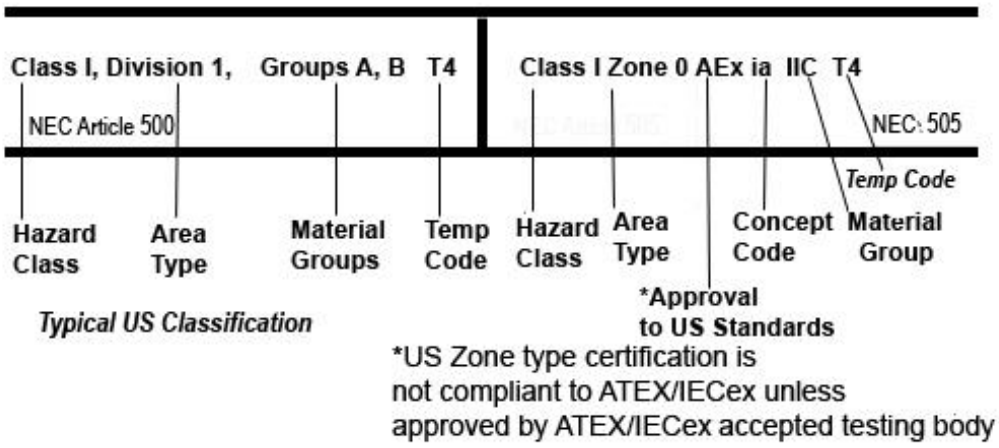


FIGURE 6-2

NEC Article 500 Hazardous Area Classification Hierarchy

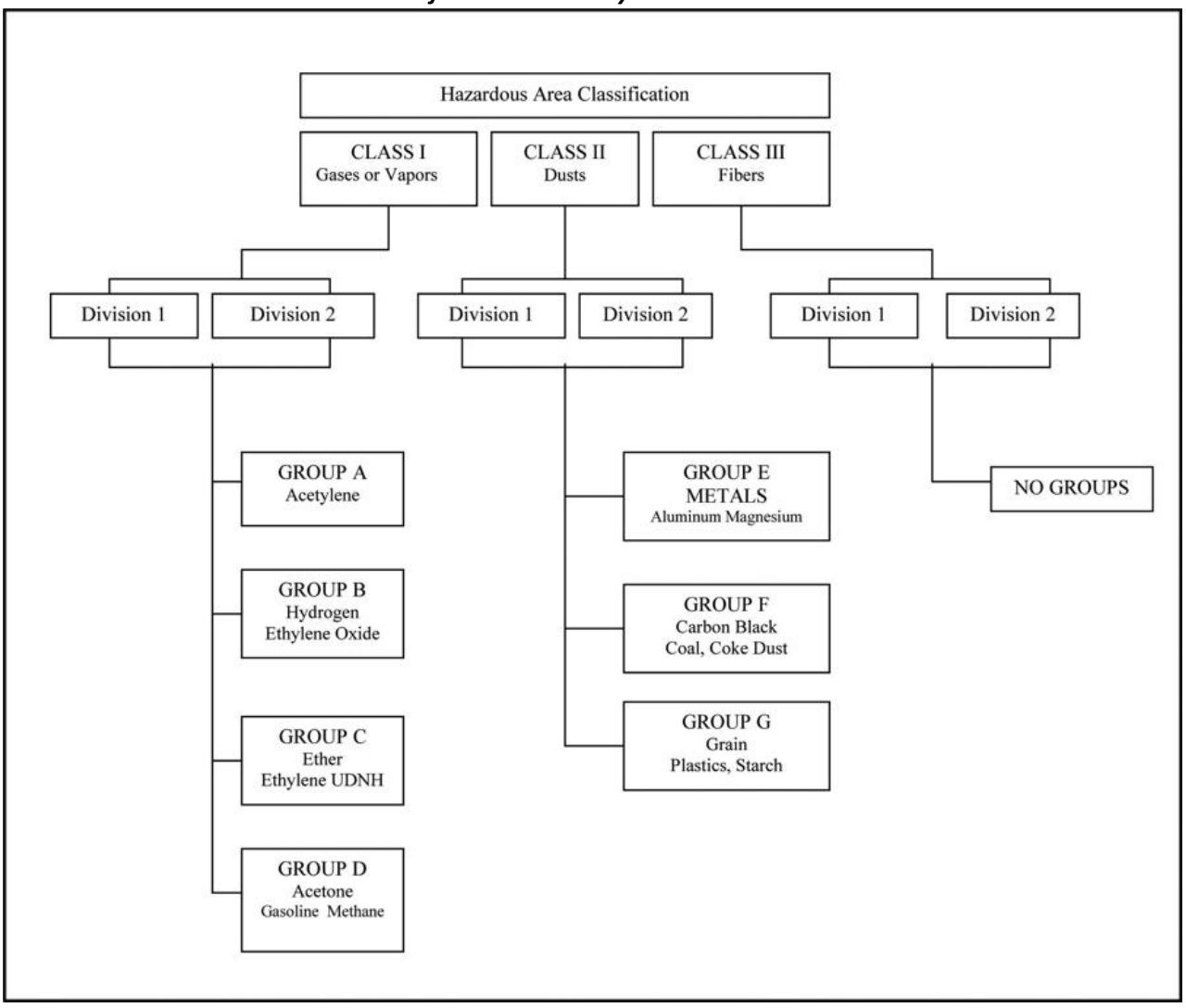
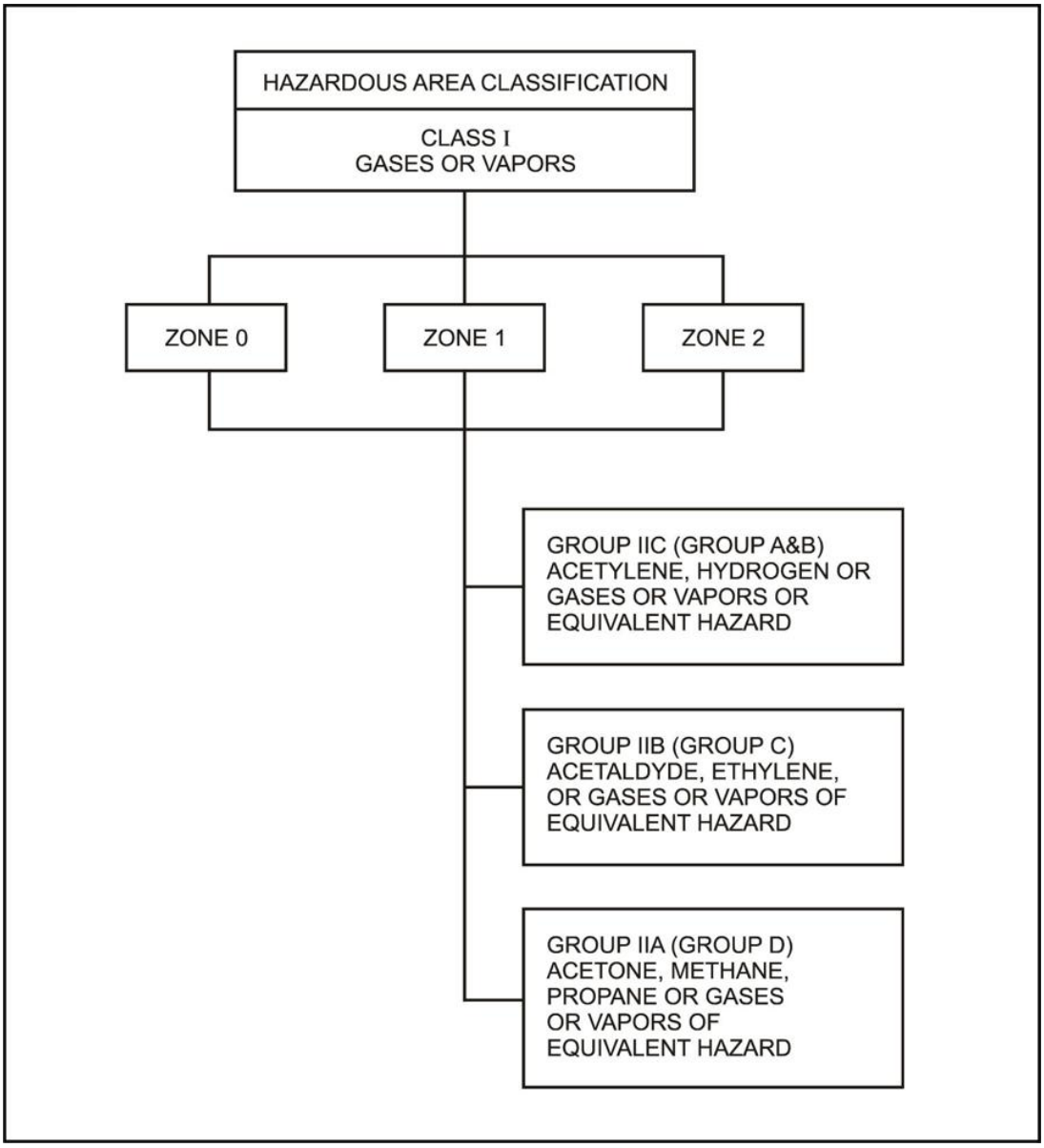


FIGURE 6-3

NEC Article 505 Zone Area Classification Hierarchy



Comparison Between Zones and Classes/Divisions		
Standards	NEC 505	NEC 500
Atmosphere		
Gases, vapors	Zone 0 Zone 1	Class I, Div. 1
	Zone 2	Class I, Div. 2
Dusts	Zone 20 Zone 21	Class II, Div. 1
	Zone 22	Class II, Div. 2

6.4.2 Area Classification Standards

The main recommended practices for area classification for Class I (flammable gases or vapors) areas are NFPA 497A, Classification of Class I Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas, and the American Petroleum Institute API RP 500, Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities. For Class II (dust) areas, the recommended practices are NFPA 499, Recommended Practice for the Classification of Combustible Dusts and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas and ISA-12.10-1988—Area Classification in Hazardous (Classified) Dust Locations.

Hazardous material classifications related to area classification are covered in NFPA 497M, "Classification of Gases, Vapors, and Dusts for Electrical Equipment in Hazardous (Classified) Locations, and NMAB 353-5, Classification of Gases, Liquids, and Volatile Solids Relative to Explosion-Proof Electrical Equipment" (National Technical Information Services).

6.4.2.1 CLASS I, DIVISION 1

There are four basic methods for installing equipment in Class I areas. They are explosion-proof enclosures, intrinsic safety systems, purging, and using approved equipment. All equipment must be approved for the specific area classification that they are placed in. The instrument technician must have a basic understanding of each of these means and how to interact with them successfully. Remember that there may be devices that can serve as ignition sources in the enclosure where you are troubleshooting under normal conditions, and also that your troubleshooting actions may serve as an ignition source.

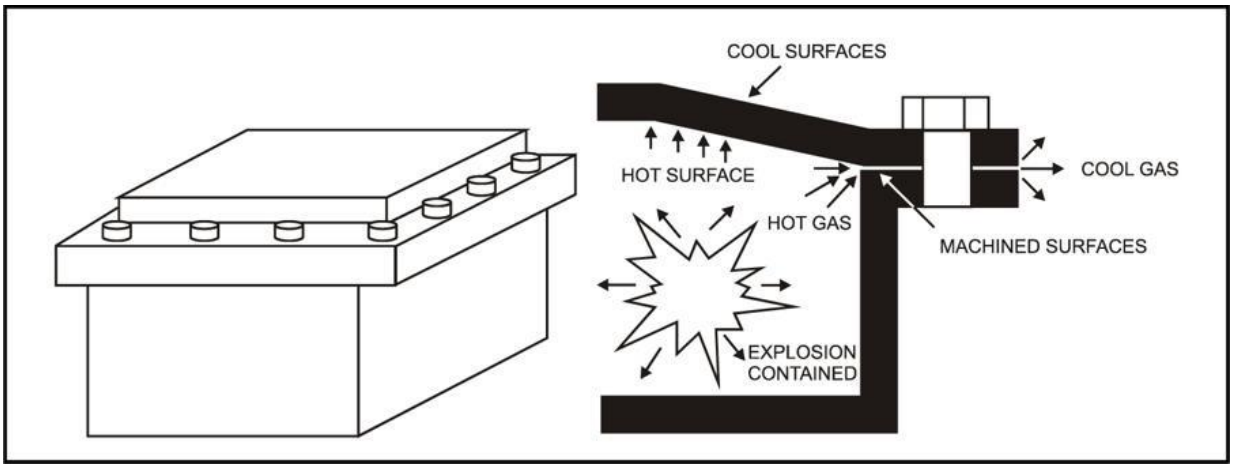
Explosion-proof enclosures (Figure 6-4) prevent external explosions by containing internal explosions and venting hot gases in such a manner as to cool them to a level at which they cannot ignite an outside flammable mixture. The enclosures are also designed so that surface temperatures under these conditions cannot serve as ignition sources. They do not prevent explosions from occurring inside the enclosure, only outside it. Explosion-proof enclosures must be approved for the specific area that they are used in, and should be approved by a nationally recognized testing laboratory such as Underwriters Laboratories (UL) or Factory Mutual (FM). When approved, the agency's label should appear on the enclosure, identifying the approved area and the agency that approved or listed it.

There are two basic types of instrument explosion-proof enclosures, flanged (which have two machined flanges clamped together with a multitude of bolts) and threaded (which vent through their threads).

The effectiveness of flange-type explosion-proof enclosures will be compromised if all bolts are not installed and torqued to the manufacturer's rating, or if the machined flange surface integrity is not maintained.

FIGURE 6-4

Explosion-Proof Enclosure (courtesy of Control Magazine)



In addition, if a bolt is replaced, it must be replaced with an equivalent bolt. Because of the large number of bolts around the flanged surface, it is common for the bolts to be left loose or for some of them to be removed. Both these practices invalidate the explosion-proof rating of the enclosure.

For a threaded enclosure or threaded connections, all the required threads (normally at least five) must be engaged. All conduits entering an explosion-proof enclosure must have a minimum of five threads engaged and be wrench-tight. Explosion-proof seals must also be installed where required.

In Europe, explosion-proof enclosures are called flame-proof enclosures. While built on similar design principles, they can be somewhat different due to the way field wiring is done in Europe, and they may be tested differently by approval agencies of the country of their origin. Do not interchange such enclosures unless the enclosure is approved for an American area classification, preferably by an American approval agency such as UL or FM, or by the authority having jurisdiction (AHJ).

Intrinsic safe systems are the second method of installing electrical equipment in Division 1 areas. These are systems that by design cannot (under both normal or abnormal conditions) release sufficient energy to ignite the most easily ignited mixture in the hazardous area. This means that if there is a short or open circuit, or a grounding of cables, wires, or components in the intrinsic safe circuit in the hazardous area, there will not be an arc or spark big enough, nor a hot surface temperature high enough, to serve as an ignition source. Intrinsic safe systems must be approved for the specific area in which they are used. They can be approved as a complete system by an approval agency, or by the user in the system on a component-by-component basis (entity concept). In this case, the individual components are approved and have stated intrinsic safe parameters that must be used in the design to get a complete intrinsic safe system.

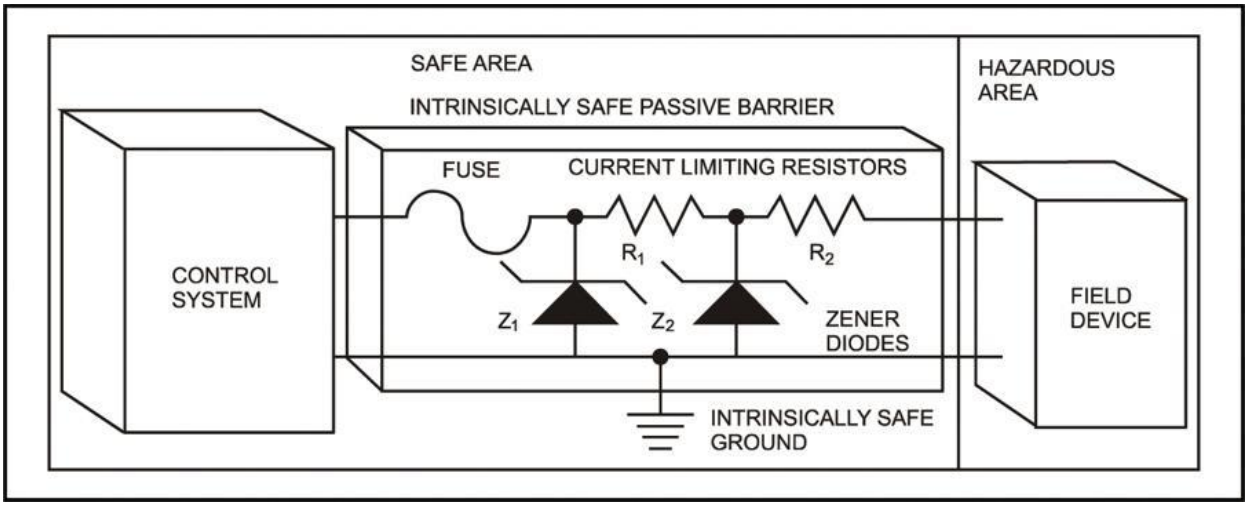
It is permissible for intrinsic safe wiring to be run in hazardous areas using nonhazardous wiring means. However, there are separation requirements between intrinsic safe systems and nonintrinsic safe systems, as well other requirements unique to intrinsic safe systems. Protection of the wiring must also be considered. Sealing is required for intrinsic safe wiring if flammable gases can be transmitted to Division 2 or nonhazardous areas via the intrinsic safe wiring.

The typical intrinsic safe system (Figure 6-5) consists of an intrinsic safe barrier in the nonhazardous area that prevents sufficient energy from reaching the hazardous area. The instruments and wiring in the

hazardous area are also designed so as not to be able to store sufficient energy to serve as an ignition source.

FIGURE 6-5

Barrier (courtesy of Control Magazine)



In addition, maintenance of intrinsic safe systems, including troubleshooting, must be done in a manner that ensures that the integrity of the system is maintained. This normally requires special training for the maintenance technicians, and additional administrative controls on maintenance activities involving intrinsic safe systems. Administrative controls may include requirements such as the following:

•Special work permits

•Engineering approval for modifications

•Post maintenance inspection

•Scheduled inspections

•Grounding verification

Engineering controls are also typically applied, such as:

•Design verification

•Documentation

•Installation controls such as labeling, color coding, and postconstruction inspection

By their very nature, intrinsic safe systems are limited in energy, so high-energy circuits and equipment cannot use this method. Examples of these are 120VAC circuits and motors.

Intrinsic safe apparatus and circuits are covered by the following standards: NEC Article 504, Intrinsic Safe Systems; ISA-RP12.06.01-2003 - Recommended Practice for Wiring Methods for Hazardous (Classified) Locations Instrumentation Part I: Intrinsic Safety; and ANSI/UL 913, Standard for Intrinsic Safe Apparatus and Associated Apparatus for Use in Class I, II, III, Division I Hazardous (Classified)

Locations. Two common breaches of the integrity of intrinsic safe systems are deterioration of the intrinsic safe ground and cross-contamination with non-intrinsic safe systems. In troubleshooting an intrinsic safe system, care must be taken so as not to compromise it by contaminating it with non-intrinsic safe connections.

Purging and pressurization is the third method for placing electrical equipment in a Division 1 area. Purging refers to the process of sweeping potential flammables out of an enclosure with a purge gas. Pressurization refers to maintaining a positive pressure on an enclosure with a purge gas to keep flammables out. The term Purging is commonly used in the industry to refer to both purging and pressurization, and will be used in this way in this discussion. Purging is covered by the following: NFPA 496, Purged and Pressurized Enclosures for Electrical Equipment, and ISA RP12.4-1996—Pressurized Enclosures.

The basic principle of purging is the reduction of an enclosure's internal area classification to one suitable for the equipment inside the enclosure. This is done by: (1) maintaining a positive pressure in the enclosure to prevent the entrance of flammable gases, vapors, and dusts; (2) by not allowing any surface temperature (either inside or outside the enclosure) to exceed 80% of the autoignition temperature (AIT) in degrees Celsius of the flammables involved (or a temperature determined to be safe by test); and (3) by assuring the integrity of the purge system. In addition, purging of the enclosure is provided at equipment start-up to sweep away any residual gases or vapors before power is restored. This is generally provided (though not required) during normal operating conditions.

Purges come in three types. Type X is a reduction in classification from Division 1 to Non-hazardous. Type Y is a reduction in classification from Division 1 to Division 2. Type Z is a reduction in classification from Division 2 to Non-hazardous. Note that only X and Y apply to Division 1 areas.

The exposed temperature of any device in the enclosure must be below 80% of the AIT because the power may be removed and the enclosure may be opened with a source of ignition still available due to the hot surface temperature before the hot surface cools.

General requirements that apply to all enclosure purges are:

•Sufficient mechanical strength

•Positive pressure (minimum 0.1 IN. WC)

•Identification labels and instructions

•Visual indication or alarm of failure to maintain positive pressure

•Purging of enclosure for hazardous gases before energizing of equipment.

Each type of purge has its own individual requirements. Purge requirements are also determined by the following:

•The volume to be purged

•Whether hazardous gases are brought into the enclosure

•The area classification where the enclosure will be located

•The area classification rating of the equipment in the enclosure. The purged systems must be designed for the specific area classification in question.

Higher voltage equipment, such as motors, may also be purged. These motors may be ventilated by positive pressure, or pressurized with inert gas. Homemade purging of motors should be done only with great care, if at all.

Purges have many times been considered install and forget systems, with only a local indication of purge. This is certainly not a good practice, as anything can and will deteriorate over a period of time.

Purge alarms are a good practice and are recommended. Many purge systems are homemade and are certainly acceptable if designed and maintained properly. Today, however, third-party purge assemblies are available on the market, and the use of these is recommended to help assure safe and consistent design of these safety systems.

When troubleshooting equipment with purges, it must first be determined if it is safe to open the enclosure. As the troubleshooting takes place, it must also be determined if it is safe to continue. This generally requires some form of hot work permit (discussed later in this chapter), and a sniffer—a portable device that can detect flammable mixtures that may occur in the area. Sometimes a firewatch is also required.

Approved equipment is the fourth method of installing equipment in a Division 1 area. Approved equipment is equipment approved by the AHJ. This may be some equipment that has been tested and found safe to operate in a Division 1 area and which has been approved for use by the AHJ.

6.4.2.2 CLASS II, DIVISION 1 AREAS

The means for installing electrical equipment in a Class II, Division 1 area are similar to those for a Class I area. In general, equipment must be approved for a Class II, Division 1 area. Dust ignition-proof equipment is used in a Class II area instead of explosion-proof equipment. This equipment is similar to explosion-proof equipment in that it is generally massive in construction, but it has a different function.

Where the explosion-proof equipment must contain an explosion and not serve as a source of ignition, dust ignition-proof equipment only must prevent any source of ignition such as an arc, spark, or hot temperature from being available to ignite an outside dust cloud or layer. The surface temperatures for equipment in dust areas are further limited to values below the dust layer ignition temperature to prevent carbonization of dust, which might allow ignition of the dust at a lower temperature. Many explosion-proof enclosures are dual-rated for Class I areas and Class II areas.

Purging is also used in Class II areas; however, the requirements are slightly different. The purge pressure may also be higher (0.5 IN. WC) if the material involved has a density greater than 130 lb/ft^3 (2082 kg/m^3).

Intrinsic safety can also be used in Class II areas.

6.4.2.3 CLASS I, DIVISION 2 AREAS

The basic requirement in these areas is that the equipment not provide an ignition source under normal conditions. There are five basic means by which electrical equipment can be located in a Division 2 area.

They are: (A) Division 1 means, (B) nonincendive systems, (C) no source of ignition rules, (D) purging, and (E) approved equipment.

Division 1 means approved for the same class and group can be used in Division 2 areas. Type X and Y purges are replaced with a Type Z purge, though a Type X can be used.

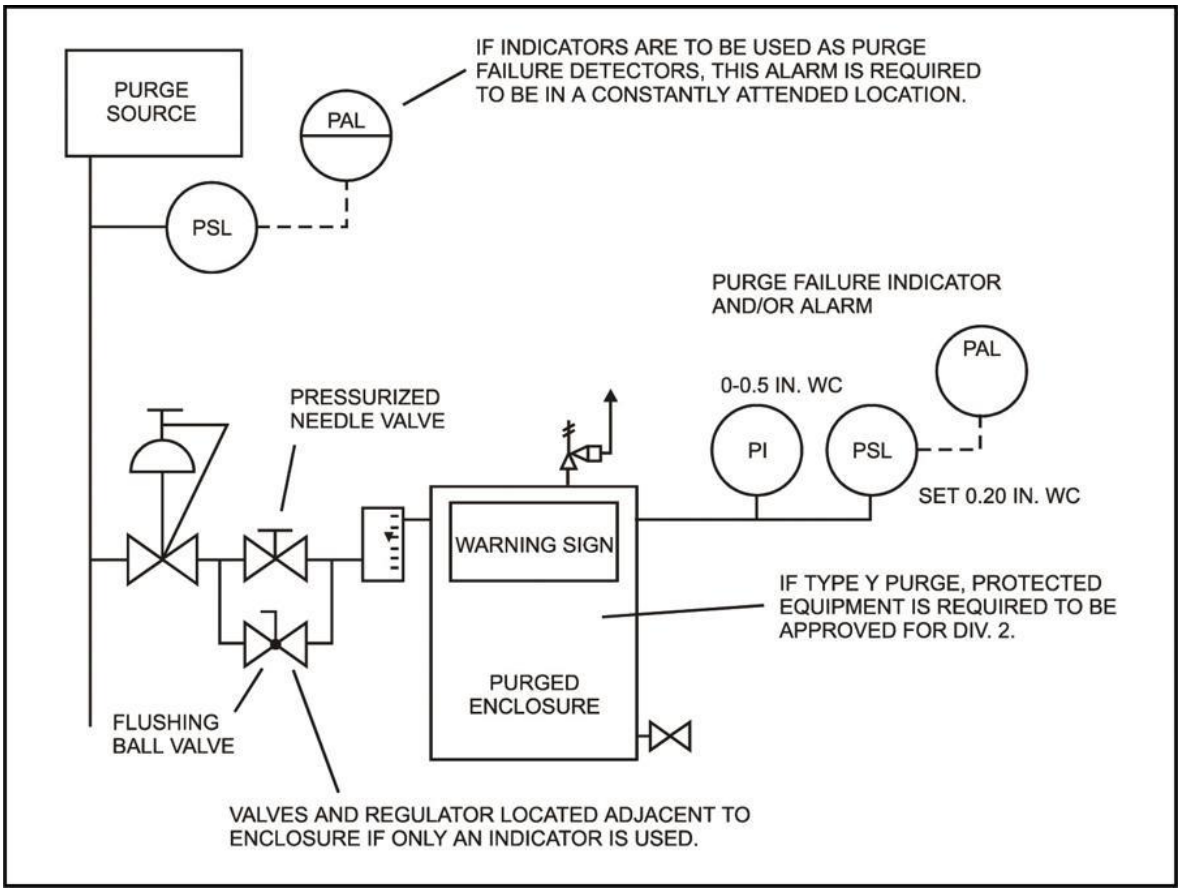
The Type Z purge can be used in a Division 2 area to reduce the area classification in the enclosure from Division 2 to Non-hazardous. Type Y and Z purges are illustrated in Figure 6-6.

Nonincendive equipment in a Division 2 area is similar in concept to intrinsic safe systems in a Division 1 area. Intrinsic safe systems require that the system in the hazardous area not provide sufficient energy to serve as a source of ignition under both normal and abnormal conditions. Nonincendive systems, on the other hand, require only that the system in the hazardous area not serve as a source of ignition under normal conditions. Both nonincendive and incendive wiring can be used.

Nonincendive-rated wiring must not serve as a source of ignition if it is opened, shorted, or grounded. Nonincendive wiring can be run with non-hazardous means. Nonincendive wiring may have sealing requirements if the wiring can transmit flammables to non-hazardous areas. Nonincendive equipment is covered in ANSI/ISA-12.12.01-2000 Nonincendive Electrical Equipment for Use in Class I and II, Division 2 and Class III, Divisions 1 and 2 Hazardous (Classified) Locations.

FIGURE 6-6

Type Y and Z Purges (courtesy of Control Magazine)



The "*no source of ignition rules*" in the NEC allow the placing of electrical equipment in a Division 2 area if there is no source of ignition available under normal conditions. This can be accomplished by hermetically sealing or oil-immersing current-interrupting contacts. Also, equipment with no current-interrupting contacts may qualify. Any exposed surface temperature must be considered—it may not exceed 80% of the AIT in degrees Celsius—or be tested to be safe. These rules, for example, allow instrument termination boxes and three-phase induction motors that under normal conditions lack an ignition source to be put in a Division 2 area. Motors do, however, have temperature limits. Many different types of instrumentation also fall under these rules.

Approved equipment is equipment that has been tested and is known to be safe for the area and approved by the AHJ.

Just as there are requirements for equipment in a Division 2 area, there are also requirements for such things as wiring means, raceways, sealing, and grounding. Refer to the NEC for these requirements.

6.4.2.4 CLASS II, DIVISION 2 AREAS

In Class II, Division 2 Areas, all the Division 1 means for Class II areas are acceptable. Nonincendive equipment is also allowed. In general, dust-tight equipment and enclosures without surface temperatures that exceed limits set in the NEC are allowed.

6.4.2.5 CLASS III AREAS

For Class III areas, see NEC Article 503.

6.4.2.6 ADJACENT AREAS

While an area may be classified as non-hazardous within an industrial facility, the area may still require more than the ordinary means allowed for non-hazardous areas. It should be remembered that a Division 2 area is normally adjacent to the non-hazardous area. The means used for this area can include the use of Division 2 means for installing electrical equipment and wiring, as well as other industrial-strength installation practices.

The National Fire Codes, and in particular, NFPA 70, the NEC, are the bible for installing and modifying electrical equipment and wiring in hazardous areas. Do not leave home without them.

6.4.2.7 ZONE AREAS

Zone means of placing electrical equipment in hazardous areas are similar to division means with some new methods recognized and covered in NEC Article 505. The biggest distinction is that the Division 1 area is divided into two zone areas, 0 and 1, which require different means. Several new techniques are allowed in zone areas such as increased safety, powder filling, restricted breathing, and encapsulation.

6.4.3 Troubleshooting in Electrically Hazardous Areas

This section will discuss the principles for troubleshooting in electrically hazardous areas.

6.4.3.1 GENERAL REQUIREMENTS

Troubleshooting activities in hazardous areas must meet the same requirements as equipment. In other words, troubleshooting activities cannot provide a source of ignition when a flammable mixture is present and must also meet any special requirements for a specific area. For example, in a dust or fiber area, dust and fiber accumulations occurring during troubleshooting activities must be cleaned up before re-energizing the equipment. Another example would be conductive dust areas, where the conductive dust must not accumulate on exposed electrical contacts or wiring.

Work in a hazardous area normally involves the use of a hot work permit, which will be discussed below. Such a permit is required where the work in the hazardous area could be a source of ignition. A hot work permit is issued once the area has been determined by Operations to be safe to work in. This determination is made based on the nature of the work, the area classification, and the presence of flammables. The area where the work is to be done is generally sniffed with a portable combustible gas detector. The gas detector (commonly called a sniffer) is left in the area where the maintenance is being done, or a firewatch stays in the area with the detector. Fire extinguishers or fire monitors may also be manned, as deemed necessary. A hot work permit is really a temporary reduction of area classification based on inspection to allow maintenance activities.

Care must be taken when sniffing an area to ensure that no flammables are lurking around. Placing the sniffer in only one place during the maintenance activity may not detect all potential flammable gas sources. Having a sniffer present during maintenance is always a good practice. Also, regular calibration of the gas sniffers must to be done to ensure that they are working properly.

One aspect that is sometimes overlooked when troubleshooting hazardous areas is the use of test equipment. Test equipment must be rated for the area that it is used in, as some items may have sufficient energy available to serve as an ignition source. However, if the area has been temporarily declassified, the equipment only has to be rated for the declassified area. Tools can also be of concern, because a spark from the tool or an electrical fault caused by the tool can serve as a source of ignition. Where possible, in areas that are temporarily declassified for maintenance, sources of ignition should not be created.

This kind of troubleshooting may also involve working on equipment that can contain or control flammables. Care must be taken to ensure that flammables are not inadvertently released.

6.4.3.2 PREPARING TO WORK IN A HAZARDOUS AREA

Any special procedures should be reviewed prior to performing work in the hazardous area. Other maintenance activities and operations currently occurring in the maintenance area should also be reviewed. All tools and test equipment should be verified as being appropriate for the hazardous area in which the troubleshooting will be done.

Troubleshooting in hazardous areas requires additional safeguards to insure the safety of the maintenance personnel and the facility. No troubleshooting should be performed until the area has been determined safe for the maintenance activity.

6.4.3.3 COMPLETING WORK IN A HAZARDOUS AREA

When the troubleshooting is completed in a hazardous area, the equipment and wiring must be restored to meet the requirements of the area classification. This includes both the existing system and any modifications made to it. All items such as explosion-proof enclosures, purges, and intrinsic safe systems must be fully functional before the equipment is returned to service. When modifications have been made, a qualified inspector should inspect all of the work.

RELEVANT STANDARDS

•NEC Article 500, Hazardous (Classified) Locations, defines division-based area classification. Article 505, Class I, Zone 0,1,2 Locations defines the zone-based area classification. Articles 501-555 further explain the requirements for the use of electrical equipment in hazardous (classified) areas.

•NFPA 497A, Classification of Class I Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas.

•American Petroleum Institute API RP 500, Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities.

•ISA-SP12, Electrical Equipment for Hazardous Locations Standards.

•NFPA 496—Purged and Pressurized Enclosures for Electrical Equipment.

•ANSI/UL 913—Standard for Intrinsic Safe Apparatus and Associated Apparatus for Use in Class I, II, III, Division I Hazardous (Classified) Locations.

6.5 PROTECTION, PROCEDURES, AND PERMIT SYSTEMS

Protecting yourself from troubleshooting hazards requires a number of different mechanisms. While the company that you work for should have a primary interest in your safety, you must realize that, first and foremost, you are the person in charge of your own safety.

Your company should provide various personal protective equipment, administrative procedures, and practices to ensure your safety, **but it is up to you to follow them and to do whatever else is necessary to ensure your safety.**

6.5.1 Operations Notification

Operations notification begins with the permit system, as described in section 6.5-3 below. At the beginning of the job (in this case, troubleshooting), a permit is filled out and signed by Operations. This represents the initial notification to Operations that work will be done on a particular piece of equipment or system. However, work may not start at that time and a start of work notification to Operations may be necessary. As the work proceeds, there may be times during troubleshooting that Operations needs to be to be updated or informed of additional activities. Be sure that Operations is informed at all times of your activities. If they are not aware of what you are doing, they may respond to your troubleshooting in ways that can cause a process upset or hazard to the plant. They may also take actions that adversely affect your troubleshooting.

6.5.1.1 PERSONAL PROTECTIVE EQUIPMENT (PPE)

Personal protective equipment is equipment that you wear or use to protect yourself during troubleshooting activities.

For the petrochemical industry, the basic equipment is generally fire retardant clothing (FRC), safety glasses, hard hat, and hearing protection (where required). To this may be added such items as side shields for safety glasses, safety shoes, respirators, mono-goggles, face shields, gloves, and slicker suits.

Electrical PPE could include items such as voltage-rated gloves, flash suits, insulating mats, and insulating guards. Most PPE items have inspection dates associated with them. Inspect any PPE before use. Ensure that it appears to be functional and in good shape, and that any required inspections are up-to-date.

Each job has its required PPE to help make it safe. Your company should have determined the minimum PPE for each job—your evaluation or your supervisor's may indicate that more is needed. Do not take any chances with your PPE.

6.5.2 Maintenance Procedures

This section discusses maintenance procedures to ensure safe troubleshooting.

6.5.2.1 LOCK-OUT/TAG-OUT (LOTO) PROCEDURE

While troubleshooting often requires work on energized circuits or equipment that has stored energy, there are times when the work is on de-energized circuits. In these cases the equipment must be locked out and tagged out. Each facility is required by OSHA to have a LOTO procedure. NFPA 70E provides

guidelines for the electrical LOTO procedure. The LOTO procedure should also apply to non-electrical power sources. Follow the LOTO procedure like your life depended on it—because it does.

6.5.2.2 DO NOT OPERATE TAGS

These tags are normally placed on equipment that the Operations Department does not want anyone to use, either because it is unsafe, or for operational reasons. The Maintenance Department may also use these type of tags. If you are asked to troubleshoot equipment with a Do Not Operate tag, contact the person who tagged the equipment and verify that it is safe to work on it.

6.5.2.3 READY-TO-WORK TAGS/PERMITS

Many facilities have a ready-to-work tag or permit that is placed on equipment after Operations has determined that it is safe to work on the equipment.

6.5.2.4 SCAFFOLD STATUS AND INSPECTION TAGS

Scaffold tags are required to indicate the status of scaffolding, such as complete, incomplete, and under construction. They are placed on the scaffolding in plain sight. Inspection status and date of inspection should be on the tag. Different colored tags are commonly used to indicate status. The user should inspect the scaffolding for tagging on the day of use and at the time of use. Do not get on any scaffold that is shaky or has any instability or is missing safety railing or floor boards. Report any deficient scaffolding to Operations immediately so that it can be tagged with a Do Not Operate tag and reported to the appropriate people to get it repaired.

6.5.3 Work Permits

Work permits are part of the administrative procedures for a facility that help ensure the safety of the facility and its personnel. Each facility will have its own permitting procedure. Troubleshooting should require a permit and be part of the permitting system. Even though some troubleshooting does not require opening enclosures, taking measurements, or doing tests, it should still fall under the permitting system. Many times maintenance people and engineering personnel short-circuit the permit system when troubleshooting because it is too inconvenient. This is a serious mistake and can lead to safety or operational consequences.

All permits should be good for a specific time. They also have safety requirements, usually in the form of check-off boxes. Make sure that your permit is properly filled out, is current, and that all safety requirements are met.

Permits should not go across shifts. When in doubt get a permit. The following is a brief discussion of the permitting system.

6.5.3.1 SAFE OR COLD WORK PERMIT

Normally a permit will be required for any work at a given facility. The primary purpose of this permit is to inform responsible parties that work is being done in their area, and to ensure that any safety requirements are in place. This permit may be called a work permit, a safe work permit, or a cold work permit. In the case of troubleshooting activities, which Operations should know about in advance, this

permit should be required. This permit is used in non-hazardous areas or in hazardous areas where no source of ignition or other hazard related to the hazardous area is involved in the work. Some companies have an Intermediate permit for work in a hazardous area that does not involve an ignition source.

6.5.3.2 HOT WORK PERMIT

A hot work permit usually applies to work in a hazardous area or an area in which hazardous materials are present that the intended work could ignite or create other hazards. Remember that troubleshooting activities such as placing test equipment probes on terminals or lifting wires can provide a source of ignition. It is generally a good practice to ground equipment before attaching any other leads. The use of a tool can also inadvertently cause a spark if it causes a short to ground or to another wire.

6.5.3.3 CONFINED ENTRY PERMIT

The confined entry permit is required when the work will be in a confined area in which there is the possibility of an oxygen-deficient (insufficient oxygen for breathing) or toxic atmosphere. This type of permit may also be issued where entry is required into small or limited spaces where mechanical hazards exist.

These permits are needed for work inside such places as vessels, vessel skirts, manholes, tanks, and process and rainwater sewers. This hazard can also exist where an instrument air system is backed up by a nitrogen system, and there has been a switch to nitrogen. This can be a hazard anywhere pneumatic instruments or instrument purges are used in confined spaces.

A common place for this to occur is in analyzer houses. Some companies, for example, require a confined entry permit when an analyzer house's ventilation is down. Pneumatic instruments or purges that use nitrogen under normal conditions can be a hazard in a confined space.

When troubleshooting, do not venture into enclosed or confined spaces just because everything is running and assumed to be safe. A potentially hazardous confined space typically has one or more of the following characteristics:

•Contains or has the potential to contain a hazardous atmosphere

•Contains a material that has the potential for engulfing personnel in the confined space

•Has an internal configuration such that a person in the space could be trapped or asphyxiated by inwardly converging walls, downward slopes, smaller cross sections, the number of egress points, the location of egress points, low spots, below-grade areas, pockets, and so on

•Contains any other recognized safety or health hazard

If you must enter a confined space, make sure, as a minimum precaution, that you have a current confined space permit; a firewatch and a sniffer (for oxygen and toxic level if appropriate) should also be present; and you should wear appropriate PPE (such as a retrieval harness, a Scott Air Pack, and an air mask). Respirators do not provide air and are not appropriate for use in confined spaces. Always follow safety precautions; the life you save may be your own.

6.5.3.4 EXCAVATION PERMIT

An excavation permit is required when any digging is undertaken in a facility. Occasionally, you may need to dig up something, or you may need to go into a trench or some other area that has been excavated. As minimum precautions, make sure the excavation permit is present and current, make sure a firewatch or other person is present, and make sure the shoring is adequate.

Be careful—many people have died in trench and hole cave-ins.

6.5.4 Loop Identification and System Interaction

In modern large, complex facilities with many measurement and control loops, identifying the loop or equipment to troubleshoot is obviously important. This process begins with the loop drawing or motor schematic, and the physical identification on loop components begins with such identifiers as wire tags, instrument tags, and nameplates.

One thing that further complicates matters is that these loops are sometimes connected to a DCS or a computer where some of the connections may occur in software. Documentation of DCS is notoriously poor. Troubleshooting a cross-connected software loop, without being aware of the software connection, can cause process upsets or operator actions and may provide incorrect troubleshooting information.

It is extremely important that the loop documentation be kept up-to-date in your facility. This is the heart of troubleshooting. The same is true of any other documentation, such as wiring drawings, electrical drawings, and vendor documentation. Also, remember that field tagging, such as wire tags and instrument identification, is an important part of your documentation. If you do not know what you are troubleshooting, how can you succeed?

Complex facilities can lead to complex interactions between instrument loops. These interactions can be physical or software-related, and they may involve process interconnection—change this loop and these other loops change. A basic understanding of the process and the instrument loop interactions is necessary for successful troubleshooting. This can also help prevent you from creating hazardous conditions from such interactions. In addition, knowing about interactions can help you troubleshoot—failure symptoms sometimes result from interactions. Lack of knowledge may lead you down the wrong troubleshooting path.

6.5.5 Safety Instrumented Systems

Normally, plants have instrument systems that have safety functions. These systems are typically such things as interlocks, emergency shutdown systems, ESDs, safety systems, and safety interlock systems. The proper name for these systems, per ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod), is safety instrumented systems (SIS).

In the U.S., maintenance activities on SIS are primarily controlled by the ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod) Functional Safety: Safety Instrumented Systems for the Process Industry Sector standard. This standard is essentially the same as the IEC standard 61511, which is used outside

of the U.S. The ANSI/ISA-84.00.01-2004 standard has specific requirements for maintenance of SIS (Section 11.8 but also distributed throughout the standard). Some of the key requirements include the following:

1.Persons who work on SIS shall be trained on the SIS that they will work on.

2.SIS shall have maintenance procedures that must be followed.

3.SIS shall be periodically tested per the Safety Requirements Specification (SRS) and per specific written test procedure for each safety function.

4.Modifications to the SIS are not allowed without a Management of Change (MOC).

5.All maintenance actions shall maintain the required safety integrity of the SIS as established in the SRS.

6.When a SIS is placed back in service, it shall be in good as new condition.

7.All SIS maintenance shall be done within the Mean Time to Repair specified in the SRS.

8.Maintenance on SIS shall be documented with specific information on maintenance actions and SIS equipment failures.

9.Records of SIS maintenance shall be kept for auditing purposes for conformance to OSHA PSM and ANSI/ISA-84.00.01 (IEC 61511 Mod).

Care must be taken that troubleshooting activities on safety instrumented systems such as bypassing, forcing, and testing do not compromise the safety integrity of the safety system. If troubleshooting activities must be done, administrative controls must be in place to maintain the system safety integrity.

One of the potential hazards related to these systems is spurious or nuisance trips, that is, when the automatic safety system is tripped for no good reason. Spurious trips can create hazards, and the resulting start-up can also create hazards. Many times, however, troubleshooting may occur near a SIS, and you have to be extra careful not to cause it to trip spuriously. For example, a young instrument technician went to work on a thermocouple, but instead of working on the process measurement thermocouple, he got on the shutdown thermocouple, which was right next to it. Down went the plant, much to his dismay. He never did it again, as he always made sure that he was working on the right loop.

Troubleshooting on or near a SIS raises the risk that you could affect safety on a large scale. But even if it is not actually dangerous, a spurious trip of a SIS will not do much for your reputation.

6.5.6 Critical Instruments

Critical instruments are instruments that have been identified as critical to some aspect of the process. This aspect can be safety, environment protection, asset protection, operational, or process related. Critical instruments typically have procedures associated with them.

Critical instruments that are identified as safety or environment protection may also be identified in their associated risk assessment (commonly called layer of protection analysis [LOPA]) as independent layers of protection (IPLs) and will have specific requirements that can have maintenance considerations.

These IPLs are required to have associated procedures and will have specific proof test procedures. IPLs cannot be changed or modified without MOC. The removal of an IPL from service for troubleshooting is commonly limited to a defined mean time to repair (MTTR). IPLs are typically part of the facility mechanical integrity program.

RELEVANT STANDARDS

•ANSI/ISA-84.00.01-2004, Parts 1-3 (IEC 61511-1 to 3 Mod) Functional Safety: Safety Instrumented Systems for the Process Industry Sector.

•IEC 61511, Functional Safety: Safety Instrumented Systems for the Process Industry Sector.

•IEC 61508, Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems.

•ANSI/ISA-12.01.01-1999—"Definitions and Information Pertaining

to Electrical Apparatus in Hazardous (Classified) Locations."

•ISA-5.1-1984 (R1992)—Instrumentation, Symbols, and Identification.

SUMMARY

Safety is a primary concern during troubleshooting. Know the hazards you are dealing with and their locations. Make sure that you are on the right equipment. Follow your facility's safety procedures, but remember that these represent minimum standards. If you need more, ask for more. Use your facility's permit system. Your company should be concerned about your safety, facilitate it, and provide a safe working environment; but in the end, it is your responsibility to ensure your own safety. Make your decisions wisely.

QUIZ

1. Who is ultimately responsible for your safety?

A.your company

B.your fellow workers

C.you

D.your plant safety engineer

2. If you are going to troubleshoot de-energized equipment, you should

A.stop or turn off the equipment.

B.lock out and tag the equipment (LOTO).

C.check for voltage before working on equipment.

D.both B and C

3. What kind of permit should you get if the work you will be doing is in a hazardous area and the work could possibly serve as a source of ignition?

A.hazardous area permit

B.hot work permit

C.cold work permit

D.excavation permit

4. The damage an electrical shock can cause to a human being is determined by the

A.current path.

B.amount of current.

C.duration of current.

D.all of above

5. The vaporization of copper has the power of

A.dynamite.

B.a large firecracker.

C.a hand grenade.

D.none of the above

6. A Class I electrical area classification designator indicates a

A.dust area.

B.hydrogen area.

C.gas and vapor area.

D.lint and flyings area.

7. A Division 2 electrical area classification designator indicates

A.a dust area.

B.where the flammable hazard is present under abnormal conditions.

C.where the flammable hazard is present under normal conditions.

D.none of the above

8. Human errors occur due to

A.slips.

B.mindset.

C.lack of information.

D.all of the above

9. Purging can be used to place electrical equipment in

A.Division 1 areas only.

B.Division 2 areas only.

C.both Division 1 and 2 areas.

D.none of the above

10. A zone is the same type of area classification designator as a

A.class.

B.group.

C.division.

D.none of the above

11. All humans make errors. True/False (T/F) _____

12. Safety systems are commonly known as

A.safety instrumented systems.

B.safety interlock systems.

C.ESD systems.

D.all of the above

13. NEC stands for

A.National Electrical Council.

B.National Educational Council.

C.National Electrical Code.

D.Neutral Electrical Conductor.

14. The temperature where damage could occur due to contact with a thermal hazard is approximately

A.115°F (46°C).

B.212°F (100°C).

C.115°C (96°F).

D.32°F (0°C).

15. Which of the following could serve as an ignition source?

A.lifting an energized wire

B.hot surface

C.relay contact operation

D.all of the above

16. Which of the following can be used to place electrical equipment in a Division 1 area?

A.purging

B.explosion-proof

C.A and B

D.nonincendive

17. Going into a space that may not have enough oxygen to breathe requires

A.a hot work permit.

B.a ready to work permit.

C.an excavation permit.

D.a confined space permit.

18. The approximate level that a human will sense electrical current is

A.1mA.

B.1-3mA.

C.10-12mA.

D.80mA.

19. When operating a electrical switch or circuit breaker, the safe procedure is to

A.only use one hand.

B.only use the right hand.

C.do not stand in front of the switch.

D.only use the left hand.

20. To have a fire you need

A.oxygen.

B.flammables.

C.an ignition source.

D.all of the above

21.What is it that you should always double check before starting troubleshooting?

22.Housekeeping is not important to safety. True/False (T/F) _____

23.Planning is important to safety. True/False (T/F) _____

24.The acronym PPE stands for what?

25.Why is troubleshooting different from regular maintenance?

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7

TOOLS AND TEST EQUIPMENT

Hand tools

Contact-type test equipment

Noncontact-type test equipment

Simulators/process calibrators

Jumpers, switch boxes, and traps

7.1 HAND TOOLS

While we can use our senses when we troubleshoot, we must also use tools and test equipment to determine information not directly available to the senses. A tool is any aid, equipment, or device used in troubleshooting. Sometimes it may be tempting to use channel locks, instead of the proper wrench that is back in the shop, or the wrong size or type of screwdriver. Resist the temptation. Use the proper tool. Using the wrong tool can create immediate problems, and it can make the life of the next person that has to troubleshoot the system (which might be you) more difficult. (The companies and model numbers of test equipment mentioned in this chapter do not represent recommendations, but rather are representative of a particular class of test equipment.)

Because troubleshooting often means working on energized circuits, it is necessary to use hand tools rated for the proper voltage. This is not a big issue when working on low voltage (<50VAC/VDC) circuits, but it becomes an issue when working on higher voltages, such as 120-, 208-, 240-, 277-, or 480VAC, or greater. Hand tools wrapped in insulating tape, though common, should never be used for voltages above 120VAC. Even at lesser voltages, a voltage-rated tool is always preferable. For voltages 120VAC and above, use tools rated to American Society for Testing and Materials (ASTM) F1505 Standard, Standard Specification for Insulated and Insulating Hand Tools.

7.2 CONTACT-TYPE TEST EQUIPMENT

In order to collect information to help in troubleshooting, you will often employ contact-type test equipment—equipment that works by making physical contact with the devices, circuits, or materials being tested. The first rule for using this equipment is simple: always know what you are connecting to or making contact with, otherwise you may get a surprise that can cost you your life.

From a safety perspective, all test equipment must be suitable (rated) for the intended service. International Electrotechnical Commission (IEC) Standard 61010, Safety requirements for electrical equipment for measurement, control, and laboratory use, provides requirements for both continuous and transient ratings for test equipment. Underwriters Laboratories bases UL 3111, Electrical Measuring and Test Equipment, on IEC-61010, and ANSI has also adopted this standard. IEC-61010 describes two continuous-level voltages (600V and 1000V) and four levels for voltage transients (Categories I, II, III, and IV), with Category I representing the lowest and Category IV representing the highest. In practice, Category I describes transients at the electronic equipment level, Category II at the receptacle level, Category III at the power distribution level, and Category IV at the utility connections level. Make sure your test equipment is rated for service in the proper transient category and voltage level, and that your test leads are properly rated as well.

If the test equipment is to be used in hazardous (classified) areas without a hot work permit (local declassification for maintenance purposes), the equipment must be rated for the hazardous area in which it is to be used. Usually this means that equipment is rated as intrinsically safe for that particular area. Not all test equipment can be rated intrinsically safe, as the voltage and current levels involved may preclude this.

RELEVANT STANDARDS

•IEC-61010 - Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use.

•UL 3111- Electrical Measuring and Test Equipment.

7.2.1 Volt-Ohm Meters (VOM)

The venerable volt-ohm meter (VOM) was once a mainstay of troubleshooting. While still in use, it has been generally replaced by more modern instruments with greater functionality. The VOM metering element is electromechanical. The Simpson 260 (see Figure 7-1) and the Triplett 630 volt-ohm meters are examples of this type of instrument.

7.2.2 Digital Multimeters

Digital multimeters (DMMs) have generally replaced the VOM, adding a digital readout and additional functions such as a large display, autoranging, frequency, voltage and current peak hold, memory, and computer interface. Modern DMMs now come in a wide variety of shapes, colors, forms, and functionalities to meet most any need.

Examples of these are the Fluke Series 80 (Figure 7-2), Tektronix TX-DMM, and the Entech 285 digital multimeters. Other manufacturers of DMMs are Ideal, Amprobe, Aemc Instruments, Simpson, and Triplett. Another key feature that many DMMs have is the ability to measure a true root mean square (RMS) value rather than a calculated RMS value. Most of the older meters and some new meters measure the average voltage or current, then calculate or scale the RMS value based on a simple sine wave. If the wave is more complex than a simple sine wave, this type of meter will read incorrectly.

The true RMS feature is important because often the loads of new equipment on which we need to perform troubleshooting are nonlinear, or the equipment operates in power systems that have nonlinear loads, which leads to the possibility of harmonics in the power system. Again be careful, though: not all DMMs can read true RMS values.

FIGURE 7-1

Simpson 260 Volt-ohm Meter (VOM) (courtesy of Simpson Electric)



FIGURE 7-2

Fluke Digital Multimeter (courtesy of Fluke Corporation)



In selecting the proper DMM, consider the size of the display, the number of digits of the display, and the size, shape, weight, environmental protection, and ruggedness required. Ruggedness or durability can be important because these meters typically take a real beating.

Add-on features (external boxes that connect to the DMM) that extend the basic tool are also available. Some of these add-on features are clamp-on amp meters, temperature probes, harmonic analyzers, and features that measure insulation resistance, relative humidity, airflow and light intensity.

One variety of DMM introduced during the late 1990s is a combination of a standard DMM and a process calibrator. Examples are the Fluke 787 and the Entech CMM-15. In addition to usual DMM functions, these DMMs allow you to measure and simulate the standard 4-20mA and other process signals.

7.2.3 Oscilloscopes

Oscilloscopes, commonly known as scopes, give a visual representation on a screen of the voltage or current waveform being monitored. The range of scopes includes bench models, portables, and newer handheld models. Scopes are used to look at waveforms for shape, timing, frequency, amplitude, distortions, and noise. They can also look at DC to detect noise. As a rule of thumb, the scope's bandwidth (frequency range) should be a minimum of twice the frequency of the signals that will be measured. Most scopes come with more than one input channel (trace), and some have at least two

channels. Storage models, which allow waveforms or transient events to be captured for future study, are also available.

Common brands of bench and portable scopes include Tektronix, Hewlett-Packard, and Hitachi. Examples of handheld scopes include the Fluke ScopeMeter (see Figure 7-3), the Tektronix THS500 family, and the Entech Multiscope. While the handheld scopes are convenient, the quality of displays on CRT-based scopes is generally better than the backlit displays of late-1990s-era handhelds. Many of the handheld scopes also include DMM functionality. Certain scopes, such as the Fluke 43 Power Quality Analyzer and the Tektronix THS720P, are also designed to look at power quality.

When selecting a scope, consider where it will be used (indoors, outdoors, shop, field, hazardous area), ruggedness, ease of use, weight, number of channels, bandwidth, whether it is analog or digital, and whether or not it includes a storage function.

Take care that the scope and its probes are rated for the voltage levels on which they are to be used. Many scope probes are also attenuators (e.g., 10X, 100X), which must be taken into account when using the scope for measurements. Take care to follow the manufacturer's recommendations about how connections should be made. When making a ground connection, always do so first and disconnect from ground last.

FIGURE 7-3

Fluke Scopemeter (courtesy of Fluke Corporation)



7.2.4 Voltage Probes

Voltage probes are contact probes designed to detect the presence of voltage at a terminal or connection, indicating what is energized and what is not. They are primarily a go/no-go test, though some will indicate the approximate voltage level. Some of these are solid-state devices, with lights or LEDs for indicators, while others have electromechanical indicators. One common electromechanical voltage probe is the solenoid type, called a wiggy because of the way it indicates the presence or absence of voltage by wigging up and down. An example of the solenoid type of voltage probe is the Ideal Volt-Con. Examples of solidstate probes include the Fluke T2 and T5.

It is always good practice to test probes on a known energized circuit before testing the circuit in which you are interested. Some electromechanical probes will only operate above a certain minimum voltage, such as 80–90V, and they may not be able to distinguish between close voltages. They may also be more subject to damage when dropped than are solid-state probes.

7.2.5Thermometers

Many instruments use thermocouples or resistance temperature detectors (RTDs) that require special test equipment to measure. This test equipment may be an individual instrument or an add-on to a DMM. The temperature-measuring elements come in many different varieties, so matching up the test equipment to the element is important.

Thermocouples are classified by letter designations that indicate the kinds of metals used in their wires; common types are J, K, E, N, R, S, T, and B. Each type produces a characteristic millivoltage-to-temperature response. A thermocouple is also a dual-ended device, with a hot junction (the measurement end) and a cold junction (the test equipment end). Any device that measures a thermocouple directly must provide cold-junction compensation to provide the correct reading. This compensates for the ambient temperature at the measuring device end. When reading a thermocouple with a millivolt meter, manual cold-junction compensation must be provided using the millivolt-versus-temperature tables for that thermocouple.

The United States has standard wire color codes for most thermocouples; the red wire is always the negative lead. When a thermocouple's composition is stated, it is always stated with the positive lead first. For example, a Type J (iron/constantan) thermocouple uses an iron positive lead and a constantan negative lead. International thermocouple wire codes are different. The standard IEC wire color code typically uses white for negative. For example, in the United States, Type J is white/red, whereas in the IEC standard Type J is black/white. Many countries have their own codes, which further complicates things. For example, in Japan a Type J is red (positive) and white (negative), which is just the opposite of the United States. This may create an identification problem on equipment brought from European or other overseas suppliers.

Resistance temperature detectors also come in several types, the most common of which is the 100ΩPt (platinum). Other RTD varieties are generally for specialized applications. Examples include 200, 500, and 1000ΩPt (platinum); 10 and 20ΩCu (copper); and 100 and 120ΩNi (nickel). Copper RTDs are often seen in motor windings. The ohm (Ω)value of the RTD is its resistance at a particular temperature. For example, the 100ΩPt temperature is 0°C (32°F), or a resistance of 100 ohms at 0°C (32°F). As a further complication, RTDs can have different temperature curves within each type. These curves are identified by a temperature coefficient (a), which is the average resistance change per unit of temperature from

the ice point to the boiling point of water. The IEC/ASTM U.S. standard for 100ΩPt is α = 0.00385 Ω/Ω/0°C (32°F), but older standards are still in use. An example is the old U.S. standard of α = 0.003902 Ω/Ω/0°C (32°F).

The checking of temperature elements at a given temperature is commonly accomplished using temperature baths. Different types of baths, such as dry blocks, sand baths, liquid baths, and fluidized baths, allow you to insert the temperature element into a heated area. Suppliers of such baths include Hart Scientific, Techne, and Jofra.

7.2.6 Insulation Testers

Megohmmeters (see Figure 7-4) measure insulation resistance. Commonly, though incorrectly, known in the industry as meggers (Megger is actually a trademarked name of a megohmmeter manufactured by AVO International), these instruments are used to detect insulation problems in wiring and windings. The megohmmeter comes in line-powered, battery-powered, and hand-cranked versions, or in combinations that use voltages from 50–5000V (and even higher for highvoltage cables). For cables with service voltages under 500V, 500VDC is commonly used. When testing, take care to find out what might be connected; excessive voltage may damage sensitive equipment.

Megohmmeter readings are usually taken over a period of 1 minute. By keeping historical trend readings, deterioration of insulation systems can be detected. The readings are corrected to a standard base temperature according to tables provided by the test instrument manufacturer. The minimum acceptable reading is generally 1MΩ plus 1MΩ per Kilovolt (service voltage). For values below 1kV, the minimum is usually 2MΩ. Readings below this will most likely indicate damaged insulation. Readings between 2MΩ and 50MΩ are usually associated with long circuit lengths, moisture, and contamination; they may not necessarily indicate any permanent damage. Insulation testing is highly sensitive to temperature and moisture. The break point, at which cable or wiring installation need to be investigated further, is usually a reading of 50MΩ.

Readings are also compared based on time. An example would be the ratio of the 60-second reading to the 30-second reading. The ratio of 10 minutes to 1 minute is known as the polarization index. Voltage steps are also used, and decreasing resistance with increasing voltage is a sign of insulation weakness. Good insulation should show a continual increase in resistance with time when under power. The types of tests and the rules regarding acceptable resistance or ratio may vary, and your site or the equipment manufacturer may have somewhat different acceptable values. Insulation testing for motors is covered in ANSI/IEEE Standard 43 - IEEE Recommended Practice for Testing Insulation of Rotating Machinery. Take care to ensure that any capacitive charged equipment under test (EUT) is discharged before disconnecting the megohmmeter. Even cables can hold a charge.

FIGURE 7-4

Megohmmeter (courtesy of AVO International, Dallas, TX)



The hi-pot test measures the dielectric strength of insulation. It can uncover insulation weakness that the megohmmeter might not catch. This is also usually a go/no-go test. These testers usually apply an AC or DC voltage in the range of 500–1000V for low-voltage cables for a period of time of about 5 minutes. These tests will detect gross imperfections due to improper field handling and weaknesses that are likely to fault to ground when subject to high-voltage transients. DC voltage testing for large motors is covered in IEEE Standard 95 - IEEE Recommended Practice for Insulation Testing of Large AC Rotating Machinery With High DC Voltage.

7.2.7 Ground Testers

Ground resistance testers (see Figure 7-5) come in several varieties that typically insert probes in the ground and measure voltage drops and currents at varying distances between the ground rod (or rods) and the test probes; the ground resistance is calculated or determined from the results. The National Electrical Code requires that the resistance to ground be less than 25 ohms for a single made electrode; however, instrumentation specifications are commonly in the range of 1Ω – 5Ω.

FIGURE 7-5 ***Biddle Ground Tester (courtesy of Transmation Inc.)***



7.2.8 Contact Tachometers

Contact tachometers measure the speed of moving equipment. This requires contact between the moving equipment and an element that moves with the equipment. Measurements using this kind of tachometer can encounter mechanical limits on the upper end of the scale due to speed and the required mechanical contact.

7.2.9 Motor/Phase Rotation Meters

These meters indicate the rotation direction of three-phase motors. You will normally use them to determine the rotation direction of a motor before it is connected to prevent potential damage.

A similar meter is used to determine the electrical phase rotation of three-phase power circuits to insure that the proper phase rotation is maintained throughout a system.

7.2.10 Circuit Tracers

Circuit tracers trace circuits by placing a device at one end that inserts a high-frequency signal; a detector is then used at the other end to find the circuit. A simple way to find the circuit breaker for a receptacle is to make a pigtail with a plug and a light that blinks. Plug in at the receptacle; at the circuit breaker end use a clamp-on current meter to find the circuit that matches up with the receptacle. Current will cycle at the rate that the light blinks.

7.2.11 Vibration Monitors

Portable vibration monitors are used to touch the external case of moving equipment in order to measure vibration amplitude. They are also used to connect to installed vibration probes. They can be spot monitors that provide a single-frequency readout, or they may provide a scope or recorder that can see multiple frequencies. They are commonly used for periodic and spot monitoring of rotating equipment. Records are kept so that trends can be spotted and comparisons made.

Stethoscopes are also sometimes used to listen to moving equipment to determine problems. A poor man's stethoscope can be a large screwdriver. By placing the metal end of the screwdriver on the place to monitor and an ear on the plastic end, sounds in the equipment can be heard.

7.2.12 Protocol Analyzers

Protocol analyzers (sometimes called Data Scopes or sniffers) are used to analyze communication circuits. They are generally hooked up in parallel with the communication signals and allow the actual data transmissions to be monitored. Handshaking monitoring is also generally provided. Some of these can simulate communication signals. Some of these are dedicated boxes while others use a PC with software and cables.

Software protocol analyzers tend to be cheaper though limited to RS-232 without additional converters. There is free software on the Internet for sniffing a PC com port. The following Web site has considerable sources on serial communication information and a listing of sources of protocol analyzers and other related equipment and software: http://www.lvr.com/serport.htm.

7.2.13 Test Pressure Gauges

Many of these are simple mechanical gauges, but modern electronic versions are also being used. The electronic versions are sometimes called electronic manometers. Some add-on accessories, used in conjunction with a DMM, permit pressure measurement. Calibrators also may have pressure inputs. Gauged hand pumps (see Figure 7-6) and pumps with integral calibrators allow you to pump up pressures for testing purposes.

Pressure gauges are commonly used for leak detection. Some common precision mechanical gauge brands are Ashcroft, Heise, Transcat, and U.S. Gauge. Some common electronic gauges are Ashcroft, Druck, Meriam, and Transcat. Note that the accuracy of these can vary greatly. High-accuracy mechanical test pressure gauges require tender loving care to maintain their accuracy and should be calibrated regularly.

7.2.14 Portable Recorders

Portable recorders can help you compare signals or catch transients, either on paper or electronically, on paperless models. Comparing signals permits you to look for commonalities or relationships between the signals and to look for transient events, or events that happen during night or graveyard shifts, or on weekends. When looking for transients, the recorder must have the frequency and mechanical response (2X) capability to catch the transient. It must also not load down the circuit you are testing. Make sure that the common mode voltage (CMV) specification of the recorder is not exceeded when using the recorder or you may get incorrect data.

FIGURE 7-6

Hand Pressure Tester Pump (courtesy of Transcat)



RELEVANT STANDARDS

•IEC -61010 - Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use.

•ANSI/IEEE 43 - IEEE Recommended Practice for Testing Insulation of Rotating Machinery.

•IEEE 95 - IEEE Recommended Practice for Insulation Testing of Large AC Rotating Machinery with High DC Voltage.

7.3 NONCONTACT TEST EQUIPMENT

Noncontact test equipment does not require a direct connection or contact with the equipment being tested.

7.3.1 Clamp-on Amp Meters

Clamp-on amp meters (see Figure 7-7) allow you to clamp around a wire and measure the current flowing in the wire. Most of these only measure AC current, but some, such as the Transcat 22747E, measure DC current, including that used for 4-20mA signals. For AC measurements, use a meter that measures true RMS, so that harmonic current or distorted waves can also be measured. The typical analog clamp-on meters found in an electrician's toolbox do not measure true RMS. These are still useful for troubleshooting, for example, when you are only interested in whether current is there or not and what the approximate current level is or harmonics may not be involved. Analog meters can also be used in conjunction with a true RMS meter to get a rough indication of the harmonic content by measuring with both and comparing the readings.

FIGURE 7-7

Clamp-on Amp Meter (courtesy of Extech Instruments)



Another use for a clamp-on is to measure leakage current for AC circuits. The clamp-on current meter is clamped around both the hot wire and the neutral; it should read very close to zero (<0.5mA). If it does not, the reading represents current that is leaking to ground somewhere.

Clamp-on meters come in a wide variety of ranges and with various clamps, including flexible clamps. Some will work in conjunction with DMMs, or are even constructed in combination with a DMM.

7.3.2 Static Charge Meters

These meters read the static electrical charges on surfaces. They are useful in detecting where charges may be building up and where the charges are coming from. This type of meter can also sometimes be used to detect sources of electric field interference due to static electricity.

7.3.3 Magnetic Field Detectors

Electrical circuits and devices have magnetic fields associated with them. These fields can cause noise in other circuits and devices and in some cases may be a health concern (when personnel are exposed to magnetic fields). There are instruments available that measure these magnetic fields. Of primary interest are extra low frequency (ELF) magnetic fields generated by 60 Hz power. Magnetic Field Detectors (commonly called Gauss meters) measure magnetic field strength and direction. These come in two general types—single axis (measures only in one axis and must be moved around to measure in various directions to determine the magnetic fields) and three axis or omnidirectional (measures all axis simultaneously).

There are a few instruments that have both modes of measurement. The single axis type is generally best for determining where a magnetic field is coming from while the three axis is best for determining general exposure. These instruments also come with analog and digital readouts. This analog readout is generally best for locating magnetic field sources, while a digital readout is generally best for doing a reading of field strength. There are also some with LED lights but with an obvious loss of accuracy. These instruments typically read in milligauss. The accuracy of these instruments varies greatly particularly in the low frequency end and some are jittery at the low end. Some common brands for these instruments include: AlphaLab (TriField), Electric Field Measurements (EFM), Extech Instruments, FW Bell, LessEMF (Gauss Master), Technology Alternative Corporation (Cellsensor), and Walker Scientific. The following Web site provides selections of some of these instruments: http://www.lessemf.com/gauss.html (no recommendations - caveat emptor [let the buyer beware]).

7.3.4 Noncontact Proximity Voltage Detectors

These devices detect the electric field generated by an energized circuit and are available in small devices to detect fields from 120VAC circuits on up into the kilovolt range. They are primarily go/no-go testers, able to detect energized wire through its insulation. When using them, always test them on a known energized circuit first. The Fluke 1AC and the Ideal Voltage Alert are examples of commonly encountered brands.

7.3.5 Magnetic Field/Current Detectors

These are simple go/no-go devices used to detect current flowing in such things as solenoids, relays, and transformers to indicate that they are energized and that there is a current path through them.

7.3.6 Circuit and Underground Cable Detectors

These devices are used to detect magnetic fields generated by the circuit. There are also some that are used for underground cables that detect an inserted high-frequency signal.

7.3.7 PhotoTachometers and Stroboscopes

Stroboscopic light is a series of short-duration flashes. When the flashes occur at a rate near the RPM of a shaft or other rotating object, the object appears to slow down or stop. Most of us have seen a similar effect on a dance floor that has a strobe light. This effect is due to the eyes being overwhelmed by the flashing of the stroboscopic light so that the eyes see the moving object only in the flashes, making a moving object appear as if in frames. If the flash rate is near the speed of the object the object will appear to slow or stop. Objects that do not have a frame of reference (a moving edge or shape), such as a rotating smooth shaft, usually require a visible reflective dot or strip to provide this effect.

Phototachometers use this principle to measure the speed of an object moving in a cyclic manner. They flash a light at the object, which will appear to stand still when the flash rate is equal to the cyclic rate. Such a tachometer can be used on different types of moving equipment, including linear or reciprocating objects, to determine their speed. The phototachometer is often combined with a contact tachometer. Photo tachometers can have a range up five times greater than contact tachometers.

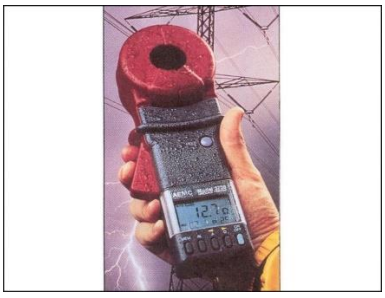
A stroboscope works on the same principle as a phototachometer, but typically is gun-shaped and puts out more light. It is used both to measure speed and to look at moving objects to analyze their motion. When the object is moving slowly, the direction of this perceived motion is the same as that of the true motion if the flash rate is slower than the true motion rate; it appears to move in the opposite direction if the flash rate is faster than the true motion rate. If the flash rate is double the object's speed, the object will appear to have a double image. If the object moves faster than the maximum rate of a strobe, the speed can still be measured by taking several readings of subharmonic rates (when the reference is moving at a consistent rate) and calculating the object's speed.

7.3.8 Clamp-On Ground Testers

One type of clamp-on ground tester (see Figure 7-8) allows you to clamp around a ground rod or cable and measure the ground system resistance. It does this by inserting a high-frequency voltage signal and measuring the resulting current.

FIGURE 7-8

AEMC Clamp-on Ground Tester (courtesy of AEMC instruments)



7.3.9 Infrared Thermometer Guns and Imaging Systems

Infrared (IR) devices allow you to detect hot spots in your electrical systems. These hot spots can be caused by malfunctioning equipment, harmonics, corrosion, and loose connections. The guns come in two varieties, those that read actual temperature and those that measure temperature rise above ambient.

Imaging equipment allows you to see the actual IR image and in some cases retain these images for future reference in pictures, on tape, or digitally. Many sites have periodic inspections with IR guns or imaging equipment to detect problems before they develop into larger problems. This same equipment is also used to detect mechanical and process problems.

7.3.10 Leak Detectors

There are two general types of noncontact leak detectors. The first detects ultrasonic sound waves generated by the leak. The second actually detects the chemical that is leaking with a chemical-specific detector. And, of course, there is always the old standby, soapy water.

7.4SIMULATORS/PROCESS CALIBRATORS

Quite a few devices on the market will calibrate and simulate 4-20mA, voltage, and pneumatic signals, as well as the functions of thermocouples and RTDs. Some of these tools have multiple functionalities, with different plug-in modules providing different functions. They range from luggable units to small ones the size of a DMM that can be carried on a tool belt. These types of meters are replacing the DMM in many cases because they provide DMM functionality as well as the simulation and calibration functions. Fluke Models 710 and 744 are examples of this type of calibrator. Other suppliers for these are Altek, Beta, Promac, Transcat, and Transmation. An example of a calibrator/simulator is shown in Figure 7-9.

FIGURE 7-9

Transmation Simulator/Calibrator (courtesy of Transmation Inc.)



7.5 JUMPERS, SWITCH BOXES, AND TRAPS

Not all test equipment comes off the shelf or from a catalog. A number of useful test items are often improvised, though some are also available commercially. One of the most common is the simple jumper, a wire with alligator clips on each end. Jumpers are a useful item for the troubleshooting toolbox. Remember, however, that the jumper must be rated for the voltage and current involved, care must be taken when attaching it, and that jumper must be removed before the equipment is returned to service.

A switch/light box also can be useful. This is a box with toggle switches and/or lights that have either a terminal strip to connect to or a pigtail to attach to devices that have digital inputs or outputs. These allow you to test switch legs and input/output cards on PLCs or a DCS. Sometimes the PLC or DCS vendor has switch/light boxes that plug right into their input/output rack.

Traps are devices or circuits that you connect to a logic circuit. They trap signals to confirm that a contact is operating or obtain information about what is causing a system to trip. Traps are used in systems where there is no parallel or shadow indication that a contact is operating properly.

Three simple traps that you can make yourself employ lights, fuses, and relays. For the first type, install a light in parallel with the suspect contact. The light will be off when the contact is closed and on when the contact is open (assuming the downstream circuit is complete). The light should be designed to handle the voltage involved and not allow the load to energize when the contact is open. This device (see Figure 7-10) works in a manner similar to a blown-fuse indicator.

Another way to do this is to install a small fuse that is incapable of carrying the load around the suspect contact. If the contact opens, the fuse will blow. The fuse-type trap (see Figure 7-11) will catch transients, whereas the light-type trap will not.

A third type of trap uses a relay that you can manually latch and wire through one of the relay's normally open contacts (see Figure 7-12). The relay is connected at the test point and the circuit return and then manually latched. If the system trips and the relay unlatches, the problem is located upstream of the test connection. If it remains latched, the problem is located downstream of the test connection.

FIGURE 7-10

A Simple Troubleshooting Trap

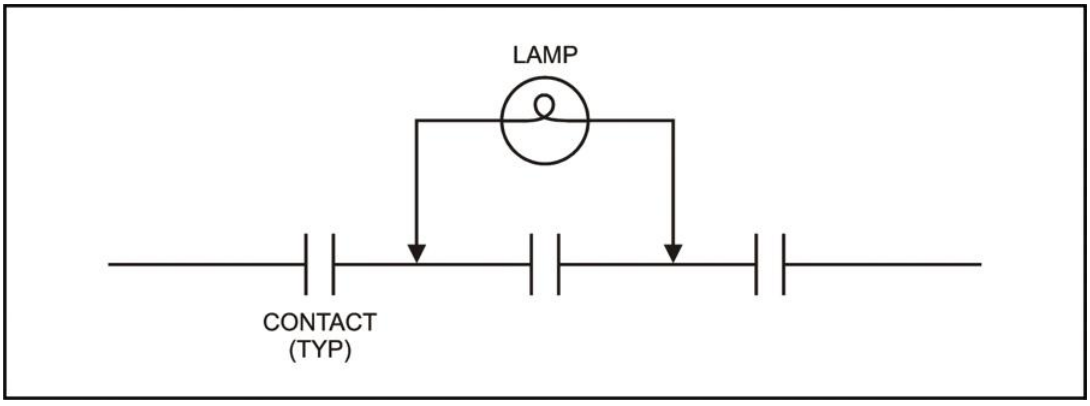


FIGURE 7-11

Fuse-type Trap

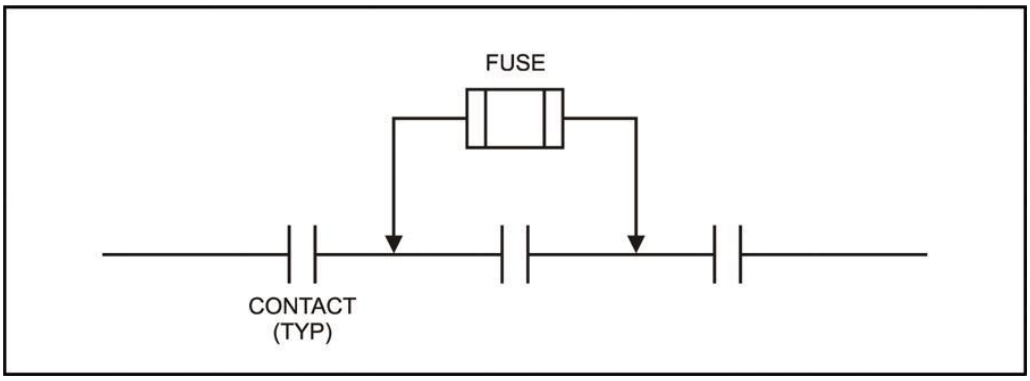
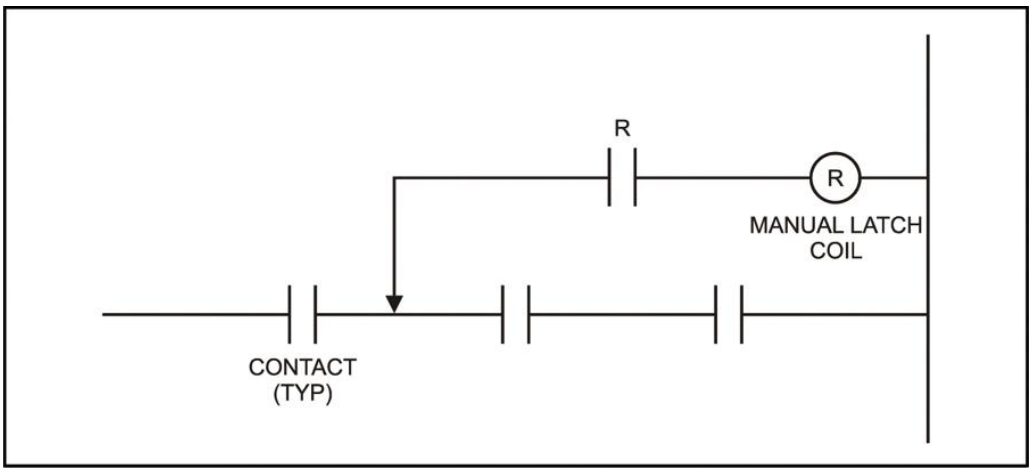


FIGURE 7-12

Latching Relay Type Troubleshooting Trap



7.6DOCUMENTING TEST EQUIPMENT AND TESTS

At times, troubleshooting means looking for deterioration in a system, unreliable instruments, or failure patterns. To do this, the troubleshooter must have records of prior tests on the system so as to compare them. Records can be kept manually or on a computer-based maintenance management system. These are supplied by companies such as Beta, Druck, Fluke, Hathaway, Honeywell Loveland, and Transmation.

Much of the test equipment on the market now has the ability to keep records itself or store data for downloading to a PC (personal computer). Take care also to evaluate the PC software associated with this type of test equipment. Some of the things to consider include the PC's requirements, ease of use, interface capability with existing computer systems, report-writing capabilities, long-term storage capability, use with smart transmitters, compatibility with industry standard databases, user configurability, and the capability for user customization.

An example of record keeping might be when a new uninterruptible power supply (UPS) system is installed. All the voltages, currents, and waveforms might be recorded at the start and used for troubleshooting. Another application is valve signatures where the valve is tested prior to installation to get a baseline for downstream maintenance. Other areas where records can be useful include cable resistances, motor tests, calibration records, and instrument failure data.

With the emphasis on reliability-based maintenance, these records can help determine the overall reliability of your instrumentation, the amount of maintenance required per instrument, and personnel loading and scheduling.

7.7 ACCURACY OF TEST EQUIPMENT

Accuracy is not always required for troubleshooting, because many times we are applying go/no-go tests—either it is there or it is not. For example, if we are tracing a 120VAC circuit, we really do not care if the voltage is 120.1V or 119.9V. We do care whether the voltage is consistent or repeatable. We do care that it is not 80V or 30V, and to check that requires only that our equipment be fully functional. All test equipment should be functionally checked on a periodic cycle.

There will, however, be cases where accuracy is important, so all of your test equipment should also be calibrated or verified on a regular basis. The minimum accuracy for the standard that you test your equipment against is twice the required accuracy (2×) or better. For most things, 3–4× should be adequate; a standard of 5× is appropriate for high accuracy, and 7–10× for very high accuracy. The standard used in the testing of your test equipment should be traceable back to the National Institute of Standards and Technology (NIST), formerly known as the National Bureau of Standards (NBS).

SUMMARY

Use the right tool for the job. It can provide the access and information you need. The wrong tool may provide the wrong information and lead you astray. It may also cause safety problems or damage the system or equipment you are troubleshooting.

You generally get what you pay for when buying tools. A simple economic analysis of downtime cost versus test equipment cost generally shows that test equipment can easily pay for itself after it's used a few times to solve a problem. All tools should be approved by an appropriate testing agency such as ASTM, UL, or FM.

A wide variety of tools and test equipment is available to assist in troubleshooting. All tools and test equipment should be rated for the service for which they are used. When you use nonrated tools and test equipment, not only are you risking your life, but failure to have the appropriate test equipment can also severely hamper your ability to troubleshoot equipment in a timely manner. Do not scrimp on your tools and test equipment. And take care of them—they are your livelihood.

QUIZ

1. All tools should be

A.rated for the service for which they are used.

B.1000V.

C.insulated.

D.all of the above

2. IEC-61010-rated DMMs are rated for

A.service voltage.

B.transient protection.

C.maximum voltage and current.

D.both A and B

3. Noncontact instruments can measure

A.voltage presence.

B.magnetic field presence.

C.static electricity.

D.all of the above

4. The test equipment to measure speed of rotating equipment is a

A.megohmmeter.

B.tachometer.

C.voltmeter.

D.thermoimager.

5. Test leads should

A.be 2 ft (.61 meters) long.

B.have alligator clips.

C.be rated for their service.

D.none of the above

6. DMM stands for

A.digital megohmmeter.

B.digital micrometer.

C.digital multimeter.

D.all of the above

7. Noncontact voltage probes sense voltage by sensing the

A.magnetic field.

B.static field.

C.current field.

D.electric field.

8. Noncontact photo tachometers sense speed using

A.a strobelight effect.

B.magnetic fields.

C.a rotating shaft.

D.none of the above

9. ELF stands for

A.extra low frequency.

B.extra low field effect.

C.electric low frequency.

D.none of the above

10. A wiggy is a

A.current detector.

B.magnetic field detector.

C.static electricity detector.

D.voltage detector.

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4.Shen, E. Multimeter Safety: Ultimately, It's Up to You. *I&CS*, February 1998.

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8

TROUBLESHOOTING SCENARIOS

Mechanical systems

Process connections

Pneumatic systems

Electrical systems

Electronic systems

Valves

Calibration

Programmable electronic systems

Communication circuits

Transients

Software

NOTE: *The troubleshooting scenarios in this chapter are drawn from actual experience. They are brief and may not apply to the specific equipment you will be working on. Also, they do not include safety precautions.*

Always follow company and industry safety procedures and standards.

8.1 MECHANICAL INSTRUMENTATION

8.1.1 Mechanical Field Recorder, EXAMPLE 1

PROBLEM: Pressure recorder chart reading zero.

ACTION	RESULT
1. Examine chart.	Chart indicates reading went to zero suddenly.
2. Remove chart and cover.	Examine instrument. Mechanical linkage between bellows and recorder pen arm has come loose.
3. Reinstall, then check calibration.	Problem solved.

8.1.2 Mechanical Field Recorder, EXAMPLE 2

PROBLEM: Pressure recorder chart reading incorrectly.

ACTION	RESULT
1. Examine chart.	Examination indicates nothing unusual.
2. Remove chart and cover, then simulate signal to input bellows.	At the high end, the recorder is off by about 10%.
3. Examine linkage arm between bellows and pen arm.	Looks slightly bent, probably due to an overpressure event.
4. Replace link, reinstall, and test.	Problem solved.

8.1.3 Mechanical Field Recorder, EXAMPLE 3

PROBLEM: Pressure recorder chart did not read properly.

ACTION	RESULT
1. Examine chart.	Chart is not turning.
2. Remove chart and cover to examine chart motor.	Motor not wound following last chart replacement.
3. Wind motor.	Problem solved.

8.2 PROCESS CONNECTIONS

8.2.1 Pressure Transmitter, EXAMPLE 1

PROBLEM: Pressure transmitter is not reading correctly.

ACTION	RESULT
1. Check recorder in control room.	Pressure signal is not changing and indicates no noise. (Process signals are normally noisy.)
2. Check in field.	Field instrument reading same as control room. (This verifies that the receiver instrument is not causing the problem.)
3. Check process pressure connection.	Connection is plugged.
4. Rod out pressure connection.	Problem solved.

8.2.2 Pressure Transmitter, EXAMPLE 2

PROBLEM: Pressure transmitter is not reading correctly (frozen).

ACTION	RESULT
1. Check DCS trend in control room.	Pressure signal is not changing and does not indicate any noise.
2. Use loop drawing and P&ID to locate instrument, then check in field.	Field instrument reading same as control room.
3. Check process pressure connection.	Root valve is closed.
4. Check maintenance records.	Night crew worked on transmitter and failed to open root valve when done.

5. Verify instrument piping integrity and open valve.	Problem solved.

8.2.3 Temperature Transmitter

PROBLEM: Temperature signal is slow to respond. (NOTE: Temperature signals are normally slow to respond, so the first task is to rule out a process problem.)

ACTION	**RESULT**
1. Check trend recording history, then verify that signal response is abnormally slow.	Slow response verified.
2. Review loop drawing and P&ID, and check signal by simulating signal from transmitter to DCS.	Checks OK.
3. Remove field transmitter and test in shop.	Checks OK. Proposed cause: coated thermowell.
4. Verified next outage by removing examining thermowell.	Problem solved

8.2.4 Flow Meter (Orifice Type)

PROBLEM: Flow meter is not reading correctly.

ACTION	**RESULT**
1. Check DCS Trend in control room.	Flow signal is changing but operator says it is not consistent with other flow signals in system and does not indicate noise.
2. Check in field.	Field instrument output same as in control room.
3. Check process pressure connections.	Low sides of orifice flange taps are plugged.

4. Rod out and blow down to clean impulse lines.	Problem solved.

8.3 PNEUMATIC INSTRUMENTATION

8.3.1 Pneumatic Transmitter, EXAMPLE 1

PROBLEM: Flow transmitter is sluggish.

ACTION	**RESULT**
1. Check trend recordings.	Flow signal is slow to respond as indicated by trend recordings.
2. Check signal.	Signal does not indicate any noise (normally noisy).
3. Check in field.	Signal-pressure gauge reading same as control room.
4. Check flow taps.	Taps check out OK.
5. Examine transmitter.	Manual damper has been tightened down to damp signal. (An operator on another shift did this to smooth out flow signal in the control room.)
6. Adjust damper.	Problem solved.

8.3.2 Pneumatic Transmitter, EXAMPLE 2

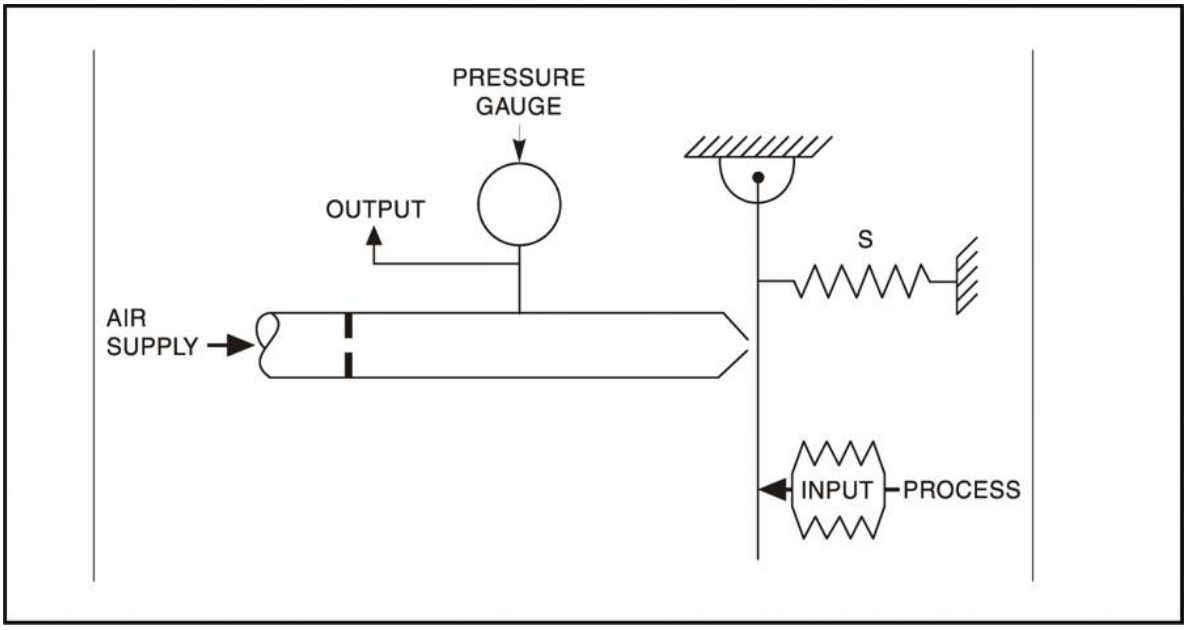
PROBLEM: Flow transmitter reads full scale (See Figure 8-1).

ACTION	**RESULT**
1. Verify transmitter output on local panel.	Full-scale output verified.

2. Locate transmitter, remove cover, examine.	Flapper is away from nozzle (normally the flapper would be have to be against nozzle to cause full-scale output).
3. Check nozzle.	Plugged.
4. Clean nozzle. (Clean restrictor as a preventive measure.)	Problem solved.

FIGURE 8-1

Use of a Flapper



8.3.3 Pneumatic Transmitter, EXAMPLE 3

PROBLEM: Pressure transmitter reads zero (see Figure 8-1).

ACTION	**RESULT**
1. Verify transmitter field output.	Zero output verified.
2. Examine air restrictor.	Plugged (very common).
3. Clean restrictor. (Clean pneumatic relay and replace air filter as preventive measure.)	Problem solved.

8.3.4 Pneumatic Transmitter, EXAMPLE 4

PROBLEM: Flow transmitter reads zero.

ACTION	RESULT
1. Verify field output.	Zero output verified.
2. Examine restrictor.	Appears to be OK. No air out restrictor port.
3. Examine transmitter supply.	None.
4. Check filter.	Plugged.
5. Replace filter.	Problem solved.

8.3.5Pneumatic Transmitter, EXAMPLE 5

PROBLEM: Temperature transmitter reads zero.

ACTION	RESULT
1. Verify output.	Examination of transmitter field output verifies output.
2. Examine restrictor.	Appears to be OK.
3. Examine transmitter supply.	Appears to be OK.
4. Examine pneumatic relay.	Dirty relay seat and plug.
5. Clean relay seat and plug. Replace filter as a preventive measure.	Problem solved.

8.3.6 I/P (Current/Pneumatic) Transducer

PROBLEM: Valve is sluggish.

ACTION	RESULT
1. Verify output.	I/P in field output verifies output and tracks control signal.
2. Examine I/P supply.	Reads 20 psi (138 kPa)—OK.
3. Examine filter.	Almost completely plugged.
4. Replace filter element.	Problem solved.

8.4 ELECTRICAL SYSTEMS

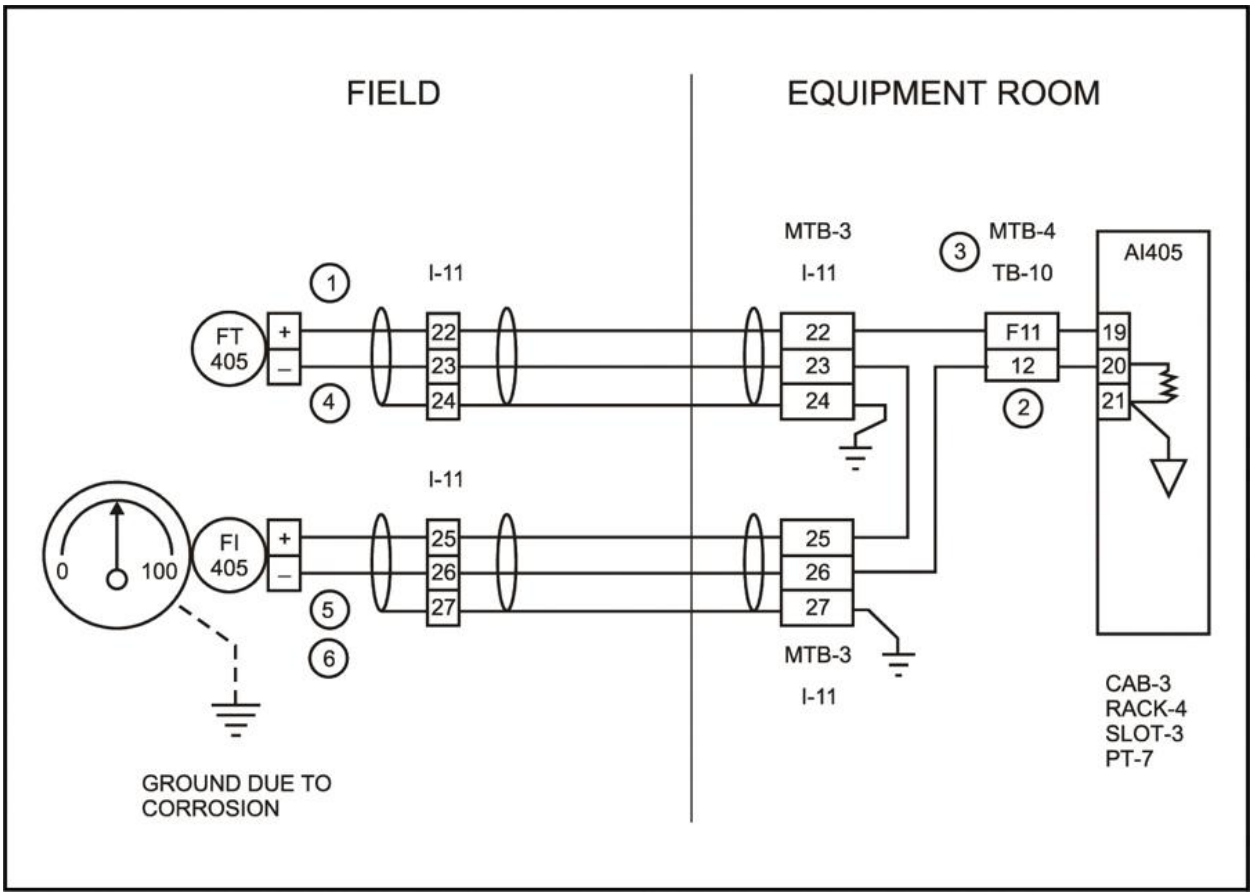
8.4.1Electronic 4-20 mA Transmitter

PROBLEM: Transmitter reads zero (see Figure 8-2).

ACTION	RESULT
1. Examine transmitter field output to verify output.	Zero output verified—no voltage present at transmitter, possibly due to loss of power.
2. Review loop drawing to find loop fuse location, then check fuse.	Fuse status indicator shows a blown fuse.
3. Replace fuse (F11).	Blows again.
4. Examine transmitter. (NOTE: Start at one end where short is most likely, i.e., in this case in the field.)	No indication of short to ground.
5. Examine field indicator located at control valve.	Cover loose—water and corrosion in indicator are causing a short to ground.
6. Replace local indicator and install a new fuse	Problem solved.

FIGURE 8-2

4-20 mA Loop



8.4.2 Computer-Based Analyzer

PROBLEM: Lab analyzer resets (reboots) randomly.

ACTION	**RESULT**
1. Review analyzer manual and run diagnostic program.	No problems indicated.
2. Examine electrical power drawings and wiring.	Small pump found to be on same circuit.
3. Cycle the pump. (This technique is called "fault insertion.")	Analyzer resets due to current transient at pump startup.

4. Relocate pump power onto another circuit.	Problem solved.

8.4.3 Plant Section Instrument Power Lost

PROBLEM: An instrument technician shorts 120VAC in a field instrument; all instrument power in that section of the plant is lost.

ACTION	RESULT
1. Trace wiring via electrical drawings and loop drawings to find the fuse upstream of the loop fuse that protects the section of the plant in which power is blown, and then check that fuse.	Size is correct per drawings, but the fuse type is not.
2. Consult engineering.	Engineering checks fuse coordination curves and finds that the installed fuse will not coordinate with the loop fuse, thus blowing fuse before the loop fuse can blow. (Construction personnel ran out of correct fuses during installation and substituted without asking. No post-construction inspection was done on the loop.)
3. Replace fuse.	Immediate problem solved.
4. Examine similar loops.	Two more incorrect fuses found.
5. Replace incorrect fuses.	Problems prevented.

8.4.4 Relay System

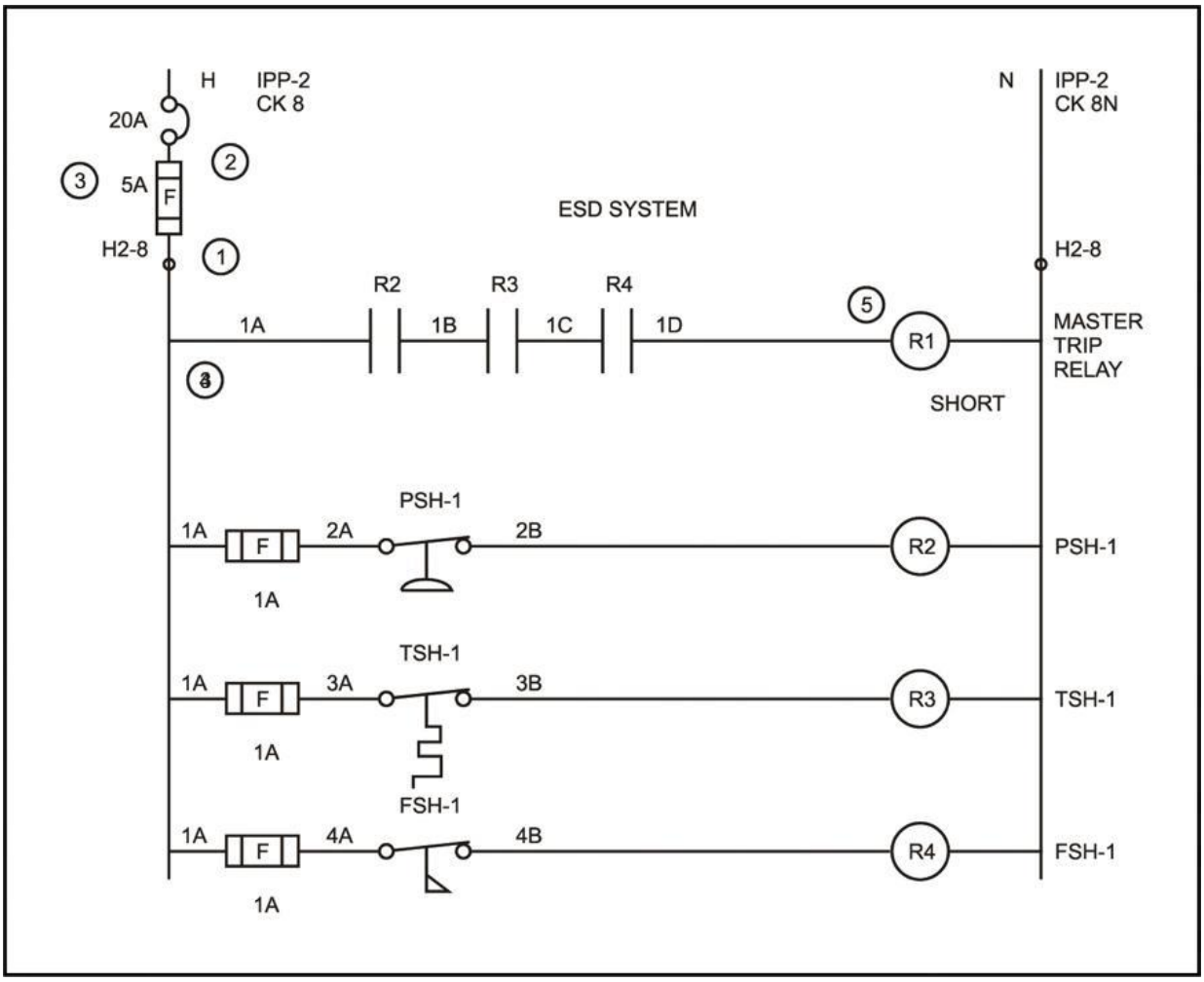
PROBLEM: A relay-based shutdown system has tripped for no apparent reason (see Figure 8-3). Loss of power is suspected.

ACTION	RESULT

1. Because the relay shutdown (ESD) system is fail-safe, measure power supply at relay system input.	No power.
2. Examine electrical power drawings to trace power back to circuit breaker/fuse combination	Blown fuse is found.
3. Replace 5A fuse.	Blows again.
4. Lift wire and tape signal input of ESD (fed by power from blown fuse), and then replace fuse.	Fuse does not blow. (Because field process switches each have fuses, a field fault is not indicated.)
5. Review drawings of ESD system to trace power and separate field power from control logic power; then trace and check logic power through each field switch logic relay contact. Disconnect signal logic power at furthest downstream point, which is at the master trip relay.	Fuse does not blow.
6. Connect to master trip relay.	Fuse blows.
7. Check resistance coil of master trip relay.	Found to be shorted.
8. Replace master trip relay and check system.	Checks out OK. Problem solved.

FIGURE 8-3

Relay Trip System



8.5 ELECTRONIC SYSTEMS

8.5.1 Current Loops

8.5.1.1 ELECTRONIC TRANSMITTER, EXAMPLE 1

PROBLEM: Operator complains that a flow transmitter is reading low.

ACTION	RESULT
1. Talk with operator about problem.	Find that F101 is reading low compared with the other flow meters on Tower 101.

2. Ask operator if anything has happened on the loop recently.	Operator says a local indicator was added at the local panel.
3. Ask operator to locate the flow meter.	Find that flow meter is a long way from the control room and the local panel is far from the flow meter.
4. Review the loop drawing.	Distances shown on drawing indicate that total resistance limit in loop was exceeded when the local panel indicator was added.
5. Remove local indicator from loop.	Loop works. (Remove and conquer.)
6. Replace local indicator with one that has a lower current input resistor (e.g., 250Ω to 100Ω).	Problem solved.

8.5.1.2 ELECTRONIC 4–20 mA TRANSMITTER, EXAMPLE 2

PROBLEM: Level transmitter stops responding (see Figure 8-2).

ACTION	**RESULT**
1. Verify by looking at trend recording and reviewing P&ID.	Does not appear to be a process problem based on flow loops in and out of the vessel.
2. Check current signal to DCS.	Current signal to DCS agrees at input card.
3. Simulate changing signal to DCS.	Faceplate value does not change.
4. Inspect DCS input.	DCS input card bad.
5. Replace card.	Problem solved.

8.5.1.3 ELECTRONIC 10–50 mA TRANSMITTER, EXAMPLE 3

PROBLEM: Reactor 201 has had a pressure runaway, and the pressure transmitter failed to actuate pressure protection valves at 80%, causing the relief valves to open and blow the reactor contents to flare.

ACTION	RESULT
1. Review transmitter output on the trend recording.	Indicates that the transmitter output went to 65% and flattened out.
2. Review safety system drawings, then check out the transmitter in shop.	Checks out OK.
3. Simulate a signal to shut down system. (Tests trip system and wiring)	Valves actuate correctly.
4. Check the power supply. (The question is what else could cause the signal not to get the trip level?)	These are stand-alone transmitters with their own power supply, and rather than the required 70VDC they show only 55VDC, enough to work under normal conditions but not under abnormal conditions.
6. Replace power supply and test.	Problem solved.
7. Check maintenance records.	Due to operational time constraints, this system was not tested during the last turnaround, a costly mistake that could have had safety implications.

8.5.1.4 ELECTRONIC 10–50 mA TRANSMITTER/RECORDER

PROBLEM: Newly installed surplus transmitter to a multipoint recorder in research lab does not read right, even though the current signal is correct.

ACTION	RESULT
1. Verify that 10–50mA current signal is correct.	Checks OK.
2. Check other recorder points.	All work correctly.

3. Simulate signal to recorder.	Checks OK.
4. Reconnect signal.	Not OK.
5. Check recorder manual.	Common-mode voltage (CMV) rating is only 25V
6. Look at wiring drawings.	Find that the 70VDC power supply is connected first to the recorder rather than the transmitter, allowing the recorder to see 65–70V, causing a ground leak and a bad reading.
7. Rewire, putting the transmitter first.	Problem solved.

8.5.1.5 ELECTRONIC TEMPERATURE TRANSMITTER

PROBLEM: Transmitter output over 100%.

ACTION	**RESULT**
1. Look at trend recording to verify up-scale reading.	Reading verified.
2. Examine loop drawing.	Loop drawing indicates transmitter uses a thermocouple and has an upscale burnout as a failure mode if thermocouple is open.
3. Check thermocouple.	Found to be open.
4. Replace thermocouple.	Checks OK. Problem solved.

8.5.2 Voltage Loops

8.5.2.1 ELECTRONIC TC/V TRANSDUCER

PROBLEM: Temperature readings are too low on a new loop.

ACTION	RESULT
1. Verify on the trend recording.	Readings verified.
2. Review loop drawing, locate transducer, then check thermocouple (starting at one end of the loop and walking forward).	Checks OK.
3. Simulate thermocouple to transducer.	Checks OK.
4. Simulate transducer to receiver instrument.	Checks OK.
5. Check wiring.	Notice that wiring distance may be too great.
6. Check receiver instrument specifications.	Receiver input impedance too low for application.
7. Install E/I converter on output of TC/V transducer and run signal with 4-20mA (one possible solution; another could be replace TV/V with one of higher input impedance).	Problem solved.

8.5.2.2 ELECTRONIC LAB FLOW INSTRUMENT

PROBLEM: The flow meter on a new loop reads 100% all the time.

ACTION	RESULT
1. Verify on the trend recording, then disconnect receiver instrument and check flow meter output.	Trend verifies 100% output. Flow meter output checks OK.

2. Simulate flow meter to receiver instrument.	Checks OK.
3. Check output voltage with high-impedance voltmeter at receiver instrument to check for receiver low input impedance as a possible cause.	Reads same as receiver instrument.
4. Check flow meter and receiver instrument specifications.	Flow meter output is 0–10V, but receiver instrument is 0–1V.
5. Install correct input card in receiver instrument.	Problem solved.

8.5.2.3 TEMPERATURE INSTRUMENT

PROBLEM: Temperature thermocouple (T/C) input erratic

ACTION	**RESULT**
1. Check the trend recording.	Erratic reading verified.
2. Review loop drawing and check for loose terminals that may be vibrating or expanding/contracting, starting at the field T/C head and walking through the loop into the control room.	Several terminals are slightly loose on one of the field junction boxes and at marshalling cabinet.
3. Tighten terminals.	Problem solved.

8.5.3 Control Loops

8.5.3.1 ERRATIC CONTROL LOOP, EXAMPLE 1

PROBLEM: Control loop behavior erratic, put on manual.

ACTION	RESULT
1. Examine loop behavior on the trend recorder.	Loop oscillates when a process transient occurs. Operator says this is not normal.
2. Examine loop-tuning parameters and compare to records.	Parameters changed. (Midnight shift operator changed them to where he "liked" them.)
3. Change parameters to original values.	Loop behavior OK. Problem solved.

8.5.3.2 ERRATIC CONTROL LOOP, EXAMPLE 2

PROBLEM: Control loop behavior erratic, put on manual.

ACTION	RESULT
1. Examine loop behavior on the trend recorder.	Loop slow to respond and then overshoots when process transient occurs.
2. Examine loop-tuning parameters. Compare to records.	Everything OK.
3. Make a small-step change in set point.	Valve slow to respond.
4. Make a small-step change in set point while observing valve in field.	Valve is hanging up.
5. Examine valve packing.	Find debris in packing.
6. Replace packing.	Problem solved.

8.5.4 Ground Loops

8.5.4.1 TRANSMITTER CABLE SHIELD

PROBLEM: Temperature reading too noisy.

ACTION	RESULT
1. Verify on the trend recording.	Noise verified.
2. Review loop drawing, then check current signal at DCS input.	Fluctuations found in signal.
3. Simulate receiver instrument to transmitter in field with loop calibrator.	No noise detected.
4. Simulate transmitter in field to receiver instrument.	Noise detected.
5. Lift shield ground in equipment room.	Noise goes away.
6. Check shield at field junction boxes.	Looks OK.
7. Check shield at transmitter.	Looks OK.
8. Check shield at field indicator at grade.	Shield grounded to case, causing a ground loop in shield.
9. Tape back shield in indicator.	Problem solved.

8.5.4.2 TEMPERATURE TRANSMITTER

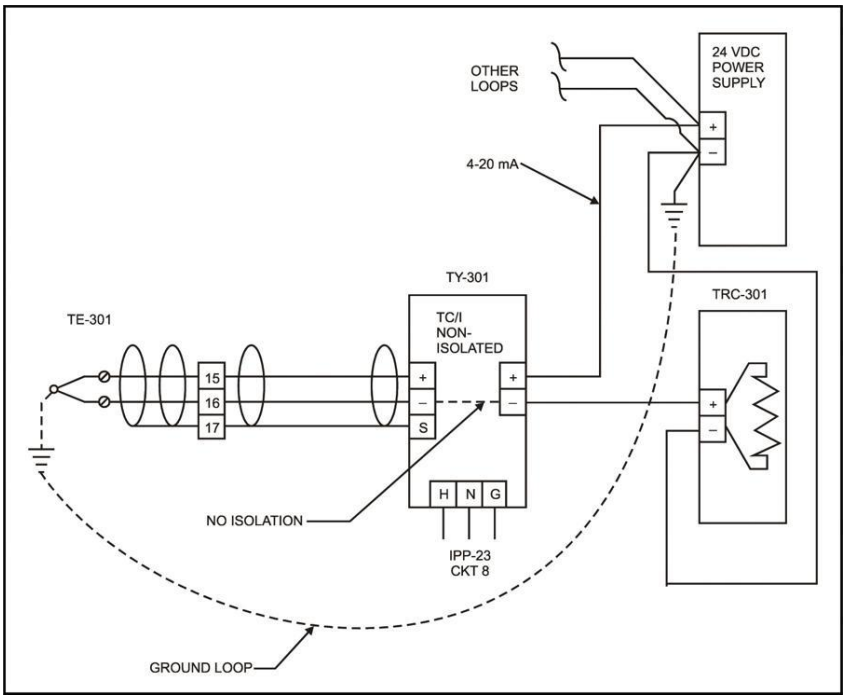
PROBLEM: Temperature readings are unstable; some other loops behaving erratically and put on manual (see Figure 8-4).

ACTION	RESULT
1. Examine temperature loop drawing.	Loop is a thermocouple input to a transducer whose output is 4–20mA.

2. Check transducer output.	Output matches receiver instrument.
3. Check 4–20mA output.	Output correct compared to thermocouple reading when disconnected from loop.
4. Reconnect.	Output bad.
5. Simulate signal to receiver.	Checks OK.
6. Check voltage across input/output terminals (thermocouple plus side to output side plus).	Voltage at 18V. (no isolation)
7. Engineer reviews loop drawing, determines that thermocouple is an ungrounded type, transducer is non-isolating type; suggests checking for ground in the thermocouple circuit. Check thermocouple.	Circuit grounded at the thermocouple, causing a ground loop.
8. Look at maintenance records.	Thermocouple failed on graveyard shift. A replacement part was not available, so a grounded thermocouple was fabricated and installed, which caused a ground loop in the DC power system, thus affecting other instruments.
9. Replace with ungrounded thermocouple.	Problem solved.

FIGURE 8-4

Ground Loop Problem



8.6 VALVES

8.6.1 Valve Leak-By, EXAMPLE 1

PROBLEM: Operator says control valve is shut but flow meter is still reading a flow.

ACTION	RESULT
1. Check flow reading on DCS.	Indicates flow.
2. Locate field transmitter and valve.	Field flow transmitter indicator indicates the same as DCS.
3. Look at control valve and stem indicator.	Indicates closed.
4. Inspect valve installation.	Bypass valve might be open slightly.
5. Close bypass valve.	Flow stops. Problem solved.

8.6.2 Valve Leak-By, EXAMPLE 2

PROBLEM: Operator says valve is shut, but flow meter is still reading a flow.

ACTION	RESULT
1. Check flow reading on DCS.	Indicates flow.
2. Check field transmitter indicator.	Field transmitter indicator indicates flow.
3. Look at valve and stem indicator.	Indicates closed.
4. Check bypass valve.	Closed.
5. Put valve on bypass and remove valve for examination.	Erosion and wire drawing apparent on valve seat.
6. Talk to operator for history.	Valve had been operated very close to its seat, causing high velocities, erosion, and wire drawing.
7. Install smaller trim to raise valve off its seat.	Problem solved.

8.6.3 Valve Oscillation

PROBLEM: Control valve is oscillating slightly.

ACTION	RESULT
1. Put loop in manual.	Oscillation still present.
2. Check air supply.	Steady.
3. Replace I/P (current to pneumatic transducer).	Problem solved.

8.7 CALIBRATION

8.7.1 Low Reading on Flow Transmitter

PROBLEM: Operator complains that after night crew worked on flow transmitter, it reads low.

ACTION	RESULT
1. Examine trend recording and current operating conditions to verify that this is not a process problem.	Verified.
2. Review loop drawing, then check the loop current.	Transmitter is sending what the control system is seeing.
3. Check calibration.	A 200 IN. WC transmitter was installed where a 100 IN. WC was required, giving a reading that was low by the $\sqrt{2}$ (1.414) factor.
4. Look at transmitter data sheet.	Loop drawing and the DCS do not agree with the data sheet that the night crew used.
5. Verify that the 100 IN. WC was the correct range. Recalibrate. Reconcile loop drawing and DCS.	Problem solved.

8.7.2 Inaccurate Pay Meters

PROBLEM: Input pay meters at ethylene plant are reading low when com- pared to feeder plant meters. Feeder plant is complaining (a common com- plaint, since plant pay meters determine plant performance).

ACTION	RESULT
1. Talk to process engineer about plant material balance.	Find slight favor to receiving plant.

2. Examine input pay meters.	All zeros turned down slightly. (Apparently, someone tried to "improve" the plant's performance.)
3. Rezero.	Problem solved.

8.7.3 Plant Material Balance Off

PROBLEM: Some of the distillation tower material balances are off for the month.

ACTION	**RESULT**
1. Talk to process engineer to identify potential problem areas.	
2. Examine some of the meters.	Calibration is off by an average of 3%.
3. Check the maintenance management system.	Meters were calibrated in the last month as part of an accuracy maintenance program.
4. Talk to people in the accuracy maintenance program.	All used the same calibrator.
5. Check the calibrator.	Calibration is off. (A common cause failure)
6. Recalibrate all meters it has been used on since its last calibration.	Problem solved.

8.8 PROGRAMMABLE ELECTRONIC SYSTEMS

8.8.1 PLC

PROBLEM: PLC loses its memory.

ACTION	**RESULT**

1. Reload memory.	Everything appears OK, indicating a possible power transient
2. Because memory is not in EPROM, check battery.	Checks OK.
3. Examine battery terminals.	Corrosion apparent.
4. Clean battery terminals, then cycle power.	Memory holds. Problem solved.

8.8.2 PLC Card

PROBLEM: New additional PLC input card does not work.

ACTION	**RESULT**
1. Check input data table with simulated inputs.	No inputs from new card recognized.
2. Check vendor manual.	On this type of PLC, the rack must be reset or power cycled for the PLC to recognize new cards.
3. Cycle power.	Problem solved.

NOTE: This is also true of configuration DIP switch changes. DIP switches are typically only scanned on power-up or when reset. Also, replacement cards must have their DIP switches and jumpers set the same as the original card or the card will not work properly.

8.8.3 PLC Pump Out System

PROBLEM: A PLC-controlled pump out system reached the High-High level limit. The PLC controls a pump out system on a knock out drum where an output pump is turned on at a High level limit and turned off at a Low level limit.

ACTION	**RESULT**
1. Check the DCS log and see if the K.O. drum High level alarm is activated. (This checks the High level switch operation	Alarm activated.

and the input to the PLC and to some extent the PLC.)	
2. Get with Operations and verify that it is OK to turn the pump on.	OK.
3. Connect a PLC programming panel to the PLC and force the High level switch input that turns on the pump and verify that the pump (and PLC logic and output circuit) is functional.	Pump fails to start.
4. Verify that the PLC output has activated.	OK.
5. A review of the loop drawing indicated that there is an interposing relay. Verify that the interposing relay has energized. (Note: interposing relays on the outputs of PLCs that go to motor starters are a common practice.)	Not energized but has voltage on coil.
6. Replace interposing relay.	Problem solved.

8.9 COMMUNICATION LOOPS

8.9.1 RS-232, EXAMPLE 1

PROBLEM: New RS-232 communication link does not work.

ACTION	**RESULT**
1. Review communication system drawing, then put a breakout box on link.	Handshaking appears OK; master transmit light is blinking, receive lights are not indicating that the transmit and receive wires may be wired backwards.

2. Use breakout box jumpers and switches to switch wiring.	Works OK.
3. Rewire.	Problem solved.

8.9.2 RS-232, EXAMPLE 2

PROBLEM: New RS-232 communication link with short-haul modem works intermittently.

ACTION	**RESULT**
1. Review system wiring drawing, then examine wiring and routing for wiring mistakes, loose terminals, or interference from other wires.	Wiring OK; wire length to modem is less than 50' (15 meters).
2. Check specifications on data source for possible grounding problem.	Data source output is not isolated.
3. Examine short-haul modems.	They are non-isolating type.
4. Replace modems with isolating type.	Problem solved.

8.9.3 RS-485, EXAMPLE 1

PROBLEM: RS-485 link between computer and eight motor drives stops working.

ACTION	**RESULT**
1. Check unit.	LED communication lights flash intermittently; computer error says I/O timeout, indicating RS-485 was not getting any acceptable response from the drives.
2. Check diagnostics.	None of the drive diagnostics indicate anything is wrong.

3. Ask if anything had been changed lately.	No changes.
4. Check wiring.	No visible problem.
5. Turn off all the drives and turn them back on one at a time.	When the sixth drive comes on-line, the link stops working.
6. Turn sixth drive off and bring on seven and eight.	Both come on-line OK.
7. Replace sixth drive's communication board.	Problem solved.

8.9.4 RS-485, EXAMPLE 2

PROBLEM: New RS-485 communication link between control rooms is intermittent.

ACTION	**RESULT**
1. Because communication link is between buildings about 300' apart, suspect a grounding problem; check wiring for termination resistors.	Checks OK.
2. Examine drawings.	Unit is four-wire full duplex version of RS-485, not isolated, and master/slaves do not provide isolation.
3. Evaluate possible solutions: 1) buying isolating-type modems and running new cable with a ground wire; 2) a fiber-optic link.	Decide on a fiber-optic link.
4. Install fiber-optic link.	Problem solved.

8.9.5 Fieldbus

PROBLEM: Several loops fail at the same time.

ACTION	RESULT
1. Review system drawings to look for common point.	All loops are connected to C101 concentrator.
2. Check concentrator.	H2 communication card to control house not working.
3. Replace card.	Problem solved.

8.9.6 Programmable Logic Controller, Remote Input-Output (PLC RIO)

PROBLEM: New remote PLC rack was added and now has random rack faults.

ACTION	RESULT
1. Recycle main processor.	Problem clears temporarily but occurs again.
2. Ask construction crew if they moved termination resistor to last rack.	Yes.
3. Replace serial I/O card in new rack.	No change.
4. Replace serial I/O card in main process.	No change.
5. Check all cables.	Cables OK.
6. Double-check termination resistor.	Termination resistor in new rack is there, but the one in the old rack is also still there.
7. Remove old termination resistor.	Rack faults stop. Problem solved.

8.9.7 Communication Loop Has Noise Problems

PROBLEM: Communication loop has been extended and has noise problems.

ACTION	RESULT
1. Check to see if the termination resistors are installed after extension (termination resistors make the communication line appear infinitely long and no reflections will occur when signal hits the end of the cable).	Termination resistors are installed.
2. Chase down the path of the cable.	Find that cable has been run under the floor next to some power cables.
3. Re-route the communication cable.	Problem solved.

8.9.8 Communication Loop Has Noise Problems

PROBLEM: Communication loop has been extended and has noise problems.

ACTION	RESULT
1. Check to see if the termination resistors are installed after extension (termination resistors make the communication line appear infinitely long and no reflections will occur when signal hits the end of the cable).	Termination resistors are installed.
2. Chase down the path of the cable.	Cable does not go near any power cables or other cables that could couple noise into the communication cable.
3. Check new cable connections.	Find several bad connections (loose, dirty, poorly made, moisture, corrosion, etc.).
4. Replace connections.	Problem solved.

8.10 TRANSIENT PROBLEMS

8.10.1 DCS with PC Display

PROBLEM: DCS PC display unit randomly resets itself and has to be reloaded.

ACTION	RESULT
1. Because problem is most likely with the power supply, put a Dranetz power recorder on power.	Transient occurs when DCS printer prints large report.
2. Trace power.	Not wired per drawings; PC on regular receptacle power with printer rather than uninterruptible power supply (UPS).
3. Switch PC to UPS circuit.	Problem solved.

8.10.2 PC Cathode-Ray Tube (CRT)

PROBLEM: PC CRT randomly flickers and is distorted.

ACTION	RESULT
1. Try a new CRT.	No change.
2. Put Dranetz power recorder on power.	Power appears to be OK.
3. Draw "imaginary sphere" around system to help visualize possible causes; because CRT is next to a wall, look on other side of wall. (Circle the wagons.)	Find power distribution panel.
4. Move PC CRT away from distribution panel.	Problem goes away.
5. Relocate CRT.	Problem solved.

8.10.3 Printer Periodically Goes Haywire

ACTION	RESULT
1. Examine printer printout.	Find that the printer periodically (but at a random interval) prints nonsense. Occurs mostly on day shift.
2. The printer is replaced.	Does not solve the problem.
3. Drawing a circle around the printer, there are two primary inputs that can cause the problems – the DCS that is feeding the printer or power supplying the printer.	Since the DCS shows the printer to be OK and there are no other DCS problems, the power is suspected.
4. Place a recording power monitor on the power to the printer.	Sure enough, when the printer prints nonsense, there is a power disturbance.
5. Since the problem occurs several times a day, an instrument tech was assigned to monitor the power on the power monitor.	When the printer started printing nonsense, the instrument tech reviewed all the actions occurring in the control room and associated offices with the occupants. It was discovered the problem occurred when the control room copier was operated.
6. The control room electrical drawings were reviewed.	No obvious connection was noted and it was found that the copier branch circuit did not appear on the drawing.
7. The power circuits were physically traced down.	It was found that the copier and the printer shared the same branch circuit. Further investigation found that the copier outlet had been added by a contractor at the request of the Operations Supervisor at the last minute on another job and the drawings had not been updated. The circuit breaker panel that was added was full and the copier circuit was "added" to the printer branch circuit breaker on the same panel.
8. Placed the copier on its own dedicated branch circuit on another power panel.	Problem solved.

8.11 SOFTWARE

8.11.1 PLC-Controlled Machine Trips

PROBLEM: PLC-controlled machine trips on low-lube oil pressure, but there does not seem to be any problem with lube oil system. Machine restarts OK.

ACTION	RESULT
1. Examine PLC logic to find out what can trip machine.	Low-lube oil pressure wired in series with several other shutdown switches.
2. Look at logic.	Appears to be OK.
3. Test lube oil switch.	Tests OK, and calibrated to value on loop drawing.
4. Check vendor manual for recommended setting, then verify.	Verified.
5. Examine lube-oil running pressure (not too far from trip setting).	Pressure signal is noisy; momentary pressure drops due to noise may be causing trips.
6. Discuss with vendor.	Vendor recommends 3-sec time delay on trip contact in PLC.
7. Program time delay.	Problem solved.

8.11.2 PLC Relay "Race" Problem

PROBLEM: Program change does not operate correctly.

ACTION	RESULT
1. Analyze PLC ladder logic.	Appears to be logically correct.
2. Check input and output circuits.	Circuits OK.

3. Review PLC programming manual.	Logic is scanned in such a way that one contact outraces another, causing the outraced contact to be ignored.
4. Reorganize logic.	Problem solved.

8.11.3 FORTRAN Interface Program

PROBLEM: FORTRAN Driver interface program craters when an additional device (# 17) is added on the communication link. Error dump indicates out-of-range variable.

ACTION	**RESULT**
1. Run program in List mode to find where error occurs.	The array DEVICE (Unit) occurs where error is listed.
2. Check array definition.	Defined for 16 devices; new device is # 17.
3. Adjust array limit to 32; adjust data arrays.	Adjustments made.
4. Compile and test.	Tests OK. Problem solved.

8.12 FLOW METERS

8.12.1 Flow Meter, EXAMPLE 1

PROBLEM: Flow meter reading low on orifice-type flow meter.

ACTION	**RESULT**
1. Verify low reading.	Verified.
2. Check calibration of transmitter.	Calibration OK.
3. Check impulse lines and orifice run flange taps.	Impulse lines and taps OK.
4. Check maintenance records.	Orifice plate has not been checked in 2 years.

5. Remove orifice plate.	Worn down.
6. Ask engineering to check plate material for a plate less likely to wear down.	Engineering recommends different material.
7. Replace with new plate with improved materials.	Problem solved.

8.12.2 Flow Meter, EXAMPLE 2

PROBLEM: Orifice meter does not agree with new turbine meter. (NOTE: Turbine meters are normally more accurate than orifice meters, but they should agree within limits.)

ACTION	**RESULT**
Examine orifice meter and associated drawings.	Appears to work OK and is installed properly.
Examine turbine meter and associated drawings.	Appears to work OK and is installed properly.
Compare flow compensation calculation done by DCS for each meter.	Some parameters do not agree between meters.
Correct parameters.	Meters agree. Problem solved.

8.13 LEVEL METERS

8.13.1 Level Meter (D/P), EXAMPLE 1

PROBLEM: Operator states that the level transmitter is reading incorrectly.

ACTION	**RESULT**

1. Discuss with operator.	He indicates that the level gages and the level transmitter do not agree.
2. Look at DCS trend.	Verify that the level appears to be varying (bottom port does not appear to be plugged).
3. Check transmitter.	Functioning properly.
4. Ask operator to pull a sample and send to lab to determine specific gravity.	Lab results indicate a significant change in specific gravity, enough to account for the indication difference.
5. Ask operator to check operations to determine where the specific gravity problem is coming from.	Find that wrong material was mixed into the tank.
6. Tank cleaned out and proper material lineup made. Level checks out.	Problem solved.

8.13.2 Level Meter (D/P), EXAMPLE 2

PROBLEM: Operator states that the level transmitter is reading incorrectly.

ACTION	**RESULT**
1. Discuss with operator.	Operator indicates that the level gages and the level transmitter do not agree.
2. Look at DCS level trend.	Observe that the level does not seem to be varying (flatlined).
3. In the field, block out the transmitter and bleed both ports.	Level goes to zero on the DCS, which indicates that the bottom port is plugged (if it were the top port the level would still vary but may give an incorrect reading depending on the variation of the head pressure above level).
4. Clean out bottom port either by rod out or by emptying the tank cleaning out the port. Other procedures may be	Tests OK. Problem solved.

available depending on the materials involved and acceptable plant procedures. Place back in service.	

8.13.3 Level Meter (Radar)

PROBLEM: Operator states that the level transmitter readings are varying when they shouldn't.

ACTION	**RESULT**
1. Review DCS trend.	Level is noisy and wandering.
2. Run level device self-diagnostics.	Checks OK.
3. Check Level Device power supply with a scope to see if the level power supply is varying or noisy.	Checks OK.
4. Have operator draw a sample to see if product properties have changes.	Indications of foaming, which could cause level variations.
5. Have Operations check the upstream operations.	Find problems with anti-foaming agent addition.
6. Operational problems fixed.	Problem solved.

8.13.4 Level Meter (Ultrasonic Probe)

PROBLEM: Operator states that the low level ultrasonic probe sensor did not detect a low level.

ACTION	**RESULT**
1. Run level device self-diagnostics	Checks OK.

2. Have Operations notify you when you can pull the probe.	Find probe coated so that it always indicates level.
3. Research problem and find that this can be a common occurrence based on the materials in the tank. This type of probe is inappropriate for this service.	Replace with an ultrasonic sonar type level device mounted on the top of the tank. Problem solved.

9

TROUBLESHOOTING HINTS

Mechanical systems

Process connections

Pneumatic and electronic systems

Grounding

Calibration systems

Programmable electronic systems

Valves

9.1 MECHANICAL SYSTEMS

This chapter contains troubleshooting hints drawn from experience with typical problems. Though they are important to remember, they are not complete descriptions or explanations.

Mechanical systems have links and levers that can become loose, bent, or fall out of calibration. Check mechanical instruments for the following:

•Make sure that all links and levers are straight and secure.

•Make sure the doors on mechanical instruments are always closed tightly.

•Keep all instruments away from vibrations, particularly switches, gauges, and mechanical instruments.

•Watch out for damage caused by over-range transients (bent links or levers or overextended bellows).

9.2 PROCESS CONNECTIONS

Process connections and impulse lines are a major cause of field instrument problems. Here are some tips to bear in mind:

•Small ports can become plugged up, even by apparently clean liquids.

•Keep process tap purges running.

•Arrange connections so that it can be easily determined whether they are plugged, and so that they can be easily rodded out or cleaned out.

•Bubbles, improper fill, and temperature variations in the wet legs of transmitters can cause recurring problems.

•Make sure that instrument manifolds are installed properly and do not get plugged up. They typically have small ports.

•Transmitters too close to hot service will fail sooner rather than later.

•Do not use mechanical snubbers on transmitter inputs. They will plug up.

•Make sure that all tubing is properly supported and that piping is not used to support instruments.

9.3PNEUMATIC SYSTEMS

The reliability of pneumatic systems depends on the quality of the air supplying them. Always precede pneumatic instruments with a coalescing filter (water and particulates). Here are some other hints:

•Pneumatic signals are normally 3–15 psig (21 - 104 kPa) with a supply pressure of 20 psig (138 kPa). The air supply pressure for control valves may range up to 100 psig (689 kPa). Incorrect supply pressure, long lines, leaks, rust, particulates, and insufficient supply capacity will cause pneumatics to malfunction. Lines that are too short can also sometimes cause problems.

•Pressure gauges on field pneumatics should be maintained. They can be of great use during troubleshooting.

•One common problem is a plugged restrictor, which generally causes output to go to zero. Most pneumatic instruments have a removable restrictor that you can clean. Some have filter screens that can become plugged.

•Many pneumatic instruments have a control relay, which has small ports, a diaphragm, and control stem. These commonly get crudded up and stop functioning.

•A plugged relay vent typically causes full-scale output.

•Mud dauber wasps and other insects can build nests in vent holes, which can cause problems for pneumatic systems.

•Many pneumatic systems need a minimum volume in the downstream output tubing. Close coupling may cause oscillations.

•Long pneumatic signal or air supply lines will slow response.

9.4 ELECTRONIC SYSTEMS

With electronic systems, check for simple things first, such as loose terminals, which can cause erratic behavior or loss of power. Here are some additional hints for these systems:

•Problems that occur regularly at certain times of the day may be related to ambient temperature changes.

•Ambient corrosion is a common cause of field problems but can occur in buildings if they use fresh air makeup. Sulfur compounds and moisture are a common cause of corrosion in instrumentation.

•Make sure that all the associated and configurable parameters in a system agree.

•Electrolytic capacitors fail as they get older, and are common problems in electronic equipment.

•Heat and cold are the enemies of all electronic equipment. Avoid exposure to high process temperatures, high ambient temperatures, low temperatures, and direct sunlight. Operating an instrument below its temperature range is also harmful. Avoid cycling temperature.

•Wiring lives by its connections. Loose, corroded, or improper connections are the bane of a wiring system.

•Transients, particularly high dV/dt (fast voltage spikes), will damage thin semiconductor sections, as will high temperatures (either ambient or service).

•Static electricity is dangerous to electronic equipment. It can damage, destroy, and cause electromagnetic interference (EMI).

•Using blown-fuse indicators is a good practice.

•Make sure fuses are coordinated so that the fuse closest to the fault blows first and no upstream fuses are damaged during the fault. Know where your fuses are and what size and type they are. You cannot

troubleshoot a power system failure without knowing where your fuses are.

•Isolate, isolate, isolate! The use of non-isolated equipment and I/O is asking for trouble.

•Dip switches and jumper on cards and modules need to be checked when replacing or installing new devices. Many devices only scan their dip switches and jumpers on start-up (power on) or when a reset has been done.

•Poor power (quality) can have significant effect on sensitive electronics.

9.5 GROUNDING

Some points to keep in mind about grounding are:

•All grounds are not equal. The farther apart the grounds are, the more the difference there will be. The purpose of bonding metal parts and connecting all grounding electrodes together is to minimize this difference.

•At low frequencies, more than one ground in a system can cause problems. Extra grounds create ground loops that cause strange and erratic problems and expose the system to potential damage from lightning strikes and other transients.

•When multiple systems are affected or strange effects are encountered, it's wise to suspect that the ground is involved.

•If you suspect a ground problem, safely lift the known ground and see if the problem improves. This will not help if you have more than two grounds in the system. Always reconnect the ground; if double-grounded, remove the incorrect ground.

•Ungrounded thermocouples being replaced by grounded thermocouples may cause a ground loop if non-isolated transducers or input cards are being used.

•Large amounts of noise on a signal cable may be due to more than one ground in the cable shield system or to the shield being grounded in the wrong place.

•Using an isolating transducer can isolate two grounds in the system.

•Never share a ground return path.

•Never double-ground the neutral.

•Never use the ground as a current-carrying return path.

9.6 CALIBRATION SYSTEMS

Calibrators should be calibrated regularly, at least once a year. The inspection schedule can be lengthened or shortened depending on a calibrator's tendency to drift.

Out-of-calibration calibrators can cause errors in any instruments that they are used to calibrate and thus have the potential of causing widespread errors.

Calibrators should be in good condition before use, and the date they were last calibrated should be shown on the calibrator.

9.7 TOOLS AND TEST EQUIPMENT

Having the appropriate test equipment and tools to troubleshoot a problem is the first step to an efficient troubleshooting process.

Test equipment should be in good condition before use, and the date they were last calibrated should be shown on the test equipment.

All tools should be appropriate for the task. The use of inappropriate tools may damage the instrument system and be a safety hazard.

Develop custom tools and test equipment if necessary to make troubleshooting easier. Make sure, however, that these custom tools are safe to use. Never develop a custom tool if an off-the-shelf one will do the task successfully.

If you don't have the tools or test equipment, ask your company to supply them. If you really need them, it is typically not difficult to justify the tools or test equipment if you can show the same

troubleshooting or maintenance time, which translates into better availability for the instrumentation hence better or more production and reduced maintenance costs (remember, your time is valuable).

Remember - Use the right tool or test equipment for the right job.

9.8 PROGRAMMABLE ELECTRONIC SYSTEMS

Hints to remember about programmable electronic systems include:

•Modern programmable devices typically have diagnostic capabilities that can be read in registers or error message logs. Know what these are. Error messages and register information can be cryptic so if they are not decoded in the manufacturer's documentation then talk with the manufacturer and get the appropriate documents that explain them. These can be invaluable in troubleshooting problems.

•Many PLCs have historical capability (can provide historical trace of PLC I/O and internal contacts and variables) and may also have timing chart capability (trace the timing between contacts). Know what diagnostic tools are provided by the PLC or its software.

•In many programmable electronic systems, the configuration switches and rack configuration (I/O cards) are only scanned on start-up or when reset. For example, if you install a new I/O card, the system may not see the card until you cycle the power on the rack or you reset the rack.

•In programmable systems, things are shared, and a change in one place can affect other places in the system.

•In digital systems, all related system parameters must agree, or be consistent. For example, a range change on a flow transmitter must be reflected into the DCS engineering unit parameters for that flow point, i.e., Transmitter (0-1000 gpm) → Signal 0-100% (4–20mA) → DCS (4–20mA) → DCS (0-1000 gpm).

•Make sure the fail positions are programmed properly. Put them on the loop drawings and P&IDs so that you may recognize a failure to fail position, which will give clues as to what may be wrong. Modern smart transmitters can have a configured failure position, either above 20 mA or less than 4 mA when transmitter diagnostics indicate a transmitter failure. Namur standard NE-43 is one standard that manufacturers use to define failure modes which is shown in the table below.

NAMUR STANDARD NE-43	
4-20 mA	**Normal Operation**
3.8 – 4.0 mA	**Normal Under range**
20.0-20.5 mA	**Normal Over range**
3.6-3.8 mA	**Transmitter failure**
20.5-22.0 mA	**Transmitter failure**

0-3.6 mA	**Wiring problem (open)**
> 22.00 mA	**Wiring problem (short)**

•Digital systems are much more sensitive to power problems than traditional analog electronic instruments.

•Annotated program listings are a must in troubleshooting PLCs, other programmable devices, or computer programs. Comments and functional descriptions are also necessary for efficient program troubleshooting. This can be of particular importance in this day and age of changing workforces. When you document the system, always remember the next guy or the fact that you may be out in the middle of the night two years down the road troubleshooting the system.

•Always back up your work. This is of great importance!

•Document, document, document!

9.9 SERIAL COMMUNICATION LINKS (LOOPS)

There is a considerable amount of information on the Internet on serial communication in the form of tech notes, white papers, articles, and so on.

One way to start to troubleshoot communication loops and most anything for that matter is to ask, Is this a new loop or an existing one? For an existing loop, a good question to ask is, What if anything has changed?

9.9.1 General Considerations

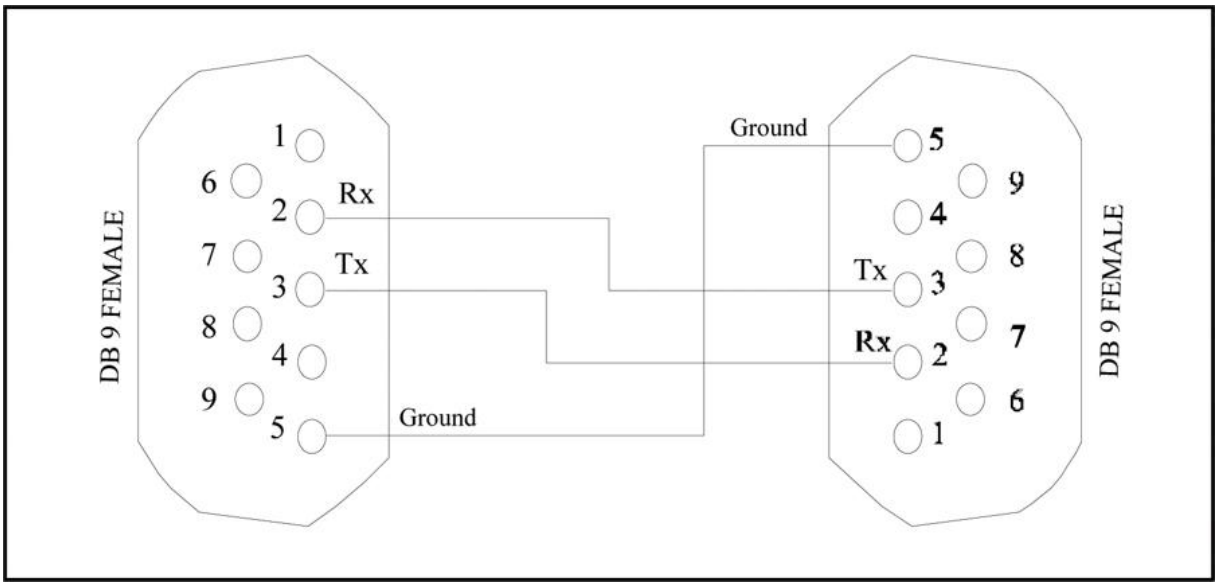
If a malfunctioning communication loop is a new loop, rather than an existing loop that was working and now is not, a multitude of things might be the source of the problem.

For new communication links, some of the common problems include:

Transmit and receive wires are crossed — Different manufacturers connect these differently. For a RS-232 loop, this can be verified using a null modem (see Figure 9-1). A null modem is wired to connect the transmit line of the sending device (DTE) to the receive line of the receiving device (DCE) and the receiving line of the sending device (DTE) to the transmitting line of the receiving device (DCE). There are also inexpensive in-line devices that allow you to re-wire the wiring connections and to monitor voltages on the lines.

FIGURE 9-1

Simple Null Modem Without Handshaking



Incorrect handshaking – RS-232 requires handshaking to control the data flow. The handshaking is commonly provided by the Ready to Send (RTS) and Clear to Send (CTS) lines. This is provided in two ways – 1) The two devices' handshaking lines are connected together (common for modem connections) and 2) The device's handshaking lines are connected back on each other (RTS to CTS) making the loop always ready to receive when ready to transmit (common for PLC connections). The key here is to have the right handshaking per the manufacturer's wiring diagrams. Another type of handshaking in communication loops is software handshaking using Xon and Xoff signals. Problems with this type of handshaking are typically in the devices themselves such as in the software drivers or their configuration. Inexpensive in-line devices are available to provide LED indications of handshaking and transmit/receive signal presence.

Wrong Baud Rate – All the devices on a communication link must be talking at the same transmission speed; otherwise, you get garbage. The Baud Rate is sometimes set by dip switches on the device or in the software configuration of the device.

Wrong Parity – Serial communication protocols commonly use a simple error-checking method called *parity checking* (also known as a Vertical Redundancy Check or VRC). Parity checking is based on adding a 1 bit to the data stream based on whether the data string has an even or odd number of 1s or 0s, or in the case of none parity, of adding no bit. The key here is that all devices that talk to each other must have the same parity setting — odd, even, or none. Even parity is common for asynchronous transmissions, which most serial communication links are.

Wrong number of start and stop bits – Some protocols have start and stop bits that can be configured.

Failure to provide correct cable termination devices – Many communication loops (e.g., RS-485 and RS-422, which are balanced transmission loops, but not RS-232, which is not balanced) have termination devices at both ends of the cable, commonly a resistor with a resistance equal to the characteristic impedance of the cable (not the impedance of the cable but the intrinsic

impedance property of the cable itself). Sometimes a resistor and a capacitor is used. These devices make the cable appear infinitely long and minimize reflections that can cause transmission errors. While

it is always a good practice to provide the recommended termination device, this is primarily a function of cable length and transmission speed. The longer the cable and/or the higher the transmission speed, the more important the termination devices are. Some good information on this can be found at:

http://www.maxim-ic.com/appnotes.cfm/appnote_number/763.

Cables or drops too long – Each communication standard has a maximum cable length, which is generally dependent on transmission speed and cable characteristics. RS-232 is commonly quoted at 50 ft. (15 meters) but is actually dependent on cable capacitance and speed and much higher lengths can be achieved with the right cable. RS-485 is commonly quoted at 4000 ft. (1200 meters) but the length is really a function of the transmission speed, cable characteristics, and transmitter characteristics. The manufacturer will provide guidelines on cable types and length. Sometimes in communication loops, a multidrop arrangement will be used that has drop cables off of the main truck cable. Follow the manufacturer's guidelines on these and make sure that the drop connections are properly made up.

Improper grounding – Bad grounding will get you into trouble anywhere. RS-485 can be the worst as many people consider it a differential input (reads voltage between two lines and not ground) and only two wires are required but this is not the case. For transmissions of any distance a ground wire is required. See the application note at

http://www.robustdc.com/library/san005.html for a good description of this issue.

Wrong address – Communication protocols are in the business of transmitting data from devices to other devices. Each device generally has a device address that the protocol uses to identify what device it is talking to and what device is talking. If one device (say address #1) sends a request to device #3 but there is no device #3 or it really wanted to talk to device #2, a communication failure will occur. The same thing applies to data addresses down inside a device. Talk to the wrong address, and you get the wrong data or no data at all.

Driver Mismatch – Manufacturers seem to take great pleasure in tinkering with standard communication protocols and wiring standards. They use non-standard terminal connections, different voltage levels, strange addressing, different timing, etc. This makes connecting devices of two manufacturers a crapshoot sometimes. In these cases, the best thing is for Engineering to have done their homework up front so you won't have this problem. However, that is not always going to be the case (this is where front-end loading input from Maintenance can be very important). The only thing you can do is make sure everything is OK on your end and go to each manufacturer for help. Typically, however, it is common for one manufacturer to blame the other manufacturer. This is where you need a firm hand. Don't let either manufacturer off the hook.

Babble – The devices that need to talk to each other not only must have the correct electrical signals but must also talk the same data interchange protocol (and in the same way). They must also respond to the data requests and send back the proper data. For example, some communication bridges and network interface devices commonly have buffers where you have to store data from the source (with its protocol and tag identifiers) and convert it to the destination data (with its protocol and tag identifiers). If this buffer is not configured right, the source data will not get to the destination. Another example is the Modbus protocol. This is a general-purpose protocol that manufacturers sometimes implement in different ways. For example, in true Modbus, the data addresses are offset by one, but some

manufacturers of the slave devices may not use this offset. This can lead to getting data that makes no sense because it came from the wrong address.

For existing communication links without changes, it is unlikely that the loop will be miswired or that the communication parameters are wrong. Some of the problems can be:

Device failure – One of the communication devices has had a hardware failure. Lightning is a common cause of communication link damage. This type of damage is typically caused by poor grounding.

Degraded installation – This is where the installation has been degraded by corrosion, moisture, abuse, and so on. A thorough inspection of the installation including the grounding is a good place to start.

Power problems – Noise or transients from the power supply. Poor grounding (as you may note, this appears as a number of problem causes).

Noise from a changed electromagnetic environment – New wiring not related to the communication loop can couple noise into the loop. Addition of radio sources is a potential source of this. Poor shielding and grounding can affect this.

For existing communication links with changes, the potential causes are similar to a new installation but of a narrower scope related to the change. Changes are not always apparent and usually have to be dug out.

9.9.2 Modbus

Modbus is a common generic communication protocol used by many manufacturers in various forms. Sometimes the manufacturers call it Modbus and sometimes they have their own name. Sometimes the manufacturers stay pure to the original Modbus spec and sometimes they tinker with it. Modbus is a master/slave arrangement where the Master (only one) does the commanding and the Slave (multiple) responds to the Master's requests. The best place for information on Modbus is www.modbus.com. The potential problems in Section 9.9.1 all apply to Modbus installations.

Some of the problems that can be encountered with Modbus include the following:

Wrong type of Modbus – There are two types of Modbus: RTU (binary) and ASCII (based on ASCII characters). RTU is the most common but if the master and slave are talking different types of Modbus, they will not understand each other.

Addressing – Different Modbus devices sometimes use different addressing schemes. The original Modbus addressing was based on addressing in the Modicon PLC. There are two issues here. First the Modbus driver will use an addressing scheme, typically either Modicon based (Oxxxx – outputs, lxxxx – inputs, 3xxxx – input registers, and 4xxxx – internal registers) or sequential addressing (0-xxxxx for internal coils and outputs). The actual Modbus transmission frame uses a command number that identifies data type and sequential addressing (xxxx) for the actual address. Use of a Modicon based driver for non-Modicon devices may require faking the addresses to get the correct data. Second, where it gets tricky is that the original Modbus addressing is offset by one. So for an internal register if a zero is sent in the address field, a Modicon PLC will recognize this as a command related to 40001 or a non-Modicon device that uses the original style Modbus addressing (but not Modicon type identifications)

will recognize this as address 1 and not 0. Some manufacturers do not use this offset and in this case, you may configure the Modbus driver for one address thinking that there is an offset and get data from a place offset by one from the desired address.

9.9.3 Communication Information Sources

A good library of tech notes on communication links can be found at:

http://www.bb-elec.com/technical_library.asp

and

http://www.robustdc.com/techResources-appnotes.htm?a=8

9.10 SAFETY INSTRUMENTED SYSTEMS (SIS)

Working on safety instrumented systems or SIS (also known as ESD systems, interlocks, safety shutdown systems, and so on) presents additional challenges for the Maintenance department. Here are some hints regarding troubleshooting these systems:

•If you are not trained to work on the specific SIS, don't work on it!

•Make sure that you are on the right loop. Getting on the wrong loop can cause a spurious trip of that loop or can render that safety loop unavailable to perform its safety function.

•SIS loops are generally better documented historically, and this documentation may help you understand the problems that the loop in question or SIS loops in general have encountered in the past. Look in the SIS loop equipment file for this information.

•SIS loops may have more diagnostics, both from the instruments themselves and designed in by the SIS designers. Know them and use them.

•Follow your SIS maintenance procedures.

•Follow your bypassing procedure.

•Document what you found wrong. It is essential to track failures of SIS equipment to ensure that the failure rates assumed in the calculation of probability of failure on demand are appropriate and that inappropriate equipment is not used for SIS.

•Make sure that the SIS loop is returned to service properly and that all bypasses have been removed. This is commonly controlled by a checklist. Otherwise, you will have a safety system that is not in service and your plant will not be protected.

•Changes are not allowed in SIS without MOC.

9.11 CRITICAL INSTRUMENT LOOPS

Here are some hints for working on critical instruments:

•Make sure that the critical instrument loop is returned to service properly.

•Document what you found wrong. Failure tracking of the equipment used in safety-related systems is important because these loops have a qualitative requirement for dependability. Instruments that are not considered dependable should not be used in critical instrument loops.

•Changes are not allowed in critical instrument loops that have been identified as being independent layers of protection without MOC.

•Work on critical instrument loops including troubleshooting should be done in a timely manner to help assure the availability of the critical instrument loop.

9.12 ELECTROMAGNETIC INTERFERENCE

Remember the following hints about electromagnetic interference: There are four types of EMI: Electrostatic (capacitive coupled), magnetic (transformer coupled), radiated (through the air), and conducted (through wires and conductive materials). All EMI becomes conducted once it enters the circuit.

Electrostatic noise is electric field (voltage) based and is capacitively coupled into a system. Higher voltage lines close to low voltage lines can couple this EMI, for example, 120 VAC near 24 VDC or thermocouple lines. Separation, orientation (90°), or any grounded metal shield (grounded only one place at the zero potential point of the circuit at low to medium frequencies) will help shield against this noise.

Magnetic noise is based on current and is coupled into a system by inductive or transformer effect. Current in a power circuit can couple into a signal circuit. Separation, orientation (90°), magnetic material, or twisted pair will help shield against this noise.

Radiated noise can come from radios, lightning, or in some cases arcs or sparks. Generally speaking, any self-supporting metal enclosure will protect against radiated noise as long as there are no holes larger than 1/20 of the wavelength of the noise.

Conducted noise can come from the other three sources of noise or be generated internally by the electrical circuit by non-linear devices and switching transients. Once the noise is in the circuit, filters, ferrite beads, common mode chokes, and twisted pair are some of the methods that can be used to reduce the noise.

Most methods used for reducing EMI work equally well for both the source and target.

EMI reduces rapidly with distance, both in the air and when conducted. The higher the frequency the more rapid the reduction is in conducted noise due to wires appearing as inductors at high frequency.

Improper grounding of shields or the circuit is a common problem. At frequencies less than 100 Khz, ground a shield in only one place – the zero potential (reference) point of the circuit. Generally speaking, using multiple grounds at low frequency is asking for trouble. Ground does not dissipate noise nor is there any such thing as a quiet ground! Remember, noise like electricity works complete circuits.

The key to troubleshooting an EMI problem is to identify the source of the noise and its entry into the system. This can be done by identifying the amplitude, frequency, duration, timing, and shape of the noise. For example, if the noise is high frequency, then the source is unlikely to be a 120-VAC 60 Hz line. On the other hand, if the noise is a multiple of 60 Hz, then a non-linear device such as a variable speed drive is likely to be the culprit. Transients conducted for short durations are likely to be switching transients. Sixty Hertz noise goes a lot further in a system than 1 MHz noise so amplitude and frequency can give you some insight as to the general location of the noise source. EMI from a lightning storm can be on transients on the power lines, radiated through the air, and due to rapid ground potential variations.

The first step in troubleshooting a noise problem is to use test equipment to view the noise and to identify its characteristics. The second step is a good inspection of the system and wiring that is the target of the noise. Look for the target system's proximity to potential sources. The timing and duration of the noise can help pinpoint the source. If the noise is continuous, the source must be continuously inputting noise into the target system. If the noise is in transient, then the source must also be transient though not necessarily random. Regularly switching transients can be tied to devices that switch regularly, such as switches, relays, contactors, motor starters, and so on.

The following books are good resources on this subject. Later versions of these books may be available.

1.*Noise Reduction Techniques in Electronic Systems, 2nd ed.*, Henry Ott, Wiley Interscience, ISBN: 0-471-85068-3.

2.*Grounding and Shielding Techniques in Instrumentation, 3rd ed.*, Ralph Morrison, Wiley Interscience, ISBN: 0-471-83805-5.

3.*EMI Troubleshooting Techniques*, Michel Mardiquian, McGraw-Hill, ISBN: 0-07-134418-7.

4.*Grounding and Bonding*, Michel Mardiguian, Interference Technologies, Inc., ISBN: 0-944916-02-3 (on grounding in general).

9.13 VALVES

Remember the following hints about valves:

•If a valve sounds like it is passing gravel, the problem is cavitation (liquid converting to gas and then collapsing). High noise can be a symptom of flashing (liquid converting to gas).

•If you cannot get the expected output through a valve, suspect choking. This occurs when the downstream pressure is approximately one half or less of the upstream pressure. This can be caused by flashing (liquid changing to vapor) or by reaching sonic velocity (gas) — commonly called *choking.*

•High velocities in a valve can cause erosion and wire drawing. This can occur when a valve is operated close to its seat where high velocities can occur.

•Improperly sized valves cause controllability problems, as can operating the valve at its high or low extremes.

•Sticking valves are a major problem in control loops.

•Make sure valves have sufficient air capacity to operate: 3–15 psig. (21–104 kPa) instrument signals typically do not have much capacity.

•Valves that do not operate regularly need exercising. If they require oil, have the oilers on a preventative maintenance program. Always exercise shutdown valves during outages.

•New valve actuators that require oil should be exercised 10 to 15 times with an oiler before installation.

•Solenoids with small ports can be trouble. Solenoids also have temperature ranges to consider (both high and low temperature). Cracked bypass valves can be a sign of an undersized valve or control problems.

•A visual verification that a safety valve is closed is not by itself a 100% test that it is in fact closed. This does not provide assurance that the valve fully closed. Tight Shutoff (TSO) valves require further testing to assure TSO.

9.14 MISCELLANEOUS

•Good documentation is essential to successful troubleshooting. Poor documentation not only leads to difficulties but can also be dangerous. The process of field tagging instruments, equipment, and wiring is part of the documentation system and should match the drawings and be maintained in good order. A good tagging system should allow a technician to move around a system or circuit even if the drawings are not up to date or are in error.

•Changes to a system can introduce problems. Undocumented changes can cause difficulties when you are trying to troubleshoot with out-of-date documentation. Always ensure that as-builts are picked up on the drawings.

•Orifices need to be checked, even on clean service. Put them on a preventative maintenance cycle based on the type of service.

•Differential pressure types of level transmitters depend on the material's density, which is affected by temperature and the material's composition.

•Even though thermocouples are simple devices, they can go bad.

•Type K thermocouples are affected in a reducing atmosphere by green rot. All thermocouples can suffer problems at their junction due to corrosion, material migration, and damage due to vibration (smaller ones are more sensitive). RTDs are also sensitive to vibration due to their small wires.

•All instruments are subject to environmental damage if their enclosures are not secure.

•Noise and transients can come from arcing contacts, welders, lightning, ground transients, switching, faults, and other wires nearby.

•Dead time is a major problem in control loops and comes primarily from the control loop sensor being far away from the point where the loop is actually controlled (the control valve).

•Always make sure that the sensor is measuring a representative sample of what it is supposed to measure.

•The longer the lines are between the process tap and the process measurement, the more likely it is that problems will occur.

10

AIDS TO TROUBLESHOOTING

Maintainability

Drawings

Tagging/identification

Equipment files

Manuals

Maintenance management systems

Vendor technical assistance

Direct vendor access

10.1 INTRODUCTION

With today's complex and sophisticated systems, it is impossible for anyone to keep track of all the details in a facility. Many systems have documentation and other aids to help in troubleshooting. It is essential for most of these aids to contain detailed information about the system and its functions. For other aids, the key can be access to external knowledge. Knowing how to use these aids efficiently when troubleshooting can substantially increase your troubleshooting abilities and rate of success.

10.2 MAINTAINABILITY

Maintainability is an inherent characteristic of a design or installation that determines the ease, economy, safety, and accuracy with which maintenance actions can be performed. This also includes ease of troubleshooting. The design of a system for maintainability is not often under a technician's control, but the maintenance department should have considerable input in design activities. Regular feedback should be provided to the design or engineering group regarding maintainability issues. In addition, field modifications may be made by the maintenance department to improve system maintainability. Safety should always be considered when making field modifications; significant changes should go through a management of change (MOC) process.

Systems should be designed to be accessible for safe and efficient work. They must also allow efficient testing and troubleshooting. Once the cause of a problem has been determined, well-designed systems allow the repair to be done efficiently.

Remember, maintainability is not only the responsibility of engineering; it is everyone's responsibility. If you cannot work on something safely or efficiently, consider making changes in the system.

Maintainability consists of the following:

- Safety
- Accessibility
- Testability
- Reparability
- Economy
- Accuracy

10.2.1Safety

One aspect of a safe system is that it is designed so that no unsafe act is required for maintenance activities. Exposure to energized, hot, and sharp or pointed surfaces must be minimized. Head-knockers, trip hazards, awkward actions, and pinch points should be eliminated. The system should be designed with ergometrics (human factors) in mind. Analyze potential human errors during maintenance to identify and minimize potential error points.

10.2.2 Accessibility

Accessibility includes providing both adequate physical access and lighting to perform maintenance. The National Electrical Code (NEC) Article 110 provides code access requirements. Basically, you should not have to be a contortionist to get to parts of the system that need maintenance, nor should you put yourself at unacceptable risk while doing troubleshooting activities. You also need a level of lighting adequate to see the equipment. You must also consider egress—can you leave quickly if an unsafe condition occurs?

10.2.3 Testability

The ability of a system to be tested includes access to areas that are to be tested and test points that allow you to check the system. A testable system also allows ease of testing when built-in test points are not readily accessible and includes designed-in diagnostics, lights, telltales, indicators, trend indicators, and alarms that help identify where the cause of failure lies.

10.2.4 Reparability

Reparability is the ability to repair the system efficiently and effectively. Reparability can include allowing for access to remove and replace parts, access to bolts, the number of bolts, crane access for heavy parts, platform access, and access height. Availability of spare parts is also a consideration. Common parts may be kept locally or in on-site storehouse stock. On-site vendor consignment stocking may also be considered. Off-site availability of critical parts must also be considered.

10.2.5 Economy

Systems should be designed to be economical to troubleshoot and repair.

10.2.6 Accuracy

Systems should be designed to be repairable exactly as the original equipment (i.e., returned to services good as new).

RELEVANT STANDARDS

•ISA-5.1-1984/R 1992 - Instrumentation Symbols and Identification.

•ISA-5.4 -1991 - Instrument Loop Diagrams.

10.3 DRAWINGS

Drawings provide the troubleshooter with a map of the system. Just as a road map can tell you how to get somewhere, drawings can get you to places in the system you are troubleshooting. And just as an

inaccurate map can get you lost, so can incorrect drawings. Incorrect drawings are common- place, so take care, and when you find errors, turn them in to be corrected. Piping and instrumentation drawings (P&IDs) and electrical one-line drawings provide an overall view of systems. They show how the system you are troubleshooting interacts with other systems and fits into the big picture. The two primary troubleshooting drawings are loop drawings and motor control schematics (see Figures 10-1, 10-2, and 10-3). These drawings show point-to-point connections and wiring and provide equipment details.

FIGURE 10-1 ***Pneumatic Loop Drawing Example***

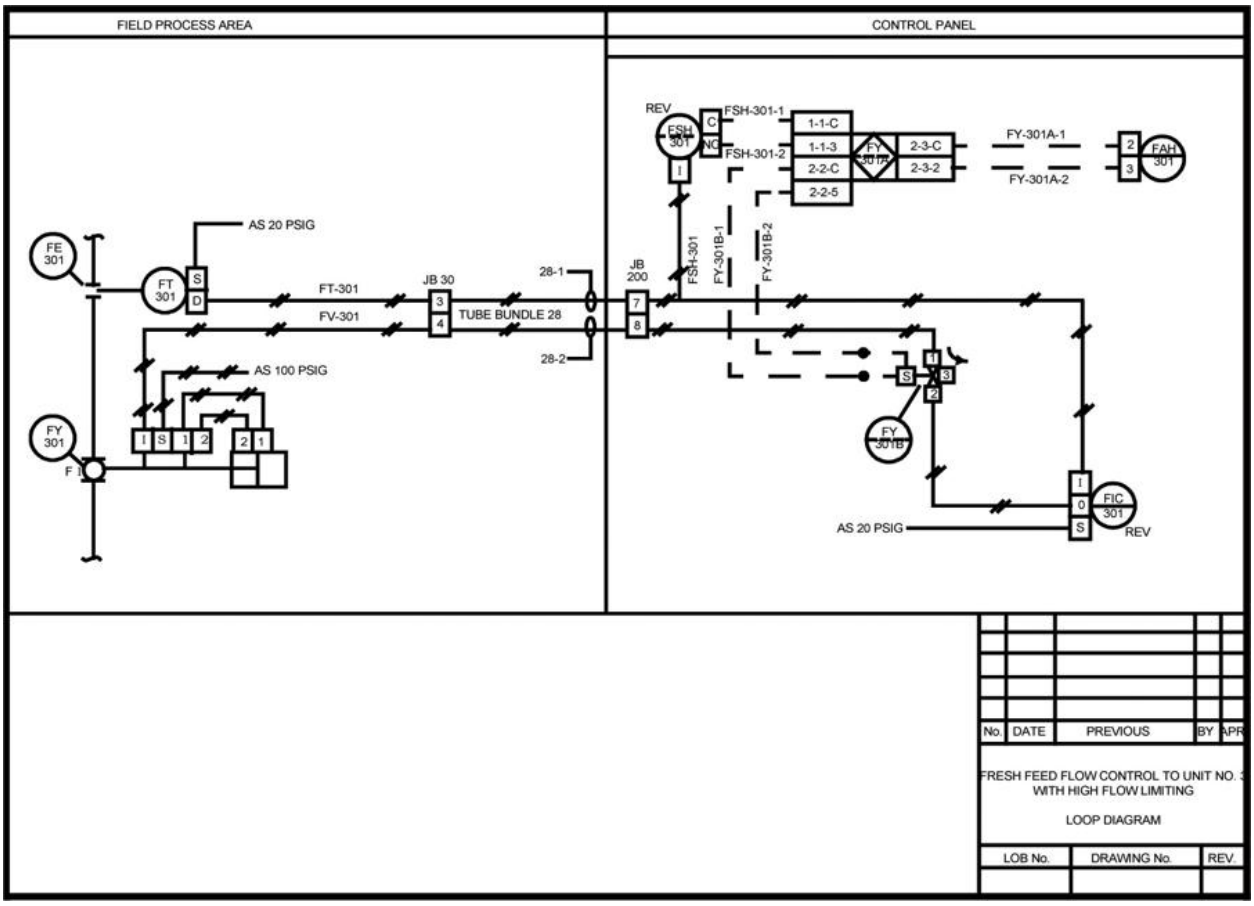
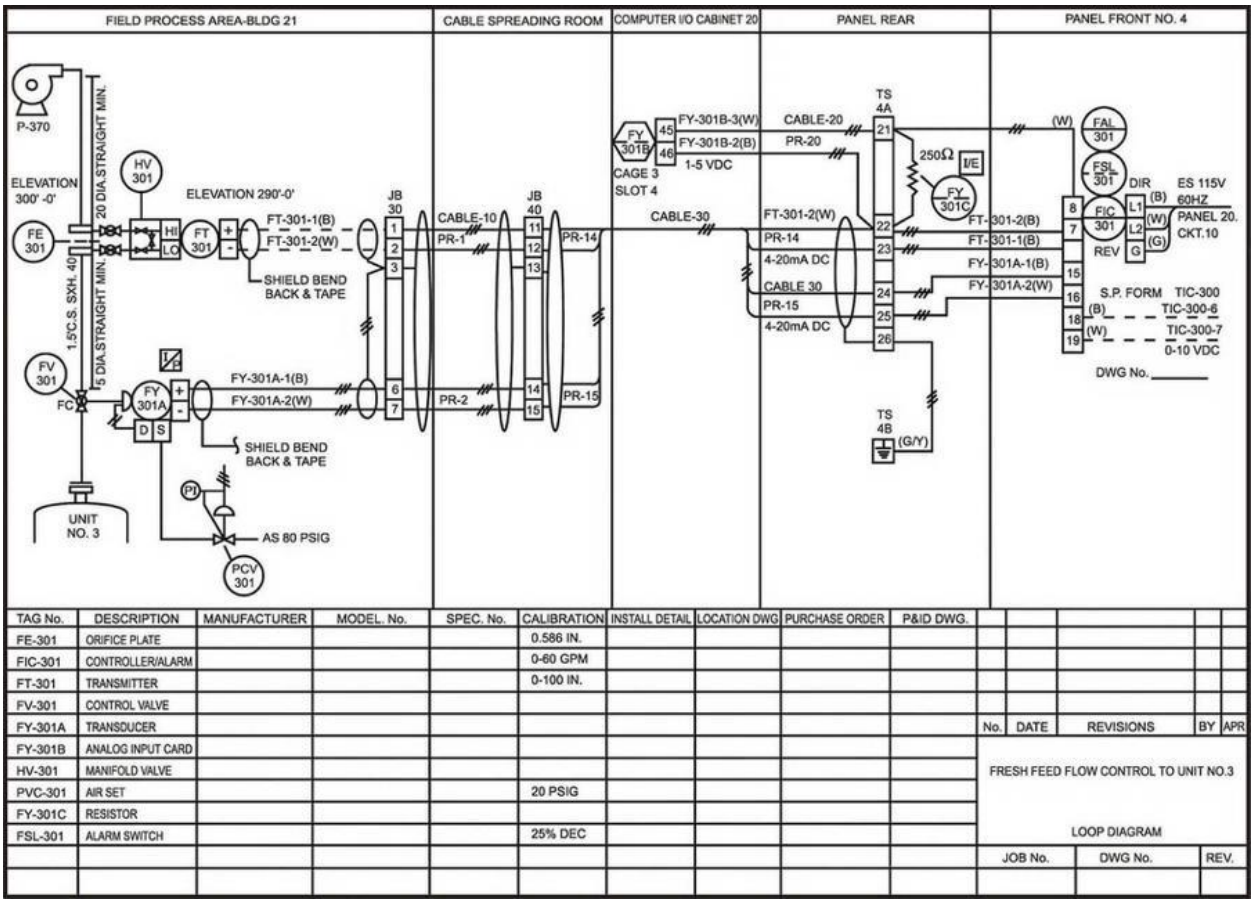


FIGURE 10-2 ***Electronic Loop Drawing Example***



TAG No.	DESCRIPTION	MANUFACTURER	MODEL. No.	SPEC. No.	CALIBRATION	INSTALL DETAIL	LOCATION DWG	PURCHASE ORDER	P&ID DWG.
FE-301	ORIFICE PLATE				0.586 IN.				
FIC-301	CONTROLLER/ALARM				0-60 GPM				
FT-301	TRANSMITTER				0-100 IN.				
FV-301	CONTROL VALVE								
FY-301A	TRANSDUCER								
FY-301B	ANALOG INPUT CARD								
HV-301	MANIFOLD VALVE								
PVC-301	AIR SET				20 PSIG				
FY-301C	RESISTOR								
FSL-301	ALARM SWITCH				25% DEC				

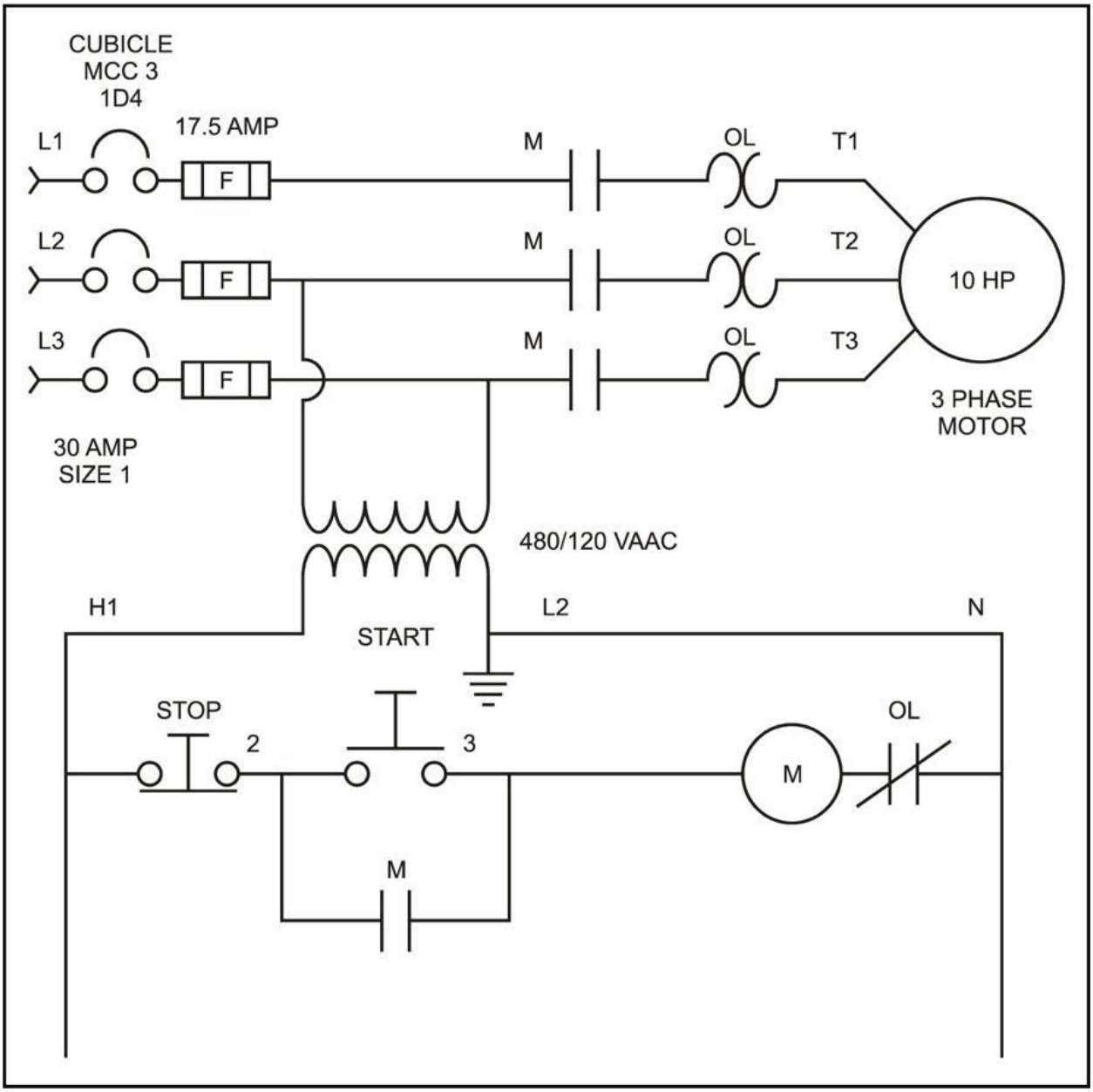
No.	DATE	REVISIONS	BY	APR

FRESH FEED FLOW CONTROL TO UNIT NO.3

LOOP DIAGRAM

JOB No.	DWG No.	REV.

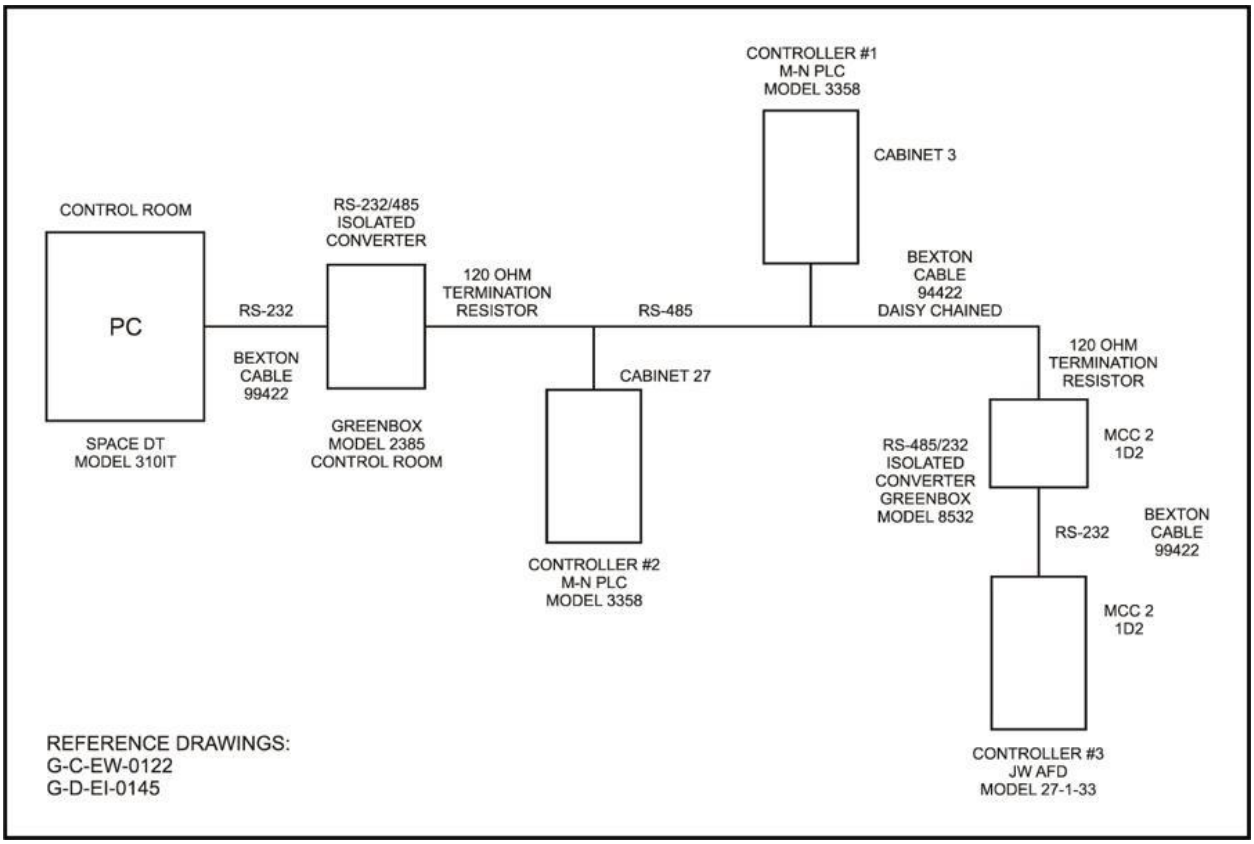
FIGURE 10-3 ***Motor Control Schematic Drawing***



Some systems, particularly those from original equipment manufacturers (OEMs), may also have other types of wiring diagrams or mechanical drawings.

Examples include compressor skid, packaged equipment, and panel fabrication and layout drawings. Complex systems may have an overall system drawing (see Figure 10-4).

FIGURE 10-4 ***System Drawing Example***



Know your way around the drawing system and how to find drawings. Spending excessive time looking for drawings is a waste and can seriously impact repair times.

10.4TAGGING AND IDENTIFICATION

Tagging (device wiring and equipment identification) is an extension of the drawing documentation system. Tags identify such things as equipment, wires, cables, switches, and boxes. In a good tagging and identification system, you should be able to use basic system knowledge to move around the system without the benefit of drawings. You should always know to which system wires or components belong. Here are some examples of tagging:

•Loop wire tag 80F301-1, for plant 80, flow loop 301 in section 300, wire 1.

•Motor loop tag GM501-3, for plant G, equipment type M, number 501, wire 3.

•Terminal strip in field junction box I-3-2, for instrument box 3, terminal strip 2, connected to a terminal strip in a marshalling or main terminal strip I-3-2.

An example of equipment identification would be a tag on an instrument identifying it as FT-301, or a power switch that is tagged with the instrument it powers and the power box and circuit number that supplies it.

Tagging and identification should match up with what is shown on the drawings. Tagging and identification help ensure that you are on the right system or part of a system and can be a great advantage in troubleshooting.

10.5 EQUIPMENT FILES

Your plant should keep equipment files on all major pieces of equipment. In some cases, they are kept down to the loop level. These can benefit you because this is where the equipment history is kept, as well as user and vendor drawings, manuals, and other associated data. A well-kept equipment file system can go a long way toward improving overall maintenance efficiency.

10.6 MANUALS

Vendor manuals are essential to work successfully on equipment. Yet many times equipment manuals cannot be found. Making sure that the manuals are acquired in the first place and that people return them after use is a matter of discipline. If equipment you troubleshoot does not have manuals, ask your supervisor to get them.

Make sure that you get all the manuals associated with the equipment, typically a user manual, an installation guide, and a maintenance manual; there may be other specialized manuals as well. Complicated systems may have a whole series of manuals. By the late 1990s, manuals were often available on the Internet, and sometimes through a fax-on-demand system as well or on CD.

Many vendors also supply drawings. In some cases, these may be certified drawings for a particular system. Strongly consider filing these drawings with the normal system drawings so they will not become misplaced.

10.7 MAINTENANCE MANAGEMENT SYSTEMS

Computerized maintenance management systems (MMSs) became increasingly popular in the late 1990s. For troubleshooting purposes, it serves as the historian for a facility. With an MMS, you can quickly find out the history of the system or instrument on which you are working. This can help determine if you need a specific or a more general solution. An MMS can help you spot failure trends and common-cause failure mechanisms. It will be hard, if not impossible, to do serious reliability improvement in a facility without a good MMS.

While an MMS may be a manual system, use a computerized system to get the most benefit out of it. But remember: an MMS is only as good as the information put into it. The old computer adage garbage in, garbage out applies to MMSs.

10.8 VENDOR TECHNICAL ASSISTANCE

Vendors can provide technical assistance remotely or on-site. Vendors suffer the same staff constraints that user companies do, and the level of available technical assistance often suffers. Finding the right person to help can be difficult. Once you find a good technician or engineer, keep his or her name in your records. If you get an on-site visit, request someone you know is good. Work with the vendor service person during the visit and get as much training as you can. Many times, vendor service persons are willing to pass along a good deal of wisdom to those who are interested.

Do not overlook vendor representatives, particularly distributors, as they may have people on staff locally who can be of assistance. Many times information beyond that contained in the manuals is available from the vendor. Always ask if they have troubleshooting guides, application notes, or other materials that may help maintain the system.

Doing business with companies that provide good technical support is an obvious good practice, but be careful that the technical support does not revolve around just one person. If that person leaves, the technical support may decrease substantially.

10.9 DIRECT VENDOR ACCESS

For today's sophisticated equipment, the vendor may be able to troubleshoot equipment by dialing in over telephone lines, through a modem, or over a wide area network (WAN). This can be very helpful in solving difficult problems. There is a risk, however: when vendors dial in on a running system or a computer system with multiple functions, the system may be unintentionally compromised. System security must be a concern here as you are giving an outsider access to your system. Use this option with great care.

10.10 MAINTENANCE CONTRACTS

For today's sophisticated equipment, maintenance contracts with the vendor or a third party are not uncommon. These are encouraged if they are cost effective, particularly for systems that are new to the facility (can allow for a learning curve). These may include on-site support, phone support, Internet support, and/or e-mail support. If you use these, learn from them about the system and how to troubleshoot them as you never know when the bean counters will do away with the contracts, and you will be stuck with troubleshooting and maintaining the system.

SUMMARY

To be successful at troubleshooting, all available resources must be used. The aids discussed here are just some of the resources you can draw on. Look for new ways to improve your troubleshooting skills. Continuous improvement is the way to go.

QUIZ

1. Maintainability includes which of the following?

A.accessibility

B.testability

C.safety

D.all of the above

2. Accessibility includes

A.egress (means and ability to leave an area).

B.lighting.

C.ability to get to work areas.

D.all of the above

3. Which of the following drawings show the big picture?

A.loop drawings

B.motor schematics

C.P&IDs

D.wiring diagrams

4. Tagging and identification are extensions of the drawing system.

A.true

B.false

5. MMS stands for

A.maintenance monitoring system.

B.maintenance management system.

C.maintenance message system.

D.none of the above

6. Direct vendor access is

A.calling the vendor on the phone.

B.the vendor accessing the equipment through a modem or WAN.

C.the vendor coming out and directly working on the equipment.

D.all of the above

7. Manuals are

A.sometimes available on the Internet.

B.sometimes available via CD and fax-on-demand systems.

C.in equipment files.

D.all of the above

8. Testability includes

A.test points.

B.accessibility.

C.diagnostics.

D.all of the above

9. A successful MMS system depends on

A.accurate information.

B.a computer.

C.drawings.

D.none of the above

10. The responsibility for a maintainable system rests with

A.engineering.

B.everybody.

C.your supervisor.

D.the MMS system.

Appendix A

ANSWERS TO QUIZZES

Chapter 1

1-D, 2-B, 3-A, 4-C, 5-B

Chapter 2

1-A, 2-D, 3-B, 4-C, 5-D

Chapter 3

1-A, 2-B, 3-D, 4-B, 5-D

Chapter 4

1-TRUE, 2-A, 3-C, 4-D, 5-C, 6-C, 7-B, 8-A, 9-B, 10-D

Chapter 5

1-C, 2-B, 3-A, 4-C, 5-D

Chapter 6

1-C, 2-D, 3-B, 4-D, 5-A, 6-C, 7-B, 8-D, 9-C, 10-C, 11-TRUE, 12-D, 13-C, 14-A, 15-D, 16-C, 17-D, 18-B, 19-C, 20-D, 21-That you are working on the right equipment, 22-FALSE, 23-TRUE, 24-PPE = Personal Protective Equipment, 25-In troubleshooting, you commonly work on energized or moving equipment

Chapter 7

1-A, 2-D, 3-D, 4-B, 5-C, 6-C, 7-D, 8-A, 9-A, 10-D

Chapter 10

1-D, 2-D, 3-C, 4-True, 5-B, 6-B, 7-D, 8-D, 9-A, 10-B

Appendix B

RELEVANT STANDARDS

American Petroleum Institute API RP 500, Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities.

ANSI/IEEE 43 - IEEE Recommended Practice for Testing Insulation of Rotating Machinery.

ANSI/ISA-12.01.01-1999—Definitions and Information Pertaining to Electrical Apparatus in Hazardous (Classified) Locations.

ANSI/ISA-84.00.01-2004—Functional Safety: Safety Instrumented Systems for the Process Industry Sector.

ANSI/ISA-84.01-1996—Application of Safety Instrumented System for the Process Industries.

ANSI/UL 913—Standard for Intrinsic Safe Apparatus and Associated Apparatus for Use in Class I, II, III, Division I Hazardous (Classified) Locations.

IEC -61010 - Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use.

IEC 61508, "Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems."

IEEE 95 - IEEE Recommended Practice for Insulation Testing of Large AC Rotating Machinery with High DC Voltage.

ISA-RP12.4-1996—Pressurized Enclosures.

ISA-RP12.06.01-2003—Recommended Practice for Wiring Methods for Hazardous (Classified) Locations Instrumentation Part I: Intrinsic Safety.

ISA-12.10-1998—Area Classification in Hazardous (Classified) Dust Locations.

ANSI/ISA-12.12.01-2000 Nonincendive Electrical Equipment for Use in Class I and II, Division 2 and Class III, Divisions 1 and 2 Hazardous (Classified) Locations.

ISA-5.4 -1991 - Instrument Loop Diagrams. NEC Article 110-127a, Guarding of Live Parts.

NEC Article 500, Hazardous (Classified) Locations, defines division-based area classification.

NEC Articles 501-555 further explain the requirements for the use of electrical equipment in hazardous (classified) areas.

NEC Article 505, Class I, Zone 0, 1, and 2 Locations defines the zone-based area classification.

NFPA-70E, Standard for Electrical Safety Requirements for Employee Workplaces, NFPA-70, The National Electrical Code (NEC).

NFPA 79, Electrical Standard for Industrial Machinery, 2002 Edition.

NFPA-101, Life Safety Code, Electrical Safety Requirements for Employee Work.

NFPA 496—Purged and Pressurized Enclosures for Electrical Equipment.

NFPA 497A, Classification of Class I Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas, and American Petroleum Institute API RP 500, Recommended Practice for Classification of Locations for Electrical Installations at Petroleum Facilities.

For Class II (dust) areas, the recommended practices are NFPA 497B, Classification of Class II Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas, and ISA-12.01.01-1999—Definitions and Information Pertaining to Electrical Apparatus in Hazardous (Classified) Locations (an excellent resource with pictures).

OSHA Code of Federal Regulations: Title 29, Chapter XVII, Part 1910,

Subpart S, Electrical.

UL 3111- Electrical Measuring and Test Equipment.

Appendix C

GLOSSARY

A

AC—alternating current, a type of electricity whose voltage varies at a constant sinusoidal rate

administrative controls—controls placed on activities through the use of permits, standard operating procedures, standard maintenance procedures or practices, supervision, etc. to insure safe operation and maintenance of the facility as well as maintain normal operations

analytical—the use of logic or other methodologies to analyze

approved—1) acceptable to the authority having jurisdiction (AHJ); **2**) tested and certified by a national testing laboratory such as Underwriter Laboratories (UL) or Factory Mutual (FM); UL lists while FM approves equipment

authority having jurisdiction (AHJ)—1) organization, office, or individual responsible for approving equipment, an installation, or procedure (NFPA); **2**) acceptable to the Occupational Safety and Health Administration (OSHA)

autoignition temperature (AIT)—the temperature at which a hot component or surface can ignite a flammable mixture

availability—the fractional uptime of a system or process, expressed as a percentage, i.e., a system is 90% available

B

bathtub curve—a reliability curve shaped like a bathtub that plots failure rate (λ) on the y-axis against operating time on the x-axis; commonly applies to electronic and electrical instruments and equipment

blowdown—the process of venting process material from a primary element, impulse lines, or an instrument; because of the hazards involved, most facilities have safety procedures governing how to do this

board—slang for a control panel; in distributed control systems it means control room display instrumentation

board operator—control room operator

breakout box—a communication troubleshooting device that connects in series with RS-232 circuits and has diagnostic lights, switches, and wire jumpers

breathing—air moving in and out of an instrument or piece of equipment due to changes in ambient temperature or pressure

bucket truck—a truck that has a lifting mechanism with a bucket than can contain people; used to lift people to a work area

burn-in—a process used by manufacturers to expose instruments to elevated and in some cases cyclical temperatures to find infant mortality failures

burnout, down-scale—directing an instrument to fail at its lower scale when it detects a failure, e.g., when a thermocouple or RTD has been detected to be open

burnout, up-scale—directing an instrument to fail at its upper scale when it detects a failure, e.g., when a thermocouple or RTD has been detected to be open

bypassing—the process of defeating the purpose or function of a device or system; a physical means to bypass a field instrument such as a control valve. Bypassing may be physical in nature (e.g., hardwired switch, a wire jumper, or a valve that bypasses a control valve or shutdown valve) or software based (e.g., a change in a program due to a bypass request to bypass a function or a software function that is forced into a state by means of a forcing function)

C

causal chain— a linked chain of causes and effects originating from a root cause

cavitation erosion— damage to a valve caused by gas generation (flashing) across the valve when the downstream pressure is sufficient to collapse the gas bubbles

CCST—certified control systems technician

chaff—irrelevant or misleading information surrounding key or desired information

charge—the quantity of excess electrons (negative charge) or protons (positive charge) in a physical system, usually expressed in coulombs

checkout—determination of the working condition of a system

choking—condition in a valve flowing a liquid or gas when the maximum mass flow is reached. For a liquid when the downstream pressure at the venta contracta (lowest pressure point) is lower than the vapor pressure of the material causing the liquid to flash into vapor, which will choke the valve. For a gas, the choking occurs when the velocity of the gas approaches sonic velocity. These conditions occur when the downstream pressure reaches approximately half of the upstream pressure. Increased flow can only occur with an increase in the upstream pressure

circuit—a complete conducting path where electricity flows from and returns to its source

class—the electrically hazardous (classified) area designator that identifies the type of combustible material involved; Class I is gas or vapor, Class II is dust, and Class III is flyings

common-cause failure—a failure of multiple components or instruments due to a common cause; temperature and corrosion are typical common causes

complex system—a system with many components, connections, interconnections, states, or arrangements

component, capacitive—capacitors such as power supply filtering capacitors and distributed capacitors between wires and ground

conductor—a material through which electricity flows easily

confined space—any space that can be deficient of enough air to breathe

contact—part of an electrical relay or solid-state switch that controls the flow of electricity; in a closed contact, electrical current can flow; in an open contact, no current can flow

corrosion—the unwanted dissolving or wearing away of a material (usually metal) due to chemical reaction

critical instrument—an instrument system or alarm considered critical to maintaining the safety of the facility, for environmental protection, or for asset protection. Some facilities also define critical instruments that are critical to maintaining operations or production. Critical instruments for safety and environmental protection typically have specific operation and maintenance procedures and testing frequencies

current electricity—the flow of electrons in complete paths, from source and return to source

D

debottleneck—to remove bottlenecks (areas that limit capacity) in a plant

deduction—drawing conclusions by reasoning

de-energize to trip—arrangement in which electrical energy must be removed to shut down a process or machine

direct current (DC)—electricity whose voltage is constant with respect to time

distributed control system (DCS)—a group of controllers that handle multiple loops connected to operator interfaces and higher-level computing and archiving devices via interconnected data highway(s) distributed throughout a facility; in general, controllers, operator interfaces, and higher-level devices are located in separate areas

diversity—the range of different types of systems in a facility or plant; use of different hardware, software, or methods to minimize common-cause failures in redundant systems

division—the electrically hazardous (classified) area designator indicating the probability of the flammable hazard existing and the physical extent of the hazard; Division 1 means the hazard is present under normal or abnormal conditions; Division 2 means the hazard is present only under abnormal conditions

DMM—see *multimeter, digital*

d/p cell—differential pressure transmitter

dust ignition-proof enclosure—a type of enclosure construction for Class II dust areas, enclosed in such a manner that will exclude ignitable amounts of dusts or amounts that might affect performance or rating and that where installed properly will not permit arcs, sparks, or heat generated or liberated inside the enclosure to cause the ignition of exterior accumulations or airborne suspensions of a specified dust on or around the enclosure (NFPA)

dutchman—a flange ring installed between the instrument and the process that allows venting or testing; used on flange-mounted instruments

E

earthing—a term used outside the United States for grounding

egress—a means to exit a location; for safety purposes, normally two safe means of egress must be provided; see NFPA-101—Life Safety Code

electrical diagram—a type of electrical drawing that shows point-to-point wiring for electrical circuits that do not have a standard format. Sometimes used synonymously with 'electrical schematic'

electrical protective equipment (EPE)—protective equipment used during maintenance and other activities on electrical equipment and circuits

electrical schematic—a type of electrical drawing that shows point-to-point wiring for electrical circuits that do not have a standard format. Sometimes used synonymously with 'electrical diagram'

electromagnetic interference (EMI)—interference due to electromagnetic fields

emergency shutdown system (ESD)—system designed to shut down equipment or part or all of a facility during an emergency

EMI—see *electromagnetic interference*

energize to trip—arrangement in which electrical energy must be applied to trip or shut down a process or machine

engineering controls—controls that are engineered into the installation to insure safe operation and maintenance of the installation; these can include interlocks, guards, signs, shutdowns, etc.

engineering units—the physical values that signals represent, e.g., gallons per minute, inches, degrees

E/P—voltage (E) to pneumatic (P) transducer

EPE—see *electrical protective equipment*

ESD—see *emergency shutdown system*

error—1) an abnormal or undesired result of an operation caused by a fault; **2**) the difference between a desired value and an actual value

erosion—the unwanted wearing away of material due to the flowing of materials and/or high velocities

evergreen document—a document that is required to be updated throughout its lifetime

explosionproof enclosure—enclosure construction for Class I areas capable of withstanding an explosion of the specified gases or vapors inside the enclosure to prevent the ignition of the specified gases or vapors surrounding the enclosure by sparks, flashes, venting of gases and which operates at an enclosure external temperature that will not ignite the specified gases or vapors surrounding the enclosure (NFPA)

F

faceplate—a DCS video construct that resembles a single loop controller front face

factory acceptance test (FAT)—testing by the user of an instrument system that occurs at a vendor or manufacturer's site before the user accepts the system

fail-safe—a failure that drives the system to a safe state (see Chapter 3 for more failure state terms)

failure—the inability of a functional unit to perform its expected tasks

failure, covert—a failure that is not noticed; also called a *latent failure*

failure, dangerous—a failure that puts the system in a dangerous state

failure, latent—a failure that is not noticed; also called a *covert failure*

failure, overt—a failure that is noticed upon failure; also called a *self-revealing failure*

failure rate (λ)—total number of failures during a specified interval divided by the total number of life units (hours, years, cycles, counts, etc.)

failure, self-revealing—a failure that is noticed upon failure; also called an *overt failure*

FAT—see *factory acceptance test* and *field acceptance test*

fault—a temporary or permanent condition in a functional unit that makes it deviate from its expected sequence of operation

fault containment—isolation of the fault as close to the fault location as possible; also known as fault isolation

fibrillation, ventricular—cardiac arrhythmia of the ventricular muscle; a frequent cause of cardiac arrest

field acceptance test (FAT)—test of instrumentation installed on a user's site before acceptance of the instruments. More commonly known as a site acceptance test (SAT)

fieldbus—a name given to a group of varied digital communication protocols and physical layers that connect field instruments to control equipment room instruments

final control element—the final control element on the output side of a control loop that modulates or controls the process; the most common final control element is a control valve

firewatch—a person assigned to monitor maintenance or construction activities in a plant, commonly an operator

flame-proof—the European name for a type of enclosure construction for Class I areas capable of withstanding an explosion of the specified gases or vapors inside the enclosure to prevent the ignition of the specified gases or vapors surrounding the enclosure by arcs, sparks, flashes, venting of gases and which operates at an enclosure external temperature that will not ignite the specified gases or vapors surrounding the enclosure; these enclosures can be built differently from the American explosion-proof enclosure due to different testing requirements and wiring methods

flame-retardant clothing (FRC)—clothing that resists fire; Nomex is one of the materials used to make this type of clothing

flashing—where the downstream pressure across a valve is sufficiently low to cause the process liquid to change into gas; occurs when the downstream pressure is roughly one half or less of the upstream pressure

forcing—software function available on many programmable logic controllers that allows a function to be forced into another state

frame—a defined receive or transmit block of commands, data, error checking, etc.

framework—a basic structure to operate or build from

FRC—see *flame-retardant clothing*

functional failure—where an instrument fails to perform its function but there is no hardware or software failure; the instrument was asked to do something it was not capable of doing

G

GIGO—computer term meaning either garbage in/garbage out or garbage in/gospel out

grounding—connecting conductors or conducting materials to earth or something that serves in place of earth

group—electrically hazardous (classified) area designator that identifies the physical properties of chemicals involved

H

handshaking—signals between communication devices that control data flow; common ones are ready-to-send (RTS), clear-to-send (CTS), data-set-ready (DSR), and data-terminal-ready (DTR); software handshaking signals (XON and XOFF) are also sometimes used

HART—Highway Addressable Remote Transducer – an older but popular de facto digital data communication standard that communicates on top of a 4–20 mA current loop using frequency shift keying (FSK) with 1200 Hz representing a binary one and 2400 Hz representing a binary zero. Considered a fieldbus

hot-cutover—the process of transferring the operation of the current instrumentation system to a new instrumentation system while the process is running

hot standby system—a type of fault tolerant system that has a standby system that is always powered up and reading I/O but is disconnected from system outputs. When a fault occurs in the primary or active system, the system switches to the secondary or backup system. This provides improved reliability but does not provide an increase in the safety of such a system

I

impulse lines—pressure-conducting lines that connect an instrument to a

process or primary element

independent layer of protection (IPL)—a protection layer identified in process hazards and risk analysis with the properties of independence, specificity, dependability, auditability, management of change, and security.

induction—reasoning to a conclusion about all the members of a class from examining a few members of the class; reasoning from the particular to the general

infant mortality period—the initial period in the operational life of an instrument in which it fails due to causes such as manufacturing or materials defects, or improper storage, handling, or installation

inHg—inches of mercury column; 2.04 inHg = 1 psi

instrument—a device used directly or indirectly to measure and/or control a variable

interlock system—a system designed to prevent specific actions or hazardous conditions; also known as *emergency shutdown system (ESD)*.

1) To arrange the control of machines or devices so that their operation is interdependent in order to assure their proper coordination [RP55.1];

2) Instrument that will not allow one part of a process to function unless another part is functioning;

3) A device such as a switch that prevents a piece of equipment from operating when a hazard exists;

4) A device to prove the physical state of a required condition, and to furnish that proof to the primary safety control circuit.

intrinsically safe—electrical system designed such that under normal or abnormal conditions, sufficient energy cannot be released in the hazardous area so as to serve as an ignition source

IN. WC—inches of water column; used in calibration of d/p cells and pressure transmitters for flow, pressure, and level measurement; 27.7 IN. WC = 1 psi

I/P—current (I) to pneumatic (P) transducer

L

lockout/tagout (LOTO)—a procedure to remove power from equipment and processes, lock the means that removes the power, identify the lockout, and provide a procedure for unlocking

loop—an instrument complete circuit; typically shown on a loop drawing; may consist of both input and outputs such as a transmitter, controller, and valve or may be an input and/or output to a DCS/PLC system where the rest of the loop is in software.

1) A combination of two or more instruments or control functions arranged so that signals pass from one to another for the purpose of measurement and/or control of a process variable.

2) Synonymous with control loop. See closed loop and open loop.

3) A complete hydraulic, electric, magnetic or pneumatic circuit.

4) All the parts of a control system: process or sensor, any transmitters, controller, and final control element.

loop drawing (sheet)—a drawing, often 11 x 17, showing a single instrument loop and providing information regarding the instruments in the loop; reference drawings are typically included

LOTO—see *lockout/tagout*

M

maintainability—the characteristics of a design or installation that determines the ease, economy, safety, and accuracy with which maintenance actions can be performed

management of change (MOC)—a formal system of managing changes for a process. In modern times, almost all changes other than superficial ones typically go through management of change to ensure that the change is appropriate and safe. All changes on safety instrumented systems, critical instruments, and instrumented independent layers of protection must go through management of change

manlift—mobile equipment used to raise workers above grade or floor level for repair or installation work

marshalling cabinet—large cabinet in an equipment or control room where multiconductor wiring cables are terminated before running to control instrumentation

material safety data sheets (MSDS)—data sheets provided by the manufacturer detailing the safety hazard data for a chemical

maximum experimental safe gap (MESG)—maximum gap (flame path) where explosive gases can be vented from an enclosure without causing an explosion or fire outside the enclosure

mean-time-between-failures (MTBF)—a measure of reliability for repairable equipment; equal to the mean-time-to-failure (MTTF) plus the mean-time-to-repair (MTTR); expressed in life units (hours, years, cycles, counts, etc.)

mean-time-to-failure (MTTF)— 1) Equal to the total number of life units (hours, years, cycles, counts, etc.) divided by total number of failures within a population during a particular measurement interval under stated conditions. **2**) A measure of system reliability for nonrepairable equipment. See MTBF for repairable systems

mean-time-to-failure spurious (MTTFs)—the mean time between spurious

trips of a safety instrumented system. The inverse of the spurious trip rate

mean-time-to-repair (restore, restoration) (MTTR)—a measure of maintainability; the mean time needed to repair a piece of equipment; the sum of the maintenance time for a piece of equipment divided by the number of repair incidents

means—a way or method of accomplishing an end or purpose

mentoring—an experienced technician helping one or more inexperienced technicians to learn job skills; not usually a formal program

meter—a field instrument such as a flow, pressure, temperature, or level transmitter

Modbus—a generic communication protocol developed by Modicon, which has become a de facto communication protocol in the process industry

Modbus/TCP—a TCP/IP protocol that encapsulates the original Modbus protocol that allows it to run on Ethernet, the Internet, etc.

monitor, fire—a fixed device that can spray water over an area for fire protection

Motor Operated Valve (MOV)—A valve whose actuator is a motor. Common in refineries

motor schematic—a type of drawing that shows the motor protection and control circuits, typically in a ladder diagram format

MOV – Metal Oxide Varistor—a common surge protection device whose resistance increases with voltage. Repeated high-voltage transients can damage the MOV leading to failure

mulitmeter, analog (VOM)—device that measures electrical voltage, current, and resistance and displays readings on analog gauges

mulitmeter, digital (DMM)—device that measures electrical voltage, current, and resistance and displays readings in a digital format

N

NEC—National Electrical Code (NFPA-70)

National Fire Protection Association (NFPA)—a U.S. national safety code body; the National Electrical Code (NEC) is probably the best known of these codes

nest—an older name for a rack

nonconductor—a material through which electricity does not flow easily

nonincendive—electrical system designed such that under normal conditions, sufficient energy cannot be released in the hazardous area so as to serve as an ignition source

null modem—a communication wiring adapter that crosses the receive and transmit lines in an RS-232 circuit

O

off-line—work or testing that takes place while the process is not running or operating

on-line—work or testing that takes place while the process is running or operating

on-the-job-training (OJT)—training method in which workers learn while working, used by some companies in lieu of more formal training

operations—the department responsible for the operations necessary to make a product in a plant or facility

operator—a person responsible for operating a plant or unit

P

parameters—the physical properties whose values determine the functions or operations of a system

pay meter—a flow meter by which a plant either buys feedstocks or sells product

personal protective equipment (PPE)—equipment worn to provide protection against safety hazards; can include safety glasses, safety shoes, flame retardant clothing (FRC), face shields, gloves, voltage gloves, monogoggles, flash suit, etc.

physical layer—the wiring, voltage, current, and other physical and electrical parameters of a device in a digital communication transmission circuit; does not define what the digital signals mean

pipeway—a support structure for pipes

piping and instrument diagram (P&ID)—drawing that shows the arrangement of piping and instrumentation in a system

plant dialect—the terms and abbreviations workers use to describe their plant or facility and the operations that occur there

port—a process connection (i.e., process port)

positioner—device mounted on a control valve that controls the position of the valve stem. A position controller, which is mechanically connected to a moving part of a final control element or its actuator, and automatically adjusts its output pressure to the actuator in order to maintain a desired position that bears a predetermined relationship to the input signal. The positioner can be used to modify the action of the valve (reversing positioner), extend the stroke/controller signal (split range positioner), increase the pressure to valve actuator (amplifying positioner) or modify the control valve flow characteristic (characterized positioner).

potential—voltage or charge level

power-line frequencies—50 or 60 Hz (cycles)

primary element—the element that is directly in contact with the process during the measurement process

probability—the likelihood of occurrence of a specified event

probability of failure on demand average—the average probability that a safety instrumented system (SIS) will fail to operate upon a safety demand

process taps—the connection point to the process

programmable electronic system (PES)—a system for control, protection or monitoring based on one or more programmable electronic devices

programmable logic controller (PLC)—a purpose-built computer control system primarily designed to do discrete and sequential logic but that is capable of continuous and other types of control

proof test—test performed to reveal undetected faults in a safety instrumented system so that, if necessary, the system can be restored to its designed functionality

protocol—a digital communication procedure that defines which data and commands will be transmitted; a protocol does not normally define the wiring and electrical parameters

proven-in-use (prior use)—a component may be considered as proven-in-use when a documented assessment has shown that there is appropriate evidence, based on the previous use of the component, that the component is suitable for use in a safety instrumented system

purge—

1) to ventilate an enclosure;

2) to use pressurization and ventilation to reduce the area classification of an enclosure or room;

3) to flow a material that is innocuous to the process at slightly higher pressure into a process tap to keep it clean

purge, Type X—a purge that reduces the area classification in the purged enclosure from Division 1 to nonhazardous

purge, Type Y—a purge that reduces the area classification in the purged enclosure from Division 1 to Division 2

purge, Type Z—a purge that reduces the area classification in the purged enclosure from Division 2 to nonhazardous

R

rack—a rectangular container that contains slots that power supply cards, processor cards, communication cards, special cards, and I/O cards slide into to create a system; an older name for a rack was a nest

radio frequency—frequencies at which electromagnetic radiation can be used for communications purposes; roughly 100 Khz to 100 Ghz

radio-frequency interference (RFI)—interference or noise originating outside a device or system in the frequency range of 100 Khz to 100 Ghz

random-failure period—the period in the life of an instrument where the failure rate is constant and failures are considered statistically random; also known as the constant-failure period and the useful-life period

ready-to-work permit—a permit indicating that the operations department has determined that equipment to be worked on is in a safe state

reboot—a process in which a computer-based device restarts; also known as a *reset*

receiver instrument— an instrument that receives a signal from a transmitter or transducer and displays or records it

remote input/output (RIO)—in programmable logic controllers, usually input/output racks connected by a serial link to a main processor rack

relay—

1) an electrical switching device that allows a low voltage to control a high voltage or current;

2) a pneumatic control device used to modulate a higher pressure with a lower pressure;

3) an intermediate instrument between a transmitter and a receiver or a controller, or a controller and a final control element

reliability—the probability that an instrument can perform its intended function for a specified interval (time) under stated conditions

reset—a process in which a computer-based device restarts; also known as a *reboot for computer systems*

respirator—a breathing device that filters out chemicals or dust but does not provide breathing air

RFI—see *radio-frequency interference*

rod-out—a process of cleaning out a hole, pipe, or process connection using a metal rod; normally done under pressure and requires protective gear

root cause—the initiating or original cause in a causal chain

root valve—the block valve closest to the process; the main block valve

RTD—resistance temperature detector, a temperature measuring device that is based on the temperature dependence of the resistance of various metals. Platinum, copper, and nickel are common metals used for RTDs

S

safety instrumented function (SIF)—an instrumented safety function that protects against a single hazard.

safety instrumented system (SIS)—an instrument system that has one or more safety functions; also known as safety systems, emergency shutdown systems (ESD), interlock systems, critical instrument systems, etc.

safety integrity level (SIL)—the reliability level required to maintain an acceptable level of safety. There are four discrete defined safety integrity levels, SIL 1, SIL 3, SIL3, & SIL 4.

safety requirements specification (SRS)—an evergreen specification that contains all the requirements of the safety instrumented functions that have to be performed by the safety instrumented systems

Scott Air Pack—brand name for a self-contained breathing apparatus (SCBA); sometimes used as a generic name for SCBA

self-contained breathing apparatus (SCBA)—portable breathing unit providing freedom of movement

site acceptance test (SAT)—testing that takes place after purchased equipment has been installed in the field. Sometimes refers to additional testing of purchased equipment on site that has been powered up but not installed or partially installed (external interfaces connected)

skin effect—electrical phenomenon in a conductor: as the frequency of the electrical current increases, the electron flow moves closer to the surface of the conductor

sniffer—a portable device that can detect flammables, toxic chemicals, and oxygen content

software safety requirements specification (SSRS)—an evergreen specification that contains all the requirements for the programming of the safety instrumented functions that have to be performed by the safety instrumented systems

spurious trip—a trip or operation of a safety instrumented system that is not due to a safety demand. Also known as a nuisance trip

spurious trip rate (STR)—spurious trips per year. The inverse of the mean-time-to-failure spurious (MTTFs)

staging—assembling an instrument system for the purpose of testing; staged systems are also sometimes used for training

standard maintenance instructions (SMI)—approved instructions for doing maintenance of equipment in a plant

standard maintenance procedure (SMP)—procedures used to standardize maintenance practices in a plant

standard operating procedure (SOP)—procedures used to standardize the operation of a plant

static electricity—accumulation of excess electrons or the shortage of electrons of a surface

stress—strain put on an instrument during its operational life; stressors can include temperature, process pressure, corrosion, abuse, misoperation, etc.

system—a collection of devices that work together for a common purpose

systematic failures—failures due to human error

T

tap—process connection

T/C—abbreviation for thermocouple

trend, trend chart—a paper or electronic recording of past values of a process variable over time

trip—any condition that causes a safety or equipment protection system to activate

trip, spurious—a trip caused by something other than the system's designed protective function

turbine meter—a type of flow meter that uses a rotating propeller or turbine to measure flow

U

uninterruptible power supply (UPS)—a system that provides power upon power loss, usually with a combination of batteries and backup generators; some also provide conditioned power during normal operations

useful life—the period in the life of an instrument where its failure rate is constant; also known as the random failure rate period

V

valve plug—a controlling element in a control valve that modulates the flow or pressure

valve seat—a stationary piece in a control valve that the controlling element modulates against

VOM—see *multimeter, analog*

W

wear-out period—period in which an instrument has reached the end of its useful life and is rapidly wearing out

wet leg—the low-pressure side of a differential pressure level transmitter that is filled with the process fluid or other suitable liquid

wetted parts—those parts exposed to the process; includes metals, O rings, seals, etc.

wiredrawing—an effect caused by high fluid velocities in a valve when the valve plug is close to the seat; causes drawing wire-like threads off the seat

Z

zone—electrically hazardous (classified) area designator indicating the probability of the flammable hazard existing and the physical extent of the hazard; there are three zones: 0, 1, and 2